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IMPACT ASSESSMENT REPORT

Accompanying the documents

COMMISSION REGULATION (EU) .../...

of XXX

implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for household tumble dryers, amending Commission Regulation (EU) 2023/826, and repealing Commission Regulation (EU) No 932/2012

and

COMMISSION DELEGATED REGULATION (EU) .../...

of XXX

supplementing Regulation (EU) 2017/1369 of the European Parliament and of the Council with regard to energy labelling of household tumble dryers and repealing Commission Delegated Regulation (EU) No 392/2012

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Glossary

Term or acronym	Meaning or definition
BAT	Best Available Technology
BAU	Business as Usual (a scenario without any further intervention)
CTP	Climate Target Plan
dB	Decibels
dB(A)	A-weighted sound levels
Ec	Energy consumption
EEI	Energy Efficiency Index
EED	Energy Efficiency Directive
EoL	End-of-Life
ESPR	Ecodesign for Sustainable Products Regulation
F-gases	Fluorinated greenhouse gases
GHG	Greenhouse Gas
GWP	Global Warming Potential
HFCs	Hydrofluorocarbons
IA	Impact Assessment
IEC	International Electrotechnical Commission; global standardisation organisation
IPCC	Intergovernmental Panel on Climate Change
LCC	Life cycle cost over the whole lifetime of a product, including purchase cost and energy costs
LLCC	Least life cycle cost; used to determine the energy efficiency requirements that minimise costs of a product for its whole lifetime.
MEErP	Methodology for Ecodesign for Energy related Products
MSA	Market Surveillance Authority
NECP	National Energy and Climate Plan
PO	Policy Option
RSB	Regulatory Scrutiny Board
SEc	Standard energy consumption
TD	Tumble dryer
TFEU	Treaty on the Functioning of the European Union
TWh	Terawatt hour
WEEE	Waste of Electrical and Electronic Equipment
WTO	World Trade Organisation

1. INTRODUCTION: POLITICAL AND LEGAL CONTEXT

1.1. EU energy and climate objectives

The EU has longstanding objectives to increase energy efficiency and to reduce its greenhouse gas emissions. These go along with other objectives to reduce its environmental impacts. In December 2019, the Commission presented the European Green Deal[1] to strengthen these objectives and as the cornerstone of its strategy to fulfil the United Nation's 2030 Agenda for Sustainable Development[2]. In September 2020, it presented a Climate Target Plan (CTP) for 2030[3], showing the need for a higher contribution of energy efficiency and renewable energy to achieve a net 55% GHG emission reduction most cost-effectively, in line with the Paris Agreement. The European Parliament and Council subsequently agreed to achieve this level of reduction in GHG emissions. The Commission followed this by adopting the EU "Fit for 55" package[4] aiming to achieve the necessary cut in GHG emissions.

One pillar of the CTP and subsequently the 'Fit for 55' package is energy efficiency. A revision of the Energy Efficiency Directive (EED)[5] to increase the overall ambition was proposed. This would require Member States to enhance their National Energy and Climate Plans (NECPs) to achieve at least an overall 9% reduction in EU energy consumption in 2030 compared to the reference[6]. The EED itself contains limited energy saving measures and in addition to Member State actions, measures at EU level are needed[7]. In this context, the ecodesign and energy labelling rules for products arise as important instruments to realise EU's energy and decarbonisation objectives.

Another pillar of the European Green Deal is a more circular economy. The new Circular Economy Action Plan[8] sets out steps to work towards this. It aims to reduce product environmental impacts for example through promoting longer product lives, greater resource efficiency and enhancing recycling and recycled content. Certain of these aspects can be addressed through measures under the Ecodesign Directive since decisions made at design can influence the product's life span and ease of maintenance, repair, reuse, upgrade, recyclability and waste handling. The Ecodesign Working Plan 2016-2019 identified ecodesign measures' potential to contribute to circular economy objectives and subsequent supporting studies have systematically considered resource efficiency aspects.

Reducing energy use and promoting the circular economy are also important for many other reasons[9]. Lower energy use reduces the EU's energy import dependence and improves energy security, aspects that are particularly relevant in the current context of continuous increase of energy prices. On 8 March 2022, the Commission published its Communication "RepowerEU: Joint European action for more affordable, secure and sustainable energy"[10], being one of its pillars

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[1] The European Green Deal, COM(2019) 640 final
[2] Transforming our world: the 2030 Agenda for Sustainable Development
[3] Stepping up Europe's 2030 climate ambition. Investing in a climate-neutral future for the benefit of our people (COM/2020/562 final)
[4] 'Fit for 55': delivering the EU's 2030 Climate Target on the way to climate neutrality, COM(2021) 550 final
[5] Proposal for a Directive of the European Parliament and of the Council on energy efficiency, COM(2021) 558 final
[6] EU reference scenario 2020 - Publications Office of the EU (europa.eu)
[7] These additional measures include not only ecodesign and energy labelling but also the EU ETS, vehicle CO_2 legislation and the Energy Performance of Buildings Directive
[8] Circular Economy Action Plan for a more competitive Europe, COM(2020) 98
https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52015DC0614&from=EN
[9] https://www.odyssee-mure.eu/data-tools/multiple-benefits-energy-efficiency.html
[10] EUR-Lex - 52022DC0108 - FR - EUR-Lex (europa.eu)

the fast reduction of EU dependence on fossil fuels at the level of homes, buildings and the industry by boosting energy efficiency gains.

Ecodesign and energy labelling contribute to improved air quality and support energy system integration and the alleviation of energy poverty. It is also expected to lead to the creation of jobs, encourage innovation and support and facilitate economic growth[11]. Both energy savings and promoting the circular economy lead to reduced environmental damage from materials extraction and resource efficiency. Most of these co-benefits are difficult to quantify, but are well known and perceived by society.

In March 2022 the Commission adopted the ESPR (Ecodesign for Sustainable Products Regulation) a proposal to replace the Ecodesign Framework Directive[12], which will expand the scope of ecodesign from energy-related products to any kind of physical good that is placed on the market, with some exceptions that are set out in the ESPR itself. Ecodesign product regulations adopted under the existing Directive would nevertheless continue in force until repealed. When the proposed legislation is eventually adopted future amendments to existing product regulations would be carried out under its framework. The adoption of the ESPR is expected to have little influence on the results of this IA, as it is further developed in section 5.1.

1.2. The contribution of Ecodesign and Energy labelling

There are market barriers that hamper the introduction and uptake of more energy efficient and more long lasting products. It is desirable for policies to address these barriers since they lead, from the point of view of the consumer and society, to unrealised economically viable energy savings. Behavioural failures refer to the cognitive limitations and biases that prevent consumers and investors to appreciate rationally the benefits of energy efficiency[13]. Market failures arise from the fact that many impacts and aspects of energy supply or resource use are not brought into market prices[14]. Market barriers such as lack of information and awareness or financing challenges result in economically rational energy savings not being realised or more sustainable choices not being made.

The EU Ecodesign Directive aims to address these barriers by setting performance requirements to remove the worst performing products from the market. The assessment aims to ensure that the minimum requirements are set at the level of Least Life Cycle Cost (LLCC). These harmonised requirements have specifically addressed energy efficiency and have led to significant reductions in energy use of household and industrial products. Being set at EU level, they have avoided subjecting industry to multiple national rules.

Energy labelling rules in turn complement ecodesign requirements, by providing information to consumers with the aim of encouraging them to purchase products that have a better energy performance than the minimum. This has helped consumers to reduce their energy bills by easily identifying and comparing more energy efficient appliances. Nearly 80 % of the EU public recognise the label and say it has influenced their purchase decision[15].

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[11] https://combi-project.eu/2018/06/22/combi-results-overview-policy-conclusions/

[12] Directive 2009/125/EC of the European Parliament and of the Council of 21 October 2009 establishing a framework for the setting of ecodesign requirements for energy-related products (OJ L 285, 31.10.2009, p.10)

[13] DellaValle N., Bertoldi P. (2021) "Toward a more situated energy efficiency policy agenda".

[14] E.g. impacts on air pollution, biodiversity, resource use, climate change and energy security

[15] Special Eurobarometer 492. "Europeans' attitudes on energy policy report. European Commission September 2019.

The impact of ecodesign and energy labelling measures is gathered in the Ecodesign Impact Accounting[16], which estimates that in 2020 a quarter of the energy savings and a third of the greenhouse gas emissions reductions to achieve the EU's 20-20-20 climate and energy policy goals was due to the application of those measures.

The benefits of ecodesign and energy labelling are not limited to the EU energy and environmental goals but they also contribute to some of the 17 Sustainable Development Goals (SDGs) adopted by the UN. By reducing in-use energy consumption and the subsequent environmental impacts such as GHG emissions or acidification, ecodesign and energy labelling measures contribute to the SDGs 7 (affordable and clean energy), 10 (reduced inequalities), 11 (sustainable cities and communities) and 13 (climate action). In particular the following targets included in the abovementioned SDGs are addressed: target 7.3 (double the improvement of energy efficiency), 8.4 (improve resource efficiency in consumption and production), 10.1 (reduce income inequalities) by reducing the consumer's energy bill and notably for those in situation of energy poverty, 11.6 (reduce the environmental impact of cities) and 13.1 (strengthen resilience and adaptative capacity to climate-related hazards). In addition, circular economy measures under ecodesign also contribute to the following SDGs and targets: the abovementioned targets 8.4 and 11.6 by lengthening the lifetime of the product, reducing waste and increasing recyclability and SDG 12 (responsible consumption and production) and in particular its targets 12.2 (sustainable management and use of natural resources) and 12.5 (substantially reduce waste generation).

Annex 3 quantifies the impact of the preferred policy option on the abovementioned SDGs.

1.3. Legal context of the impact assessment

Article 7 of the Ecodesign and of the Energy Labelling Regulations for household tumble dryers[17][18](henceforth "the TD Ecodesign Regulation" and "the TD Energy Labelling Regulation", respectively) mandate a review of the legislation on tumble dryers. In particular, the review should have regard to the verification tolerances and the efficiency of the appliances. In addition, Article 11 of the Energy Labelling Framework Regulation (EU) 2017/1369[19] mandates a rescaling of the existing energy labels to an A to G scale, in a way that no products are expected to fall into energy class A at the moment of the introduction of the label.

Consequently a review[20] of the TD Ecodesign and Energy Labelling Regulations was included in the Ecodesign Working Plan 2016-2019[21]. This noted that the review would also assess aspects relevant to the circular economy. The impact assessment builds therefore upon the current ecodesign rules on tumble dryers and is intended to keep the relevance of the existing legislation by adapting it to technical progress.

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[16] Ecodesign impact accounting annual report 2020

[17] Commission Regulation (EU) No 932/2012 of 3 October 2012 implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for household tumble driers (OJ L 278, 12.10.2012, p. 1)

[18] Commission Delegated Regulation (EU) No 392/2012 of 1 March 2012 supplementing Directive 2010/30/EU of the European Parliament and of the Council with regard to energy labelling of household tumble driers (OJ L 123, 9.5.2012, p. 1)

[19] Regulation (EU) 2017/1369 of the European Parliament and of the Council of 4 July 2017 setting a framework for energy labelling and repealing Directive 2010/30/EU (OJ L 198, 28.7.2017, p. 1)

[20] Review study on tumble dryers, Final Report, Viegand Maagøe and Danish Technical Institute, May 2019 for the European Commission

[21] Communication from the Commission Ecodesign Working Plan. COM(2016) 773 final, Brussels, 30 November 2016. (Ecodesign Working Plan 2016-2019)

2. PROBLEM DEFINITION

2.1. What are the problems?

2.1.1. The first key problem: missed in-use energy savings from tumble dryers

The EU Reference Scenario 2020 projects energy consumption to be reduced to 1.124 Mtoe of primary energy and 864 Mtoe of final energy in 2030, which is insufficient to attain the energy efficiency target under the fit-for-55 package. As indicated in section 1.1, the draft revision of the EED sets out an additional binding reduction of 9% of energy consumption in the EU from energy efficiency measures, which would bring the energy consumption down to 1.023 Mtoe of primary energy and 787 Mtoe of final energy by 2030.

The revised EED will mandate Member States to lay down measures at national level to collectively contribute to the EU binding target. Co-responsibility between Member States is necessary because only collectively it will be possible to meet the global EU energy and climate goals, but entails the risk that some Member States may not fully realise their national objectives, jeopardising the achievement of the 9% target. Ecodesign and energy labelling measures (which unlike measures in the scope of the EED have the advantage of being directly applicable across the EU) must contribute to that 9% by reducing energy consumption on top of the reductions achieved under the EED.

Energy consumption in the household sector accounted in 2019 for about 28% of the EU 27 final energy consumption[22], namely **248 Mtoe**. The in-use energy consumption from **tumble dryers** reached 1,3 Mtoe in 2020[23], equivalent to 0,5% of the EU final energy consumption from households and to 12% of the final energy consumption from household appliances. Further energy savings from tumble dryers could help to reduce energy consumption and increase the contribution of the household sector to the overall energy efficiency target. Nevertheless, the regulations for tumble dryers will not lead to further energy savings from this product group. This is indicated in section 7.2.2 of the review study[24], which explains that the two regulations currently in force have worked on pushing the EU market towards more efficient tumble dryers, but that further improvements are still possible by intensifying the presence in the market of tumble dryers that are based on the BAT (Best Available Technology).

2.1.2. The second key problem: market failures due to lack of regulatory provisions on circular economy

The lack of regulatory provisions on circular economy lead to market failures because embedded energy consumption during production, distribution and EoL lead to negative environmental effects that are not internalised in the price of tumble dryers. This makes industry not to receive sufficiently strong signals as to incorporate resource efficiency criteria in the design and manufacturing of their products. In addition, circular economy and climate and energy policies like energy efficiency, have not always been well integrated before the adoption of the Green Deal. For instance, the IPCC's Summaries for Policy Makers hardly mentioned circularity or resource efficiency as a decarbonisation option, and the own first Circular Economy Action Plan of 2015 vaguely mentioned climate change while acknowledging that "to date ecodesign requirements have mainly targeted energy efficiency". Nowadays, scientific literature has demonstrated the connection between circular economy, energy savings and climate change, thanks to seminal studies like Material Economics' and Sitra's, entitled *The circular economy – A powerful force for climate*

[22] From Eurostat Energy Balances (2022 edition)
[23] From EU Reference Scenario 2020
[24] Viegand Maagøe A/S, Review study on household tumble driers, Final report for EC DG ENER, June 2019.

mitigation (2018)[25], from the UN's International Resources Panel[2627] or even by the Commission's in-depth analysis for the Long-Term Strategic Vision to reduce GHG emissions. Now these interlinkages are acknowledged at both scientific and policy levels, notably under the European Green Deal and its initiatives (e.g. Renovation Wave, EPBD revision).

2.1.3. A third additional problem: reduced effectiveness/unnecessary complexity of some legal requirements

Some aspects of the legislation should be revisited because they are either unnecessary or inadequately addressed, creating confusion and reducing the effectiveness of the current legal framework. Although these aspects do not entail a quantifiable environmental impact, the revision of the legislation is an opportunity to improve them. The related drivers are already enumerated in this section in order to avoid over-complication of the next section 2.2, which will focus on the drivers to the two key problems.

Unnecessary scale of condensation efficiency

The condensation efficiency is a measure for the emission of moisture from the appliance to the ambient and has a small influence on energy efficiency. According to GfK market data from 2019, confirmed by EPREL (European Product Database for Energy Labelling) data from 2021, almost all models on the EU market have condensation efficiencies above 80% and half of the models have a condensation efficiency of 84% and above. As a result, the minimum value for the condensation efficiency (set in the TD Ecodesign Regulation at 70%) and the condensation efficiency classes displayed on the energy efficiency label are meaningless, since almost all new tumble dryers are concentrated in classes A and B. The condensation efficiency information included in the energy label has therefore no added value.

Complex approach to low power modes

Minimum values for energy consumption in low power modes applicable to tumble dryers are set in the Horizontal Standby Regulation[28] but they are not included in the TD Ecodesign Regulation. However, the TD Ecodesign Regulation includes factors linked to low power modes, which are used in the formula for the overall calculation of the energy efficiency. The existence of requirements on low power modes in both the TD Ecodesign and Standby Regulations results in an unnecessary complexity of the legislative framework. This approach is not in line with the legislation for other products such as washing machines and dishwashers, for which the factors have been deleted and replaced by minimum low power mode values. At the same time, those values have been removed from the Horizontal Standby Regulation in order to avoid duplicities.

Increased length of drying cycles

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[25] Material Economics and SITRA (2018) *The circular economy – A powerful force for climate mitigation.* Stockholm, Material Economics Sverige AB. In: https://media.sitra.fi/2018/06/12132041/the-circular-economy-a-powerful-force-for-climate-mitigation.pdf

[26] UNEP (2017) *Resource Efficiency: Potential and Economic Implications*. Nairobi: UN Environment. In: http://www.resourcepanel.org/reports/resource-efficiency

[27] IRP (2020) *Resource Efficiency and Climate Change: Material Efficiency Strategies for a Low-Carbon Future*. Hertwich, E., Lifset, R., Pauliuk, S., Heeren, N. A report of the International Resource Panel. Nairobi: United Nations Environment Programme.

[28] Commission Regulation (EC) No 1275/2008 of 17 December 2008 implementing Directive 2005/32/EC of the European Parliament and of the Council with regard to ecodesign requirements for standby and off mode, and networked standby, electric power consumption of electrical and electronic household and office equipment (OJ L 339, 18.12.2008, p.45).

Manufacturers interest to reach higher efficiency classes has resulted in compensating the decrease of temperature by an increase of the average length of the drying cycle. Consumer associations report complaints and irritation from users regarding the often very long time that a drying cycle takes. This issue may undermine the credibility of the label and is not in line with Art. 15(5) of the Ecodesign Framework Directive, which requires the implementing measures not to bring about a significant negative impact on the functionality of the product from the perspective of the user. APPLiA (Home Appliance Europe, main representative of the appliance manufacturing sector at EU level) consumer survey[29] shows that a maximum cycle time of 240 minutes for the standard cotton cycle at full load is the threshold for a cycle duration to be acceptable.

Phase out of gas-fired tumble dryers

Since 2020 there are no more gas-fired tumble dryers on the EU market[30]. The TD Energy labelling Regulation becomes therefore unnecessarily complex by including in the label information on a technology that is definitely phased-out.

Incomplete information on noise emission

The noise emission from tumble dryers is shown on the current energy label in dB(A), but no noise classes have been defined. The user is therefore confronted with an absolute value but is not able to know how good that value is. Noise emission classes have already been included on the labels for washing machines, dishwashers and refrigerators, but this is still not the case for tumble dryers.

Misalignment between the energy label of tumble dryers and the energy label of products with similar functionalities

Washer-dryers, an appliance that combines washing and drying laundry, have been included for the first time in the revised versions of the Ecodesign and Energy Labelling Regulations on washing machines[3132]. The energy label of tumble dryers is not aligned with the dryer-part of the washer-dryer label, thus both labels using different layouts for exactly the same functionality.

2.2. What are the problem drivers?

2.2.1. Lenient energy efficiency index (EEI)

The energy efficiency of a tumble dryer is determined by its EEI[33]. The EEI is specific for each tumble dryer model, and is calculated as the quotient between the Energy consumption (Ec) of that model (set out after testing) and the Standard Energy Consumption (SEc) of all tumble dryers on the market at a specific point in time. Ec gives the energy consumption of the specific model for which the EEI value needs to be known, and SEc gives an averaged energy consumption of all the models in the market with the same load capacity. By dividing both values, the EEI compares the energy consumption of the specific model under test with the energy consumption of all the models in the market with the same load capacity, giving an idea of how good the model under test is

[29] APPLiA. Tumble dryer usage and attitudes, March 2018 (not publicly available)
[30] Production stopped in 2019 but products in wholesale or retail may still have been sold in 2020 (or later).
[31] Commission Regulation (EU) 2019/2023 of 1 October 2019 laying down ecodesign requirements for household washing machines and household washer-dryers pursuant to Directive 2009/125/EC of the European Parliament and of the Council, amending Commission Regulation (EC) No 1275/2008 and repealing Commission Regulation (EU) No 1015/2010 (OJ L 315, 5.12.2019, p. 285)
[32] Commission Delegated Regulation (EU) 2019/2014 of 11 March 2019 supplementing Regulation (EU) 2017/1369 of the European Parliament and of the Council with regard to energy labelling of household washing machines and household washer-dryers and repealing Commission Delegated Regulation (EU) No 1061/2010 and Commission Directive 96/60/EC (OJ L 315, 5.12.2019, p. 29)
[33] More information on the EEI is given in Annex 4 to this IA.

compared to the rest of the market. It follows from this that the lower the EEI the better the energy performance of a model.

There are currently three tumble dryer technologies: air vented, heating element and heat pump (both heating element and heat pump are in turn grouped as condenser dryers). According to section 7.2.2 of the review study, there is currently a big gap in terms of energy efficiency between heat pump tumble dryers on one side, and heating element and air vented tumble dryers on the other side. Air vented and heating element dryers are the least efficient ones because they consume over 3 times more electricity than the heat pump dryers, for which reason they have lost a significant market share. Nevertheless, in the absence of action a significant number of heating element tumble dryers will still remain in the market as explained in section 5.1, causing an excess of energy consumption and release of GHG emissions. Therefore, under the current TD Ecodesign Regulation, the EEI threshold above which tumble dryers may not be placed on the market is too high and allows inefficient dryers to remain in the market.

2.2.2. Overpopulation of the highest classes of the energy label

The energy label of tumble dryers has not been adapted to technological progress. Section 7.2.2 of the review study explains that 99% of heat pump tumble dryers on the EU market are above energy class A, causing a congestion in the top three energy labelling classes (A+ to A+++). This makes it more difficult for consumers to distinguish between more and less efficient models and reduces the incentive for manufacturers to achieve higher energy efficiencies because they cannot be rewarded with a class higher than A+++. In addition, the A+, A++ and A+++ classes have shown to be less effective than the A to G scale in persuading consumers to buy more efficient products[34].

2.2.3. Current calculation methods do not sufficiently represent real energy consumption from tumble dryers

The real usage and energy efficiency of tumble dryers has changed over the years, but the EEI formula has not followed that change, becoming outdated and unable to provide a value that represents real energy efficiency.

APPLiA has signalled that the current **average load per cycle** and the **annual number of cycles** used in the Ec calculation are overestimated. This is also highlighted in section 7.2.3 of the review study. Under the current TD Ecodesign Regulation, the load is determined by the average of 3 tests in full load and 4 tests at part load. Latest surveys reported in the review study find a proportion of 1 to 3 between full load and part load as well as 107 cycles instead of 160 cycles per year. The most likely cause for this change is the progressive growth in load capacity. In 1995 the average nominal load capacity of tumble dryers was 4,5 kg and the average real-life load 3 kg for 175 cycles/year[35], in 2005 it was 5,5 and 3,4 kg respectively for 154 cycles/year (rounded to 160 in the regulation)[36] and now it is 7,7 and 4,4 kg respectively for 107 cycles. The load capacity has grown faster than the actual load.

Further, the Ec is currently based on an estimation of the average number of cycles per year with the help of consumer surveys. An energy consumption per 100 cycles instead of per year would be

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[34] Impact assessment accompanying the document Proposal for a Regulation of the European Parliament and of the Council setting a framework for energy efficiency labelling and repealing Directive 2010/30/EU, SWD/2015/0139

[35] Group for Efficient Appliances, Washing Machines, Dryers and Dishwashers, preparatory study Vol. III Technical Analysis by VHK for GEA under the SAVE programme, June 1995.

[36] Impact assessment accompanying the document Commission Regulation implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for household tumble dryers.

more informative and would allow the consumer to calculate energy consumption based on the amount of use they make of the appliance.

As far as the SEc, the current procedure overestimates the average energy consumption of the models in the market, resulting in unrealistically low EEIs. Besides lowering the value of the EEI, an overestimated SEc creates an incentive for manufacturers to increase the size of their models, because the reference SEc grows faster than the additional energy consumption resulting from an increase of the rated capacity. As it is explained in Annex 12, the rated capacity of tumble dryers has been steadily increasing since 2010 and this tendency is expected to continue in the absence of further legislative action.

2.2.4. Circular economy: insufficient availability of spare parts

A 2016 study carried out for DG Environment[37] shows that that the availability and prices of spare parts are among the major barriers to reparability. Spare parts are usually not available or it takes a long time to deliver them, creating the incentive to discard the product and replace it instead of repairing it. Some producers deliver spare parts only to their partners, thus setting market barriers for independent repairers.

2.2.5. Circular economy: insufficient repair and maintenance information

Repair and maintenance information is often not available to professional repairers, making it more difficult to diagnose and solve breakdowns and increasing the cost of repair. This problem has also been reflected in the ecodesign review study of washing machines[38] and can be assumed to be similar for tumble dryers.

2.2.6. Circular economy: insufficient information for end-of-life processing

The review study showed that information to facilitate the dismantling of components regulated in the WEEE Directive[39] (e.g. circuit boards, electronic displays) and information on the refrigerants in heat pump dryers[40] can help recyclers to maximise material recovery and reduce costs of recycling.

2.2.7. Circular economy: insufficient user maintenance information

An APPLiA consumer survey[41] shows that only 45% of the European consumers clean the **lint filter** every time they use the tumble dryer. The survey indicates that filters are cleaned less than once a month in 3 out of 10 cases, which is less often than required by manufacturers for an average use profile. Repairers[42] point at insufficient maintenance of the filters as one of the main reasons for breakdown of heat pump dryers. Besides, filters not properly cleaned increase energy consumption.

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[37] Monier, V., et. al., Study on socioeconomic impacts of increased reparability Final report, Deloitte the EC DG ENV, 2016.

[38] Boyano, A., et al., Ecodesign and Energy Label for household washing machines and washer-dryers, JRC IPTS preparatory study, 2017

[39] Directive 2012/19/EU of the European Parliament and of the Council of 4 July 2012 on waste electrical and electronic equipment (WEEE) (OJ L 197, 24.7.2012, p. 38)

[40] The review study mentions that currently the most frequently used refrigerant in heat pump dryers is R134a, a refrigerant with a Global Warming Potential of 1430.

[41] APPLiA. Tumble dryer usage and attitudes, March 2018 (not publicly available)

[42] R.U.S.Z Sepp Eisenriegler interviewed in July 2020.

Information on the filters is available in the user manual, but many users do not read the manual or only read part of it[43] [44].

2.3. How likely is the problem to persist?

As explained in section 5.1, energy consumption from tumble dryers will decrease due to the trend of the market towards heat pump tumble dryers. In spite of this positive evolution, in the absence of regulatory action less efficient tumble dryers will keep a significant share of sales and stock over the next two decades, limiting the reduction of energy consumption and GHG emissions.

As far as circular economy (second key problem), the review study indicates that the lifetime of tumble dryers has decreased from 14 to 12 years over the last 15 years. If no measures are taken the current average lifetime from tumble dryers is likely to remain or even to drop further. Measures in favour of reparability could reverse this trend.

3. WHY SHOULD THE EU ACT?

3.1. Legal basis

Articles 114 and 194 of the Treaty on the Functioning of the European Union (TFEU) are the legal base for measures related to the functioning of the internal market and therefore constitute the legal base for the Ecodesign Framework Directive, the Energy Labelling Framework Regulation[45] and their secondary acts.

In addition, the Ecodesign Framework Directive lays down a set of criteria to identify energy-related products that may be subject to ecodesign measures. In the view of those criteria, the review study concludes that tumble dryers are eligible on the basis of the potential energy savings that can still be achieved through the introduction of additional cost-effective energy savings.

3.2. Subsidiarity: Necessity of EU action

Without harmonised EU requirements on tumble dryers, Member States would be incentivised to lay down energy efficiency requirements at national level. Diverse national approaches with respect to energy efficiency would be an obstacle for the internal market, because products would have to be adapted to diverging national rules, thus risking of higher compliance costs for the industry. Those harmonised EU requirements must now be updated to technological progress to ensure future relevance of the legislation.

In addition, the contribution of ecodesign to the Green Deal objectives, in particular to the achievement of carbon neutrality by 2050 and the mobilisation of industry for a clean and circular economy, would be undermined if no EU action is taken on the efficiency of energy-related products.

3.3. Subsidiarity: Added value of EU action

Ecodesign measures contribute to the global EU energy and climate objectives by influencing the characteristics of construction of energy related products. In this respect, the approach from

[43] Life Is Too Short to RTFM: How Users Relate to Documentation and Excess Features in Consumer Products. Alethea L. Blackler, Rafael Gomez, Vesna Popovic and M. Helen Thompson. Published by Oxford University Press on behalf of The British Computer Society. July 2014.
[44] Consumers' Use of Written Product Information. Article in Ergonomics,15 September 2004, Vol 47, No 11.
[45] Regulation (EU) 2017/1369 of the European Parliament and of the Council of 4 July 2017 setting a framework for energy labelling and repealing Directive 2010/30/EU (OJ L 198, 28.7.2017, p.1)

ecodesign requirements is different than that from other energy and climate legal acts such as EED and RED[46], which are based on the implementation by Member States of measures based on existing technologies, but do not directly require the development of the technologies themselves. Direct EU action through ecodesign rules reinforce therefore the actions taken at national level. Ecodesign measures can only be addressed at EU level since they impact directly on the product features, which uniformity needs to be assured to avoid that different national rules undermine the EU internal market for that product.

Energy labelling in turn provides a transparent tool that Member States may use to incentivise the replacement of obsolete energy-consuming products with new and more efficient ones. According to Article 7 of the Energy Labelling Framework Regulation, when Member States provide incentives to products in the scope of energy labelling measures, those incentives shall aim at the highest two significantly populated classes of the energy label. It follows from this that energy labelling provides transparency to national aid programs by setting a common classification that Member States must use to grant incentives for the purchase of products, guaranteeing that those incentives are delivered on the basis of the EU environmental common interests and not on national preferences to favour a specific technology or set of technologies.

4. OBJECTIVES: WHAT IS TO BE ACHIEVED?

4.1. General objectives

The general objective is to contribute to the EU's energy and environmental objectives set in the TFEU. There are various possible synergies within this overall objective. The TFEU states that the aim of the EU energy policy includes ensuring security of energy supply and promoting energy efficiency and energy saving. Lower energy consumption is usually accompanied by a reduction in other environmental impacts including GHG and pollutant emissions as well as lower resource use and extraction.

4.2. Specific objectives

1. Capture in-use cost-efficient energy savings from tumble dryers;
2. Contribute towards a circular economy by addressing premature obsolescence of tumble dryers;
3. Increase the effectiveness of the legislation on tumble dryers by reducing its complexity.

There are possible trade-offs between the two first objectives. The main ones are:

- Measures to increase energy efficiency might require more materials to be used for the products or materials that have bigger impacts.
- Measures to reduce the material consumed in the manufacture of tumble dryers might lead to more energy consumption in the use phase if they result in a longer lifetime of less efficient appliances.

These trade-offs will be captured by the EcoReport tool and presented in the impacts for these options in section 7.

[46] Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources (OJ L 382, 21.12.2018, p.82)

The specific objectives 1 and 2 will contribute to the achievement of the climate-neutrality objective set out in Article 2(1) of the European Climate Law[47] (balancing Union-wide greenhouse gas emissions and removals regulated in Union law to net zero by 2050 and to achieve negative emissions thereafter) by reducing GHG emissions throughout the life cycle of the product, thus diminishing the amount of carbon to be removed from the atmosphere in order to achieve the balance between GHG emissions and removals mandated by the European Climate Law.

5. WHAT ARE THE AVAILABLE POLICY OPTIONS?

The available policy options focus on the implementation of measures that maximise the contribution of tumble dryers to energy efficiency without imposing significant burden to manufacturers or costs to end-users, as set out in Article 15(5) of the Ecodesign Framework Directive. The main target of those options is to accelerate the phasing out from sale of less efficient tumble dryers, so that this process is completed by 2026 rather than 2050, which would be the case in the absence of action. This will be achieved by setting reachable minimum requirements on energy efficiency, which would be just enough to phase-out heating element and air-vented tumble dryers, plus the least efficient heat pump appliances. Pushing towards more demanding energy efficiency targets would not be adequate in the view of the little perspective of improvement of heat pump technology, and would risk of slanting the market towards high-end and less affordable products.

Figure 5-1 shows the overall **intervention logic** and the links between drivers, problems, objectives and measures. The options displayed in figure 5-1 are of complementary nature and do not overlap with each other. Different combinations of such options will be assessed in section 6, although as later explained in the introductory part of that section, PO2 is a mandatory measure from the Energy Labelling Framework Regulation and must be included in any of the combination of policy options under assessment. Figure 5-1 shows that PO2 and PO3 address the same problem and contribute to the same specific objective of reducing in-use energy consumption, although they achieve this by different means that will be explained in further detail in section 5.2. PO4 in turn, is focussed on extending the lifetime of the product and will reduce the amount of embedded energy rather than the in-use energy, which in fact will slightly increase under this option.

[47] Regulation (EU) 2021/1119 of the European Parliament and of the Council of 20 June 2021 establishing the framework for achieving climate neutrality and amending Regulations (EC) No 401/2009 and (EU) 2018/1999 ("European Climate Law") (OJ L 243, 9.7.2021, p.1)

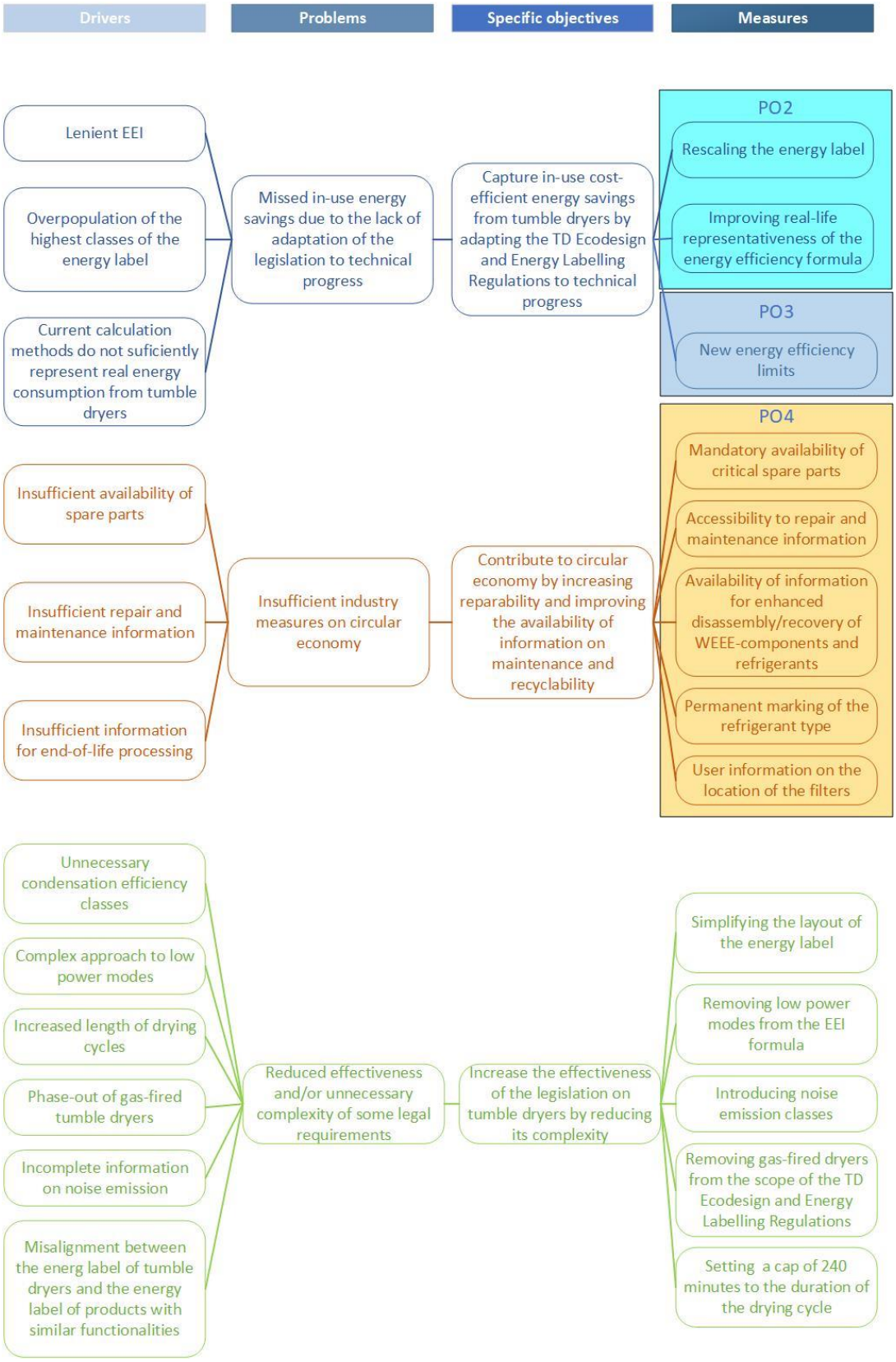


Figure 5-1. Schematic overview of drivers, problems, objectives and measures and their interlinkage

5.1. What is the baseline from which options are assessed?

The electricity prices and carbon intensity that are used as input to this impact assessment are taken from the EU Reference Scenario 2020 (REF2020). More details about further input data and analytical models are provided in Annex 4. REF2020 models EU and Member States energy

developments up to 2050, on the basis of current global and EU market trends and incorporating the effect of the "Clean Energy for All Europeans Package"[48]. Policies that are not legally implemented are not part of REF2020. In order to cater for future fluctuations of the different parameters relevant for the modelling in this IA, a sensitivity analysis based on the MIX scenario is run in Annex 8, with higher energy prices and lower carbon intensity vis-à-vis REF2020.

As indicated in section 1.1, it is expected that the effect of the future ESPR on the baseline will be modest. First, the new products in scope of the ESPR do not consume energy during their use, and therefore the increase in energy in scope will primarily relate to embedded energy (manufacture, distribution and EoL of products), which is a minor part (18%) of the energy use during the lifecycle of tumble dryers. Second, the totality of the embedded energy will not correspond to the EU but will be shared between EU and non-EU countries depending on the split of production. Third, producers are subject to ETS pricing and other regulatory measures that mean that they are likely to have less room for improvement compared to non-EU producers. Finally, the time frame to agree the ESPR, develop and adopt secondary legislation and for it to come in force will mean that substantial impacts could not be expected for the next ten years. These factors mean that any changes caused by the ESPR to the EU energy baseline for tumble dryers can be expected to be minimal. Major differences from the BAU will come from the energy prices side rather than from the ESPR.

5.1.1. Stock

Under the baseline, the market share of heat pump tumble dryers will significantly grow, but inefficient air-vented and heating element appliances will still remain beyond 2040. Gas-fired dryers have already been phased-out and do not appear in BAU.

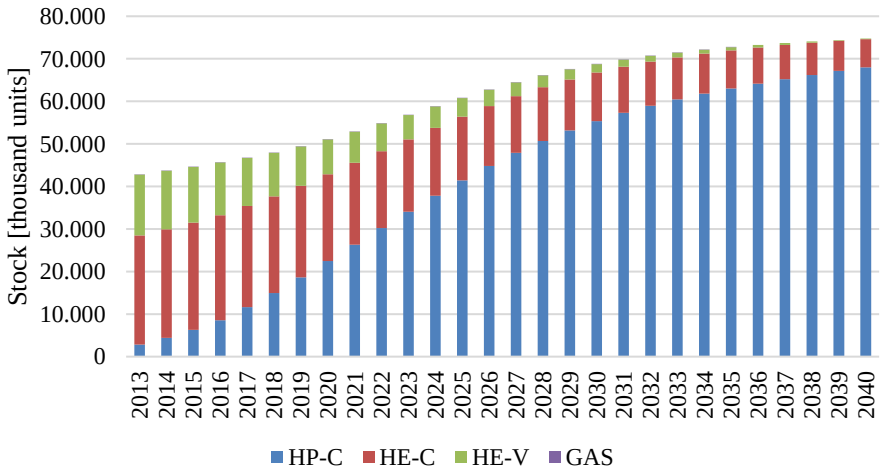


Figure 5-2.Projected stock of tumble dryers installed with shares of heat pump (HP-C), condensing (HE-C), vented (HE-V) and gas-fired (GAS) under BAU (source: based on calculations by Viegand Maagøe, see Annex 4).

5.1.2. Environmental parameters

The evolution of the stock displayed in Figure 5-2 will make final energy consumption (that includes embedded energy consumed during production, distribution and EoL) from tumble dryers decrease by 16% in 2040 with respect to 2020, from 12,5 TWh/year down to 10,5 Twh/year. The reduction of total energy consumption is smaller than the reduction of in-use energy consumption (23%) because heat pump dryers are more complex and difficult to produce and dismantle, increasing the consumption of embedded energy.

[48] Clean Energy for all Europeans Package

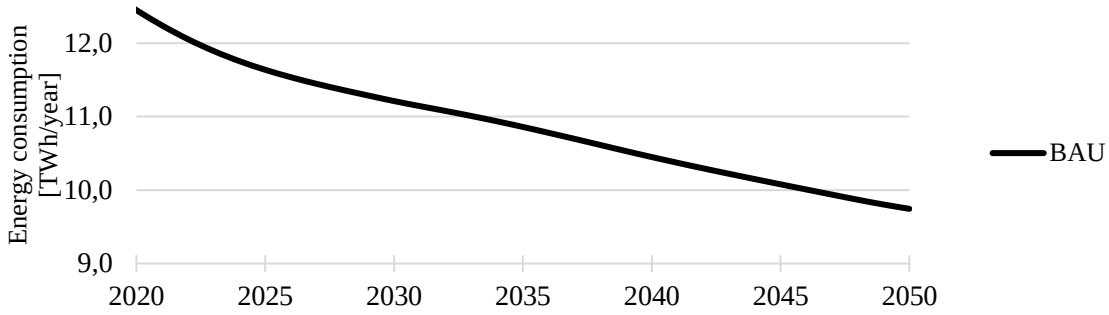


Figure 5-3. Total energy consumption under BAU (including embedded energy)

The reduction of electricity consumption results in a subsequent decrease of GHG emissions by 46% in 2040 compared to 2020 levels. The progressive reduction of the carbon intensity of the electricity mix leverages the reduction of electricity consumption, causing the GHG emissions to drop faster than the energy consumption (46% against 16%), to level off by 2040 due to the assumed stabilization of the share in the EU energy mix of carbon-free electricity generation.

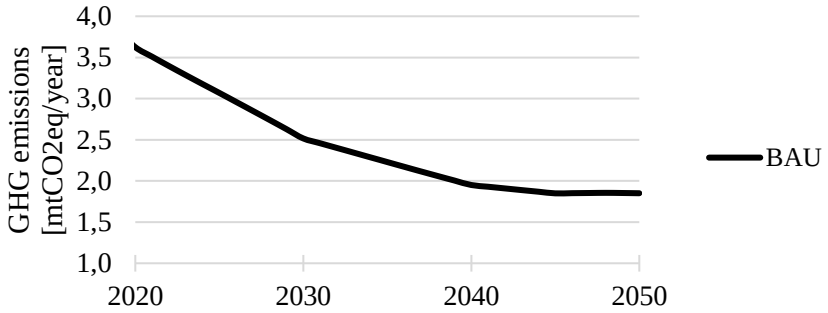


Figure 5-4. Greenhouse gas emissions (including embedded energy)

5.1.3. Economic parameters

The total turnover of the industry has been increasing since 2010 supported by growing sales and higher product prices resulting from the technology shift to heat pump dryers. This trend is expected to continue.

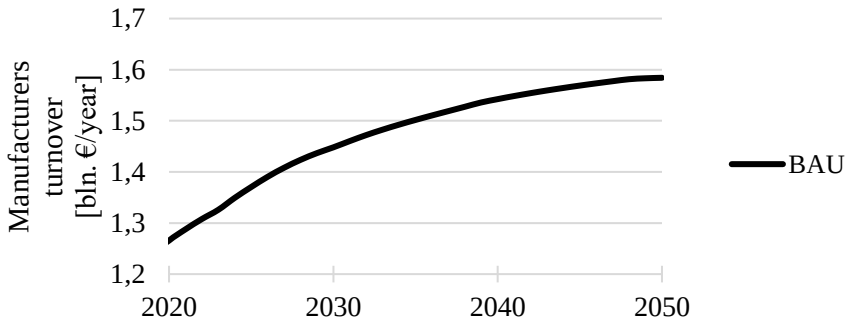


Figure 5-5. Manufacturers turnover

As far as user expenditure, it will remain constant in the mid-term because the increase of sales on the one hand and the higher purchase costs that are a consequence of the shift to the more expensive heat-pump technology on the other hand, are offset by the reduction of in-use energy costs due to the better efficiency of the new appliances. By 2045, user expenditure will grow again as the technological improvements from heat pump tumble dryers stagnate.

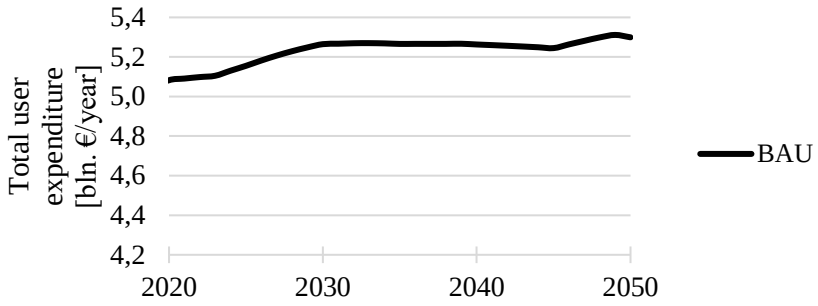


Figure 5-6. User expenditure

5.2. Description of the policy options

The following table displays the correspondence between problems, measures and policy options.

Problems	Measures	Policy options
First key problem: missed in-use energy savings due to the lack of adaptation of the legislation to technical progress	- Rescaling the energy label - Improving real-life representativeness of the energy efficiency formula	PO2: energy labelling measures (lenient option)
	- New energy efficiency limits	PO3: ecodesign measures (ambitious option)
Second key problem: insufficient industry measures on circular economy	- Mandatory availability of critical spare parts - Accessibility to repair and maintenance information - Availability of documentation for enhanced disassembly/recovery of WEEE-components and refrigerants - Permanent marking of the refrigerant type - User information on the location of the filters	PO4: circular economy
Third additional problem: reduced effectiveness and/or unnecessary complexity of some legal requirements	- Simplifying the layout of the energy label - Removing low power modes from the EEI formula - Introducing noise emission classes - Removing gas-fired dryers from the energy label - Setting a cap of 240 minutes to the duration of the drying cycle	The measures corresponding to the third additional problem will be included in all the above policy options, since they introduce administrative simplification without any additional environmental, economic or social impact.

Table 5-1. Correspondence between problems, measures and policy options

The policy options will mostly affect manufacturers, which will need to adapt their products to the new ecodesign requirements, although PO4 will also have an impact on repairers. Around 160 manufacturers are estimated to produce tumble dryers in the EU. The number of repair companies is estimated to be 747, employing an average of 2 people per company[49]. An SME test is carried out in Annex 10.

5.2.1. Policy option 1 (BAU): business as usual

The BAU corresponds to the baseline scenario explained in section 5.1.

5.2.2. Policy option 2 (PO2) – lenient option: energy labelling measures

PO2 consists of two measures, explained in the following sections 5.2.2.1 and 5.2.2.2.

[49] Annex 10 provides an explanation on the methodology used to calculate, from the data available in Eurostat, the number of manufacturers, number of repair companies and share of SMEs.

5.2.2.1. Rescaling the energy label

The energy label will be rescaled, as mandated by the Energy Labelling Framework Regulation, with an empty A class for models with an EEI equal or below 45. The adjustment makes the current A+++ class roughly equal to the new B. G will be the lowest energy class. Table 5-2 compares the current label classes on the left with the proposed new classes on the right.

The rescaling of the label could have been fulfilled by defining an A class with an EEI threshold of 50 instead of 45, as the EEI of the existing best-in-class model according to the latest data is 51. This would have ensured that the A class is empty because currently no models reach an EEI of 50, but it would also have allowed manufacturers to place new models in class A with very little energy efficiency improvement with respect to the current models. For this reason, the EEI threshold for the proposed A class has been set by subtracting 10% from the EEI of the best-in-class model, in order to incentivise manufacturers to develop appliances that are significantly more efficient than the current ones.

Current energy label at $SEc = 0.88 \times c^{0.8}$		
Label class	**EEI**	**Model distribution**
A+++	EEI < 24	17%
A++ (Top)	24 ≤ EEI < 32	29%
A++ (Bottom)		
A+ (Top)	32 ≤ EEI < 42	17%
A+ (Bottom)		
A	42 ≤ EEI < 65	1%
A		
B	65 ≤ EEI < 76	19%
C	76 ≤ EEI < 85	11%
D	85 ≤ EEI	6%

Proposed energy label at $SEc = 0.46 \times c^{0.63}$		
Label class	**EEI**	**Model distribution**
A	EEI < 45	0%
B	45 ≤ EEI < 55	6%
C	55 ≤ EEI < 70	17%
D	70 ≤ EEI < 86	33%
E	86 ≤ EEI < 100	7%
F	100 ≤ EEI < 200	20%
G	201 ≤ EEI	17%

Table 5-2. Current and proposed energy class classification[50]

Table 5-2 shows that the proposed class B is narrower than the rest of the classes. Initially it was considered to have class B as wide as classes C to E. This nevertheless shifted class A from an EEI ≤ 45 to an EEI ≤ 40, a level considered too demanding on the basis of the expected improvement potential for heat-pump systems. An alternative would have been to move all classes upwards towards less demanding values, but a detailed analysis of the existing data showed that many products that are close to the edge between classes would be promoted to a better class. In particular those on the edge between B and C, would fall into B instead of into C, thus lessening the incentive to improve. As result of all these considerations it was finally decided to narrow class B to 10 EEI points and set class A at 45. The configuration of the energy label has been intensively discussed with stakeholders.

All stakeholders supported the revision of the energy label, although not all of them agreed on the proposed approach. APPLiA, supported by some Member States, challenged the threshold for class A on the basis that it is very difficult to achieve and argued that it will not incentivise manufacturers efforts to achieve better classes.

[50] The table is based on the number of models in the APPLiA database.

5.2.2.2. Improving real-life representativeness of the energy efficiency formula

Following the conclusions of the review study, the *EEI* formula (*Ec/SEc*) will be adapted to real-use by expressing its value in relation to 100 drying cycles instead of per year, and by increasing the share of half-load cycles vis-à-vis full-load cycles.

The curve for *SEc* will be adapted to better fit the current distribution of models and to ensure that larger dryers are not unduly favoured by the metrics. Figure 5-5 shows that the current SEc curve is too steep to be representative of the market, as it is significantly above the energy consumption values of the current models, and that the distance between the models and the curve grows with the rated capacity. As pointed out in section 2.2.3, this leads to the perverse effect that increasing the size of a tumble dryer makes it easier to comply with the EEI threshold. This would be corrected with the new SEc curve proposed. Figure 5-5 also shows that the SEc proposed in the review study is steeper than the SEc proposed in this IA. This is because the latter is based on more recent data that were not available during the review study. The SEc in the review study used data from between 2013 and 2016, and the data used in this IA are extended to 2019. The new, flattened SEc curve is a consequence of the increasing presence on the market of heat pump tumble dryers with higher energy efficiency, growing from 51% of sales in 2016 to 73% in 2019.

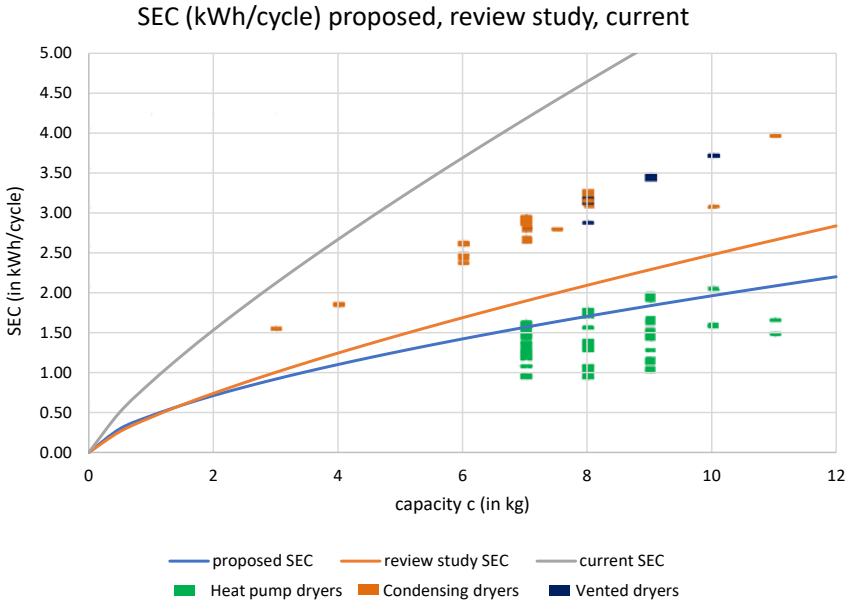


Figure 5-5. Standard Energy Consumption SEC current, review study and proposed as well as indicative distribution of kWh/cycle for the 3 main dryer technologies

Annex 5 provides a thorough explanation of the EEI formula and the proposed modifications. The EEI limit set in the TD Ecodesign Regulation would need to be adapted to the new formula[51].

The new SEc curve was subject to intense discussion at the stakeholder meetings. NGOs wanted it to be a horizontal line so that the rated capacity played no role at all in the rating, and industry wanted, on the contrary, the slope to remain steep. The new and more robust data extended to 2019 helped to achieve a final solution on which all stakeholders could agree.

[51] Current limit EEI=76 (C-class limit) is new limit EEI=205 (lowest F-class possible) at capacity c=7.7 kg

5.2.3. Policy Option 3 (PO3) – ambitious option: new energy efficiency limits

PO3 raises the level of ambition by setting a more ambitious EEI threshold (only tumble dryers with EEI ≤ 85 will be allowed to remain in the market) based on the new EEI formula described in section 5.2.2. This EEI threshold will phase out air-vented and heating element tumble dryers, as well as the worst-performing heat pump tumble dryers.

Small tumble dryers with a rated capacity equal to or under 4,5kg do not feature enough space to fit a heat pump system and are therefore less efficient, will be exempted from the scope of PO3. This is because their inclusion would eliminate such dryers from the market and thus prevent their installation where needed, namely small dwellings with little space available.

Figure 5-6 shows the level of ambition of PO3. With the new method for the calculation of EEI proposed in PO2, the current EEI threshold would be set at 181, which would correspond to class F. The proposed EEI threshold of 85 would coincide with the limit of class D with C and would phase-out not only heating element and air vented tumble dryers, but also the least performant heat-pump models. Since no tumble dryers could be placed on the market below class D, classes E to G would only be used by tumble dryers of 4,5 kg or less of rated capacity, which as pointed out above would be exempted from the ecodesign rules.

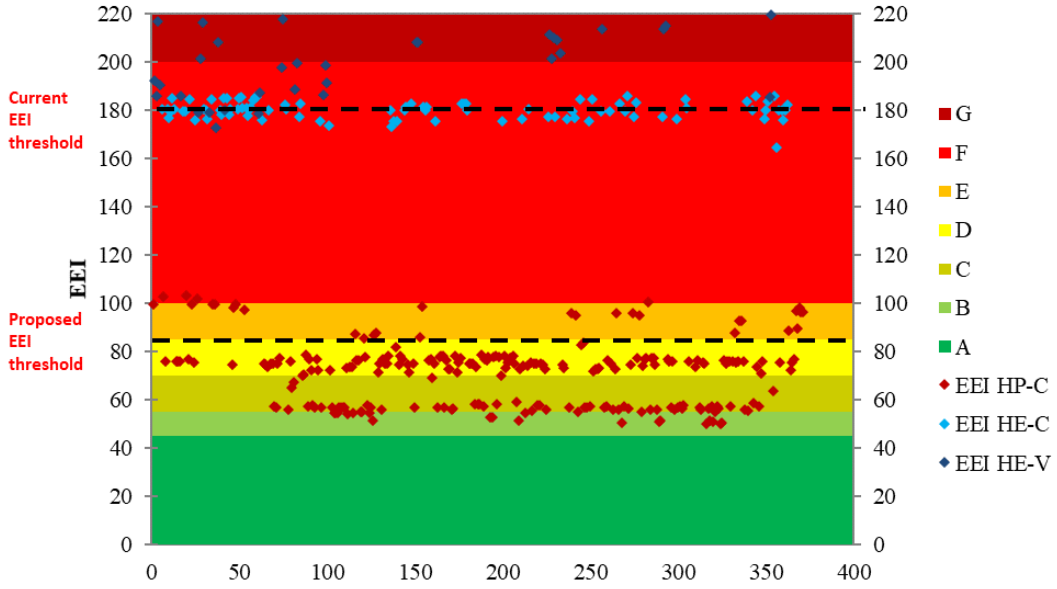


Figure 5-6. Comparison current and proposed EEI limit

In the current context of higher energy prices, energy savings become more interesting and the question raises on whether it would have been economically interesting to increase the proposed level of ambition by further lowering the EEI threshold. Section 8.4 shows that, due to the moderate expectations about future energy efficiency improvements on the basis of heat-pump technology, a more stringent EEI threshold would have been costly to achieve, making the option uneconomic.

As far as the views from stakeholders, PO3 was supported by various Member States and NGO's. The latter also proposed to raise the level of ambition by setting the EEI threshold at 80 instead of 85. Some NGO's even proposed an EEI threshold of 60, which would only allow tumble dryers placed in classes A, B and a part of class C to be placed in the market. APPLiA did not support the EEI threshold of 85 but accepted it, provided that it applies at the end of a long transitional period of five years after the entry into force of the TD Ecodesign and Energy Labelling Regulations. In this respect, APPLiA proposed a two-tier implementation, the first tier applying 18 months after the entry into force of the legislation and the second 36 months after the first tier. On the opposite side,

some Member States and NGOs proposed to have a single and short implementation date to avoid confusion.

5.2.4. Policy Option 4 (PO4): circular economy

PO4 aims to reduce embedded energy consumption by implementing the following resource efficiency measures:

- mandatory availability of critical spare parts for 10 years after ceasing the production of a model, with a maximum delivery time of 10 working days. The TD Ecodesign Regulation will include two separate lists of spare parts for the manufacturer to make available, one with spare parts available to both users and professional repairers and a second one only including spare parts available to professional repairers. Main argument to restrict specific spare parts to professional repairers are safety concerns, as indicated below.
- mandatory access to repair and maintenance information to facilitate the repair activities of independent repairers;
- availability of information for enhanced disassembly/recovery of WEEE-components and refrigerants: dismantling information must be available, on a free access website, regarding access to any of the components of the product referred to in point 1 of Annex VII of the WEEE Directive. The information shall include the dismantling steps, tools or technologies needed to access the components, and shall be available for at least 15 years after the placing on the market of the last unit of a product model;
- permanent marking of the refrigerant type in the nameplate of heat-pump tumble dryers (air-vented and heating element tumble dryers do not use refrigerant). In addition, the **refrigerant type and its GWP** shall be available in the user manual and the product information sheet;
- user information on the location of the filters and the need to clean them regularly shall be clearly visible on the appliance.

PO4 will not internalise the external costs in the purchase price, which would raise consumer expenditure, but will rather reduce the environmental footprint of embedded energy by reducing the scrapping of appliances thus slowing down the pace of production.

In general stakeholders support PO4, although there are different views on how long spare parts should be made available. NGOs and consumer associations favour 12 years, while industry recommends 7 years. Members States, NGOs and consumer organisations have proposed extensions to the proposed list of available spare parts and want them to be available for both professional repairers and consumers and not only the former. APPLiA does not support the inclusion of spare parts for the heat-pump system because it is not a cost-effective solution, and indicates that some spare parts must not be available to the general public, as there are safety issues related to their handling, for instance risk of electric shock.

NGOs and consumer associations have proposed to fit the appliance with an automatic indicator to inform the user when the filters should be cleaned. The benefit of that feature is nevertheless uncertain. Several tumble dryer models already include an indicator[52], but more knowledge about its benefits and costs is needed to be able to develop suitable ecodesign requirements. APPLiA

[52] The current sensors detecting clogged filter work by measuring the temperature changes or by counting the number of drying cycles.

points out that the sensor, its integration into the electric circuit and the **development** and **implementation** of the associated software would entail significant costs.

5.2.5. Complementary measures

The complementary measures increase the clarity of the legislation and reduce unnecessary requirements without entailing any environmental, economic or social impacts, and will be taken into the legal amendment regardless of the preferred policy option. These measures are:

- simplifying the layout of the energy label: the new label will be simplified by mirroring the dryer-part of the new washer-dryer label (see figure 5-6). Only energy class, electricity consumption per 100 cycles in kWh, load capacity in kg and cycle time in hours:minutes for cotton dry cycle will be displayed. The icon of the powering energy will disappear since gas fired tumble driers are no longer on the market and the remainder are electric, and the condensation efficiency classes will also disappear. Instead, the minimum condensation efficiency will be raised to 80%, but this will not entail any impact because all new models on the market reach that efficiency;
- replacing the low-power factors of the EEI formula by minimum values for low power modes. This will also imply excluding tumble dryers from the scope of the Standby Regulation to avoid regulatory redundancy;
- introducing noise-emission classes on the label starting with A at < 60 dB(A) and ending with D at ≥ 68 dB(A). A bandwidth of 4 dB(A) is used for the distribution of classes, as for washing machines and washer-dryers;
- removing gas-fired dryers from the energy label;
- setting a cap of 240 minutes to the duration of the cycle, to avoid consumer complaints on performance.

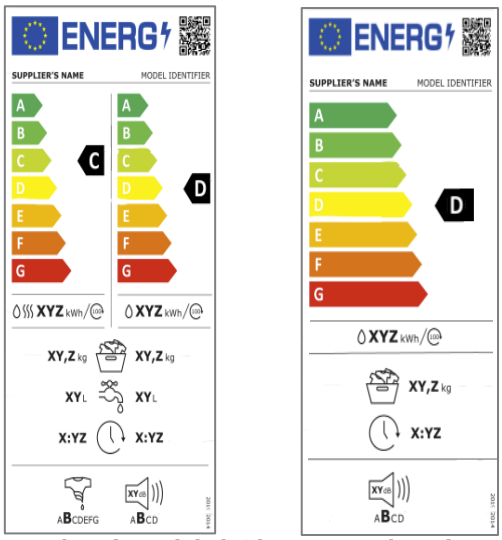


Figure 5-6. Left: current washer-dryer label (showing wash and wash&dry cycle information). Right: proposal for similar tumble dryer label (water consumption not needed for drying only cycle)

5.2.6. Transitional period for the implementation of the policy options

In order to give stakeholders (industry, standardisation organisations, retailers) time to adapt to the new requirements, a transitional period for the application of the chosen policy option has been envisaged. PO2 measures would be implemented in 2025, this is 18 months after the foreseen date of entry into force of the legislative amendment. For PO3, there would be an additional transitional

period of 18 months (mid-2026). The total transitional period would then be 36 months after the date of entry into force of the revision of the legislative amendment.

5.3. Options discarded at an early stage

5.3.1. Voluntary agreement

According to the Ecodesign Framework Directive, voluntary agreements must get priority over legislative actions, provided that objectives are met in a quicker and more cost-effective manner. A voluntary agreement has nevertheless been ruled out on the basis of the following arguments: first, the opinions within the stakeholders about some of the key aspects included in the policy options are very divergent, and in the view of those opinions a voluntary agreement is unlikely to satisfy a significant part of stakeholders, mainly a good number of Member States and NGOs. Moreover, the energy efficiency targets proposed by the industry are significantly less demanding than the targets proposed in this IA and would undermine the achievement of the objectives set out in section 4. Finally, the industry has not shown any interest in putting forward a proposal of voluntary agreement. The Commission has therefore taken the view that the industry is not ready to adopt a voluntary agreement that is at the same time representative of the vast majority of stakeholders, demanding in terms of energy efficiency, and which can be in place before the date of application of a legislative amendment.

5.3.2. Modification of the verification tolerances

From the review study and dialogue with stakeholders it follows that the current tolerances are in line with the accuracy of the current testing procedures, conclusion that has been confirmed by a round robin test run by the main manufacturers during 2017. There is therefore no reason to modify the verification tolerances.

6. WHAT ARE THE IMPACTS OF THE POLICY OPTIONS?

This section describes the impacts of each policy option. Modelling principles follow the MEErP (Methodology for Ecodesign of Energy-related Products)[53]. The analytical model has been specifically developed for impact assessments by the external consultant Viegand Maagøe, and the assumptions have been systematically crosschecked with stakeholders, in particular with the industry. Future energy prices and carbon intensities follow the EU Reference Scenario 2020. More information on the methodology followed and the assumptions underlying each policy option are provided in Annex 4.

In the view of the current context of energy crisis the use of REF2020 scenario, which features very moderate energy prices, raises the question on whether the analysis based on that scenario is realistic. Increasing electricity prices pushes up consumer expenditure and could call for further and more ambitious measures to offset the growing energy bill. Annex 8 already includes a sensitivity analysis based on the MIX scenario[54], with higher energy prices and lower carbon intensity vis-a-vis REF2020. However, even the prices from the MIX scenario are too low compared to the latest ones. Since there is no modelling available corresponding to the current situation and therefore a fully-fledged IA cannot be carried out, a preliminary affordability analysis is conducted in section 8

[53] Material-efficiency Ecodesign Report and Module to the Methodology for the Ecodesign of Energy-related Products (MEErP) PART 1: MATERIAL EFFICIENCY FOR ECODESIGN - Final report to the European Commission - DG Enterprise and Industry 5 December 2013 (https://op.europa.eu/en/publication-detail/-/publication/b7650397-32f1-436c-82c4-df39aef297a3)

[54] Policy scenarios for delivering the European Green Deal (europa.eu)

to have a first and general overview on how ecodesign and energy labelling measures could be modified to respond to a situation of escalating electricity prices.

Notes regarding the assessment of the impacts

- For the analysis of the impacts, PO2 will be included in all combinations of policy options because the rescaling of the label is a mandatory requirement.
- None of the policy options are assumed to change the total stock estimated for BAU. PO2 and PO3 only vary the share of the different types of tumble dryers.
- PO4 does not change the distribution of the stock, only the lifetime of tumble dryers, which is assumed to increase from 12 to 14 years, and the costs associated to repair.
- PO4 increases the repair costs from 5€/year under BAU to 8€/year.
- User expenditure consists of the sum of the purchase costs (which includes end-of-life costs that are normally paid by the user through a levy at purchase), running costs (energy costs during operation of the appliance), and maintenance and repair costs.
- For the calculation of turnover, a 47%/39%/14% split of the consumer price has been assumed between industry/retailer/tax, based on estimates from the sector.
- Price increase are fully carried over to the consumer.
- The impact on employment for manufacturers and retailers is estimated from the turnover, with a correction for changes in labour productivity as explained in Annex 4. Based on historical manufacturing data, an increase of 1% in turnover will lead to a 0,6% increase in workforce, and a 1% decrease in turnover leads to a 1,6% decrease in workforce.
- Only direct and indirect jobs in the production and distribution chain are considered, i.e. including employment generated in suppliers and business services but excluding induced employment effects due to spill-overs on other sectors.
- Figures on industry turnover and employment comprise both the manufacturing and repair sectors. For policy options including PO4, the repair sector figures corresponding have been disaggregated from the rest of the industry to illustrate the influence of spare parts availability on that sector.

6.1. PO2 – lenient option: energy labelling measures

6.1.1. Environmental impacts

Rescaling the energy label will have two effects, one on industry and the other on the consumer: on the industry, it will incentivise the development of better performing tumble dryers and on the user, it will contribute to a moderate replacement of purchases of air vented and heating element tumble dryers by heat pump systems. This will increase the presence of heat pump tumble dryers on the market and will subsequently yield some energy savings and a limited reduction of GHG emissions. The difference of energy consumption between BAU and PO2 is small, first because PO2 does not bring a massive replacement of other models by heat pump tumble dryers, as no new mandatory energy efficiency thresholds are put in place, and second because no strong improvements are expected on energy efficiency per unit as heat pump is already a mature technology. Even under BAU the share of heat pump tumble dryers increases over time, and the gap between BAU and PO2 will progressively diminish, by 2050 only around 300.000 out of a total stock of 76 million heat pump tumble dryers would remain.

The reduction of total energy caused by PO2 with respect to BAU (1,9% in 2040) is slightly smaller than the reduction of in-use energy (2,5% in 2040) due to the more complex production of heat-pump tumble dryers, with higher embedded energy and associated GHG emissions than the simpler air vented and heating element systems. Cumulative GHG emissions slightly increase in 2030 due to embedded energy, but are offset in the longer term by savings of in-use energy.

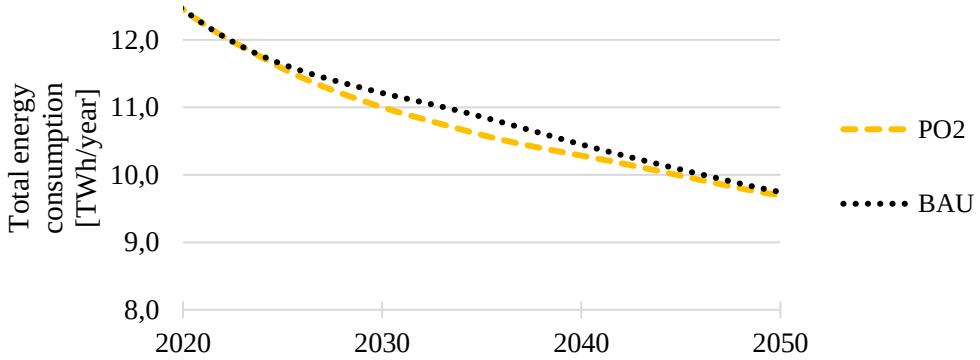


Figure 6-1. Total energy consumption for PO2 and BAU

	Energy consumption during use [TWh/year]				Savings [TWh/year]		Cumulative savings [TWh]	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	10,50	9,49	8,97	8,10				
PO2	10,50	9,42	8,75	7,90	0,23	0,20	0,96	3,60
	Total energy consumption (in-use+embedded) [TWh/year]				**Savings [TWh/year]**		**Cumulative savings [TWh]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	12,45	11,64	11,21	10,45				
PO2	12,45	11,57	10,99	10,25	0,22	0,20	0,90	3,51
	GHG emission, in-use + embedded [mtCO2e/year]				**Savings [mtCO2e/year]**		**Cumulative savings [mtCO2e]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2	3,62	3,09	2,50	1,94	0,02	0,01	-0,01	0,16

Table 6-1. Environmental impacts for BAU and PO2

6.1.2. Economic impacts

The rescaling of the energy label will not bring about a significant change on manufacturers and retailers turnover with respect to BAU, just a small increase on both due to the larger share of heat pump tumble dryers and their associated higher prices. By 2050 there will still be some turnover difference between BAU and PO2 due to the slightly higher market share and better energy efficiency of heat pump appliances.

As far as user expenditure, it presents some particularities. In the periods 2025-2028 and 2038-2050 user expenditure is bigger for PO2 than for BAU. In the first period, the short time elapsed since PO2 applies does not allow the in-use savings to offset the higher purchase prices. Between 2038 and 2050, BAU will have gained a significant share of heat pump tumble dryers thus reducing the distance with PO2, for which the small difference of in-use energy savings between both scenarios cannot offset the higher prices still paid due to PO2. Between 2028 and 2038, the in-use energy savings with PO2 will be enough to offset the higher purchase prices and will be lower than with BAU.

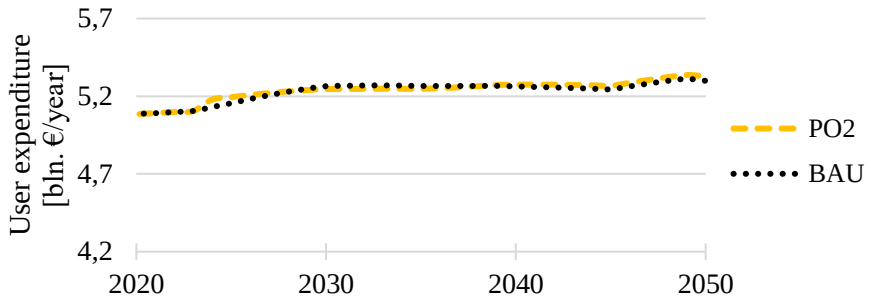


Figure 6-2. User expenditure for PO2 and BAU

	Industry turnover [bln. €/year]				Difference to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	1,27	1,37	1,45	1,54				
PO2	1,27	1,38	1,45	1,56	0,00	0,02	0,07	0,20
	Retail turnover [bln. €/year]				**Difference to BAU [bln. €/year]**		**Cumulative difference [bln. €]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	2,69	2,84	2,94	3,12				
PO2	2,69	2,88	2,96	3,16	0,01	0,04	0,19	0,52
	Total user expenditure [bln. €/year]				**Savings [bln. €/year]**		**Cumulative savings [bln. €]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	5,08	5,16	5,26	5,26			0,00	0,00
PO2	5,08	5,17	5,23	5,26	0,04	0,00	0,02	0,27

Table 6-2. Economic impacts for PO2 and BAU

6.1.3. Social impacts. Employment

Employment for PO2 follows a trend similar to turnover, namely a small increase with respect to BAU which remains more or less stable until the end of the considered period.

	Total employment [jobs/year]				Difference to BAU (jobs/year)	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**
BAU	22.023	22.977	23.651	24.531		
PO2	22.023	23.116	23.702	24.712	51	180

Table 6-3. Employment (manufacturers + retailers) in EU for PO2 and BAU

6.2. PO2 + PO3 – ambitious option: rescaling of the energy label and new energy efficiency limits

6.2.1. Environmental impacts

New energy efficiency limits will prevent the placing on the market, as of 2026, of appliances with an EEI > 85, namely heating element, air vented and the least efficient heat pump tumble dryers. This results in an important reduction of total energy consumption with respect to BAU (11% in 2040), which is nevertheless smaller than the reduction of in-use energy (14% in 2040) due to the complexity of heat-pump tumble dryers, as explained in section 6.1.1. The levelling of energy consumption around 2040 arises as heat pump tumble dryers will be close to reach the 100% of the existing stock and only minor energy efficiency gains will take place after that. After 2040, the difference between BAU and PO2+PO3 becomes smaller but remains significant because under BAU there will still be around 2 million heating element tumble dryers in use beyond 2050.

The cumulative GHG emissions displayed in table 6-4 show that the higher carbon footprint of heat pump appliances will initially produce an increase of GHG emissions over BAU, although the gap will progressively reduce due to the better efficiency of heat pump systems.

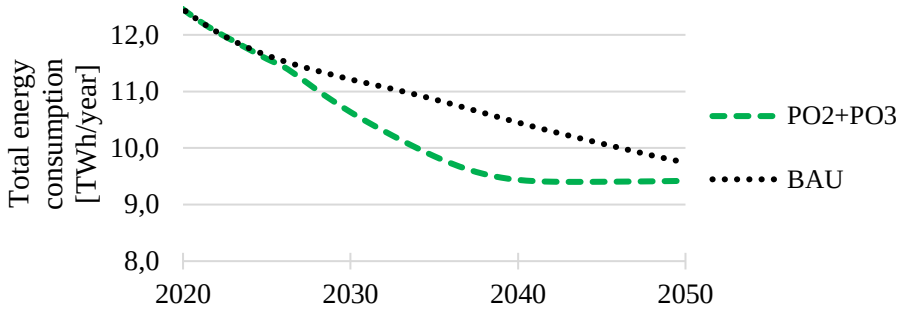


Figure 6-3. Total energy consumption for PO2+PO3 and BAU

	Energy consumption during use [TWh/year]				Savings [TWh/year]		Cumulative savings [TWh]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	10,50	9,49	8,97	8,10				
PO2+PO3	10,50	9,42	8,30	6,96	0,67	1,15	2,11	12,89
	Total energy consumption (in-use+embedded) [TWh/year]				**Savings [TWh/year]**		**Cumulative savings [TWh]**	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	12,45	11,64	11,21	10,45				
PO2+PO3	12,45	11,57	10,58	9,32	0,64	1,13	1,93	12,49
	GHG emission, in-use + embedded [mtCO2e/year]				**Savings [mtCO2e/year]**		**Cumulative savings [mtCO2e]**	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2+PO3	3,62	3,09	2,53	1,52	0,01	0,43	-0,27	0,10

Table 6-4. Environmental impacts for PO2 and BAU

6.2.2. Economic impacts

The stricter ecodesign measures will bring a steep increase in manufacturers turnover compared to BAU during 2026, when PO3 enters into application and heat pump tumble dryers make up 100% of sales. After 2026, the gap between both scenarios becomes smaller and by 2050 reaches its minimum value within the period of analysis, when sales of heat pump tumble dryers in the BAU scenario are maximised.

User expenditure will sharply rise during 2026 due to the sudden increase of prices caused by the immediate phase-out of non-heat pump tumble dryers, although in-use energy savings will quickly and progressively offset the price increase, fully offsetting it by 2031. After 2031, the savings from PO2+PO3 compared to BAU reach their maximum in 2040 and then progressively decrease until 2050, when heat pump tumble dryers under BAU will have reached 97% of the stock.

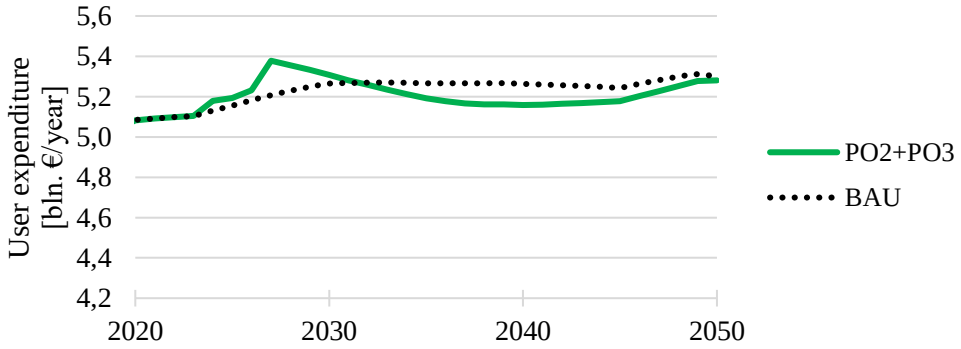


Figure 6-4. User expenditure for PO2+PO3 and BAU

	Industry turnover [bln. €/year]				Difference to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	2025	2030	2040	2030	2040	2030	2040

BAU	1,27	1,37	1,45	1,54				
PO2+PO3	1,27	1,38	1,50	1,58	0,06	0,04	0,30	0,78
	Retail turnover [bln. €/year]				**Difference to BAU [bln. €/year]**		**Cumulative difference [bln. €]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	2,69	2,84	2,94	3,12				
PO2+PO3	2,69	2,88	3,09	3,22	0,15	0,11	0,80	2,08
	Total user expenditure [bln. €/year]				**Savings [bln. €/year]**		**Cumulative savings [bln. €]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	5,08	5,16	5,26	5,26			0,00	0,00
PO2+PO3	5,08	5,17	5,26	5,12	0,00	0,14	-0,29	0,79

Table 6-5. Economic impacts for PO2+PO3 and BAU

6.2.3. Social impacts. Employment

Employment follows a trend similar to turnover. There is a significant increase with respect to BAU but the difference decreases as heat pump tumble dryers gain market share under BAU.

	Total employment [jobs/year]				**Difference to BAU (jobs/year)**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**
BAU	22.023	22.977	23.651	24.531		
PO2+PO3	22.023	23.116	24.254	24.967	603	435

Table 6-6. Employment (industry + retailers) in EU for PO2+PO3 and BAU

6.3. PO2 + PO4 – rescaling of the energy label and circular economy measures

6.3.1. Environmental impacts

Until 2031, the impact of PO2+PO4 in energy consumption comes solely from PO2 (rescaling of the energy label). From 2032 onwards, PO4 starts to yield energy savings as some of the tumble dryers sold during 2025 fail and are repaired instead of being discarded, reducing the sales of new appliances and consequently the consumption of embedded energy. The energy savings from PO4 reach a maximum in 2039, when the vast majority of tumble dryers sold in 2025 have failed because reaching the end of their lifetime (14 years) and need to be replaced. The total energy consumption will then increase due to the energy embedded in the production of new tumble dryers, which activity needs to be speeded up to replace the old devices. Energy consumption will reach a peak in 2047 and will decrease again afterwards. That peak will remain nevertheless lower than PO2, because new tumble dryers will also be repaired instead of discarded and this will be reflected in less units being produced and subsequently less embedded energy.

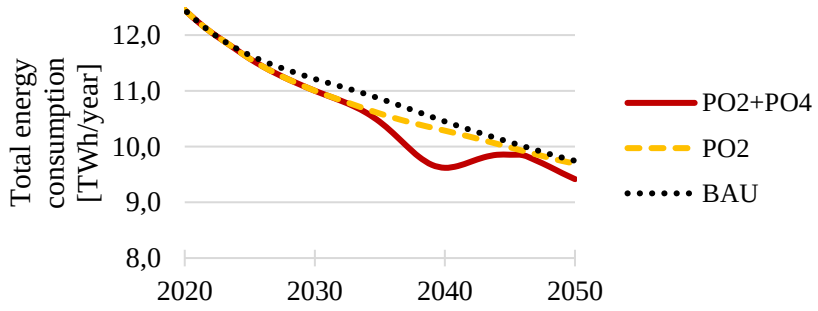


Figure 6-5. Total energy consumption for PO2, PO2+PO4 and BAU

Table 6-7 displays PO4 as standalone policy measure in order to observe the trade-off between the increase of in-use energy consumption caused by the longer lifetime of tumble dryers and the decrease of embedded energy consumption during production and EoL. PO4 increases by 0,05 TWh/year the in-use energy consumption in 2040, but this is more than offset by the 0,61

TWh/year decrease of embedded energy consumption, so that the total energy saving is 0,56 TWh/year. This same effect occurs with PO2+PO4.

	Energy consumption during use [TWh/year]				Savings [TWh/year]		Cumulative savings [TWh]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	10,50	9,49	8,97	8,10				
PO4	10,50	9,49	8,97	8,16	0,00	-0,05	0,00	-0,17
PO2+PO4	10,50	9,42	8,74	7,90	0,23	0,20	0,96	3,60
	Total energy consumption (in-use+embedded) [TWh/year]				Savings [TWh/year]		Cumulative savings [TWh]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	12,45	11,64	11,21	10,45				
PO4	12,45	11,64	11,21	9,84	0,00	0,61	0,00	2,69
PO2+PO4	12,45	11,57	10,99	9,59	0,22	0,87	0,90	6,39
	GHG emission, in-use + embedded [mtCO2e/year]				Savings [mtCO2e/year]		Cumulative [mtCO2e]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO4	3,62	3,07	2,52	1,59	0,00	0,36	0,00	1,54
PO2+PO4	3,62	3,09	2,50	1,57	0,02	0,38	-0,01	1,75

Table 6-7. Environmental impacts for PO2+PO4 and BAU

6.3.2. Economic impacts

From 2025 until 2034 industry turnover increases with respect to BAU due to two separate effects: on the one hand, the expected increase of sales revenues combined with a higher average price per unit as heat pump tumble dryers tend to dominate the market; on the other hand, the increase of repair costs per unit due to the application of PO4 (repair costs are included as part of the industry turnover in this model). In 2034 sales start to decrease as the share of failing tumble dryers that are repaired increases, bringing turnover down until 2039, in which the production speeds up to replace the tumble dryers reaching the end of their lifetime and to satisfy the new stock being needed. A new peak is reached again in 2046 and the process repeats, although the overall turnover values are somewhat higher due to the increase of sales.

The increased activity in the repair sector contributes to mitigate the turnover reduction in the industry due to the deceleration of the production. The growth of turnover in the repair sector is related to both the continuous increase of the stock of tumble dryers and the increase of the repair expenses per unit. For retail turnover, which is based exclusively on the sales of appliances, the drop of sales is not mitigated in any way, resulting in a net loss of cumulative employment by 2040.

User expenditure undergoes similar changes as industry turnover, namely a first phase of growth until 2034 due to the combined action of higher prices, increase of sales and additional repair costs, to drop by 2034 due to the reduction of sales induced by the extension of the lifetime of the appliances.

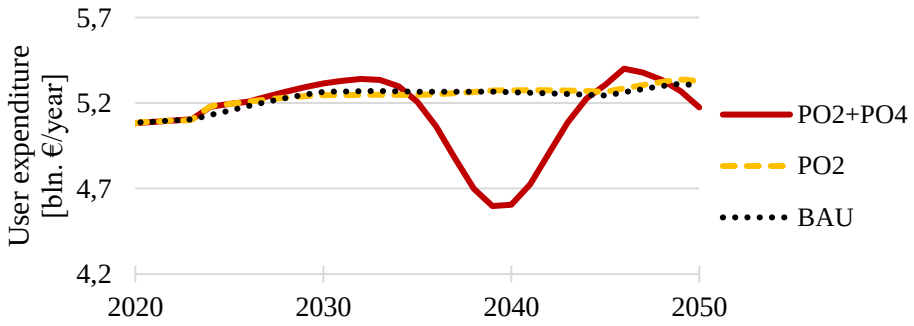


Figure 6-6. User expenditure for PO2, PO2+PO4 and BAU

	Industry turnover [bln. €/year]				Difference to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	1,27	1,37	1,45	1,54				
PO2+PO4	1,27	1,38	1,52	1,44	0,07	-0,10	0,25	0,54
Out of which in the repair sector	0,26	0,30	0,41	0,59	0,07	0,21	0,09	0,22
	Retail turnover [bln. €/year]				**Difference BAU [bln. €/year]**		**Cumulative difference [bln. €]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	2,69	2,84	2,94	3,12				
PO2+PO4	2,69	2,88	2,96	2,28	0,01	-0,84	0,19	-3,26
	Total user expenditure [bln. €/year]				**Savings [bln. €/year]**		**Cumulative savings [bln. €]**	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	5,08	5,16	5,26	5,26			0,00	0,00
PO2+PO4	5,08	5,17	5,30	4,60	-0,03	0,67	-0,10	2,34

Table 6-8. Economic impacts for PO2+PO4 and BAU

6.3.3. Social impacts. Employment

Employment for PO2+PO4 behaves similarly to turnover, this is an increase until 2034 due to a growth of repair activity that does not greatly affect sales, to drop afterwards as long as tumble dryers start to fail in a significant number. In 2039 employment grows again due to the need to increase manufacturing activity to replace the tumble dryers reaching the end of their lifetime. Nevertheless, with PO2+PO4 employment will not recover the levels under BAU due to the total reduction of the sales of new appliances. As for turnover, table 6-9 shows that repair activity contributes to mitigate the manufacturing job losses that are the consequence of the reduced production.

	Total employment [jobs/year]				Difference to BAU (jobs/year)	
	2020	2025	2030	2040	2030	2040
BAU	22.023	22.977	23.651	24.531		
PO2+PO4	22.023	23.116	24.142	18.925	492	-5.606
Out of which in the repair sector	1.602	1.906	2.594	3.680	439	1.337

Table 6-9. Employment (industry, including repairers + retailers) in EU for PO2+PO4 and BAU

6.4. PO2 + PO3 + PO4 –rescaling of the energy label, new energy efficiency limits and circular economy measures

6.4.1. Environmental impacts

Until 2031, the decrease in energy consumption follows the same curve as described in section 6.2.1 for PO2+PO3, as the benefits from PO4 are only realised in the long term. From 2032 onwards, PO4 starts to impact as described in section 6.3.1, resulting in the combination leading to the biggest energy savings across all policy option combinations.

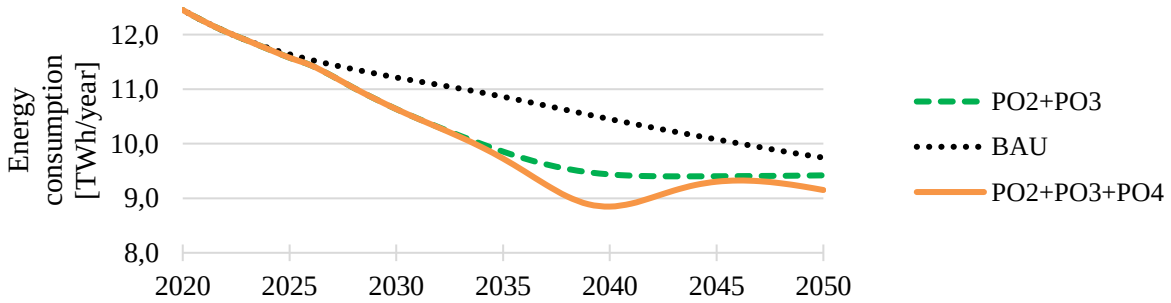


Figure 6-7. Total energy consumption for PO2+PO3+PO4, PO2+PO3 and BAU

	Energy consumption during use [TWh/year]				Savings [TWh/year]		Cumulative savings [TWh]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	10,50	9,49	8,97	8,10				
PO2+PO3+PO4	10,50	9,42	8,30	7,04	0,67	1,06	2,11	12,57
	Total energy consumption (in-use+embedded) [TWh/year]				**Savings [TWh/year]**		**Cumulative savings [TWh]**	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	12,45	11,64	11,21	10,45				
PO2+PO3+PO4	12,45	11,57	10,58	8,72	0,64	1,73	1,93	15,01
	GHG emission, in-use + embedded [mtCO2e/year]				**Savings [mtCO2e/year]**		**Cumulative savings [mtCO2e]**	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2+PO3+PO4	3,62	3,09	2,53	1,52	-0,01	0,43	-0,27	1,75

Table 6-10. Environmental impacts for PO2+PO3+PO4 and BAU

6.4.2. Economic impacts

The fluctuating evolution of turnover explained in section 6.3.2 for PO2+PO4 also takes place for PO2+PO3+PO4, although with differences linked to the greater presence of heat pump tumble dryers in the market. This causes turnover for PO2+PO3+PO4 to grow faster than for PO2+PO4 in the periods of increasing turnover, and to go down faster in the periods of decreasing turnover. This can be verified by comparing the yearly values in tables 6-10 and 6-11.

User expenditure also follows the same pattern as PO2+PO4, yet for PO2+PO3+PO4 consumer costs in the mid-to-long term are in general smaller due to bigger energy savings.

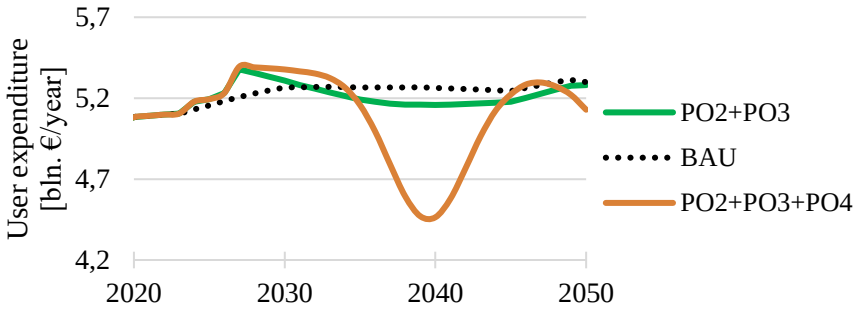


Figure 6-8. User expenditure for PO2+PO3+PO4 and BAU

	Industry turnover [bln. €/year]				Difference to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	1,27	1,37	1,45	1,54				
PO2+PO3+PO4	1,27	1,38	1,57	1,45	0,12	-0,11	0,45	1,08
Out of which in the repair sector	0,26	0,30	0,41	0,59	0,07	0,21	0,09	0,22
	Retail turnover [bln. €/year]				**Difference to BAU [bln. €/year]**		**Cumulative difference [bln. €]**	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	2,69	2,84	2,94	3,12				
PO2+PO3+PO4	2,69	2,88	3,09	2,30	0,13	-0,86	0,80	-2.32
	Total user expenditure [bln. €/year]				**Savings [bln. €/year]**		**Cumulative savings [bln. €]**	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	5,08	5,16	5,26	5,26			0,00	0,00
PO2+PO3+PO4	5,08	5,17	5,33	4,43	-0,07	0,83	-0,46	2,83

Table 6-11. Economic impacts for PO2+PO3+PO4 and BAU

6.4.3. Social impacts. Employment

The behaviour explained in section 6.3.3 with respect to PO2+PO4 applies here, with the particularity that job creation until 2035 is stronger than for PO2+PO4 because heat pump appliances are more labour intensive than other tumble dryers, although later job losses are also more significant.

	Total employment [jobs/year]				Difference to BAU (jobs/year)	
	2020	2025	2030	2040	2030	2040
BAU	22.023	22.977	23.651	24.531		
PO2+PO3+PO4	22.023	23.118	24.691	18.681	989	-6.031
Out of which in the repair sector	1.602	1.906	2.594	3.680	439	1.337

Table 6-12. Employment (industry, including repairers + retailers) in EU for PO2+PO3+PO4 and BAU

6.5. Additional impacts jointly assessed for all policy options

6.5.1. Payback period (affordability)

Payback period is calculated first by subtracting the yearly savings for each policy option (energy savings and savings due to the extension of the lifetime of the appliance) from the yearly increase of costs with respect to BAU (additional repair costs). The annual differences are multiplied by a discount factor of 4% and accumulated to the previous ones to get, for each year, the net present value of the total savings at the moment of purchase of the appliance. For each year that the difference has been computed, it is compared to the purchase costs. The payback period is given by the year in which the net present value of the savings offset the extra cost.

For PO2, the increase of the purchase price is offset by the energy consumption savings, over an average payback time of 3,4 years. This is significantly below the estimated lifetime of 12 years. For PO2+PO3 the pay-back time is 4,1 years, still significantly below the 12-year lifetime. The energy savings are considerably higher for PO2+PO3 than for PO2, although the increase of the product price delays the time of recovery of the investment. PO2+PO4 and PO2+PO3+PO4 have longer payback periods because the in-use energy savings yielded by PO2 and PO2+PO3 respectively are not sufficient to offset the additional repair costs of 1,7 €/year from PO4. The benefit of repair is nevertheless noticed after the appliance is 12 years old, because this is when the lifetime is extended by two years in the assessment, from 12 to 14. The respective payback periods are 12 and 12,1 for PO2+PO4 and PO2+PO3+PO4, which are still lower than the 14-year average lifetime considered for any combinations of policy options incorporating PO4.

	Increase in average unit price vs BAU		Average lifetime (years)	Discounted increase repair costs (€)	Discounted in-use energy savings (€)	Discounted payback period (years)
	%	€				
PO2	2,2%	11	12	0	1,9	3,4
PO2+PO3	5%	25	12	0	3,7	4,1
PO2+PO4	2,2%	11	14	1,7	0,9	12
PO2+PO3+PO4	5%	25	14	1,7	3,7	12,1

Table 6-13. Affordability of each policy option

6.5.2. Costs

6.5.2.1. Compliance costs

Compliance costs in a specific year have been computed multiplying the yearly sales by the increase in average unit price vs BAU from table 6-10. In the absence of data on costs incurred by the industry related to making available spare parts, 50% of the repair costs per unit used for the payback period have been considered to be compliance costs, considering that for each repair, 50% of the cost corresponds to the price of the component (equivalent to 13,8 €) and 50% to manpower

and travel costs. Costs to the manufacturer for making available spare parts are then 13,8 €/unit, which must be added to the total compliance costs for PO2+PO4 and PO2+PO3+PO4. This increase of compliance costs will be compensated by less manufacturing costs due to the reduction in the number of units produced.

	Compliance costs (million €/year)	
	2030	**2040**
PO2	64,6	67,1
PO2+PO3	146,9	152,4
PO2+PO4	145,7	151,2
PO2+PO3+PO4	227,0	236,6

Table 6-14. Compliance costs by measure

Other minor compliance costs which are not expected to influence the turnover are:

- Making available repair and maintenance information, as well as information on recycling following the new circular economy requirements. Manufacturers will be nevertheless able to offset the additional cost by charging a proportionate fee to repairers for the availability of repair and maintenance information.

- Updating the information on availability of spare parts and recyclability on the manufacturers websites or importers. The TD Ecodesign Regulation already requires certain inf ormation to be provided on the websites of the manufacturers and in the technical documentation, thus being assumed that the additional required information could be updated at negligible extra cost for the manufacturers.

6.5.2.2. Administrative costs

Administrative costs are defined as those costs borne by businesses, citizens, civil society organisations and public authorities as a result of administrative activities performed to comply with administrative obligations included in legal rules.

There will be extra administrative costs due to the update of the Declaration of Conformity (DoC) for the existing models. This update will be necessary following the reclassification of the products according to the rescaled energy label and the modification of the process for the calculation of the EEI. In addition, the new class will also need to be declared in EPREL. A time of 0,50 hours has been estimated for the update of the DoC and additional 0,50 hours for the update of EPREL information. Those times are multiplied by the estimated number of existing models (5.772)[55] and by the EU average hourly labour cost for 2021 (32€/h)[56], resulting in a total of 184.704 € (around 0,01% of the manufacturing turnover). For new models, the administrative costs are business as usual and they do not need to be quantified as new costs.

In addition, there will be administrative costs due to relabelling, as suppliers will have to provide two labels instead of one for a period of 6 months at a cost of 0,3 EUR to print a label. For 2,8 million household tumble dryers sold in 6 months, this means a total cost of approximately 851.550 € for suppliers to temporarily provide a second label for a transition of one label to another.

The total of these administrative costs (around €1m) are one-off costs representing around 0.05% of turnover in the year that they occur. There are no recurrent administrative costs.

[55] Figures from EPREL data
[56] Wages and labour costs - Statistics Explained (europa.eu)

6.5.3. Indirect employment

A good understanding of the employment effects of the measures on the society would require general equilibrium economic modelling. In any case, the likely effect is small enough to be difficult to identify with any certainty at the economy-wide level. The dominant economic impact is expected to arise from the avoided expenditure on energy by consumers due to the improved performance of the products, resulting in a net saving that will be spent in other areas of the economy. Although it cannot capture many of the likely interactions, a first indication of the scale of the effect is given by taking account of the differing employment intensity of the energy supply sectors and the general economy. It is assumed for the purpose of the assessment that these do not change in the coming decades. Estimates of employment intensity for the relevant sectors calculated from Eurostat data are: for the electricity, gas, steam and air conditioning supply sector (NACE D), 0,9 jobs/M€, and for the rest of the economy, 8,1 jobs/M€.

Table 6-12 provides the direct employment of the different policy options as calculated in the previous sections, the indirect employment and the net balance. The reduction of in-use and embedded energy consumption destroys employment in the energy generation sector, but the additional budget available to consumers due to both the reduction of the energy bill and the delay in the purchase of new appliances generates employment on the rest of the economy.

Employment (number of jobs)								
	Cumulative direct job creation in local tumble dryer sector (a)		**Indirect jobs in the overall economy (jobs)**				**Net impact (a-b+c)**	
			Cumulative indirect job losses in the electricity supply sector due to the reduction of energy consumption(b)		**Cumulative indirect job creation in overall economy due to the reduction of user expenditure (c)**			
	2030	**2040**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
PO2	567	1.918	160	660	1.684	6.367	2.091	7.625
PO2+PO3	3.085	8.276	364	2.445	3.717	22.833	6.438	28.664
PO2+PO4	1.675	-16.493	177	1.257	277	21.454	1.775	3.704
PO2+PO3+PO4	4.195	-11.716	378	2.954	2.310	37.343	6.127	22.673

Table 6-15. Cumulative employment on the tumble dryer sector, on electricity supply and on the rest of the economy and final net employment

6.5.4. Competitiveness

PO3 is the option with the most significant impact on manufacturers since it will them adapt their facilities to 100% heat-pump tumble dryer production. Nevertheless, as heat pump tumble dryers are already being placed in the market in large quantities it is not expected that the needed adaptations will have any impact on their competitiveness. In this context, the only issue that companies may encounter in their adaptation to the new ecodesign requirements is the time needed to convert their production facilities. This aspect is addressed in this IA by providing a transitional period.

The other sector that may be affected by the policy options, in particular by PO2+PO4 and PO2+PO3+PO4, is the repair industry, made up of SMEs with an average of two employees[57] per company. In this case however, the repair industry will benefit, as their turnover will be increased without any additional investment.

[57] Estimation from Eurostat data

In the view of the above and of the indications provided in Tool#21 of the Better Regulation Guidelines, it is considered that no further assessment of impacts on competitiveness is needed in this IA.

6.5.5. Fundamental rights

The policy options under assessment do not entail any negative impact on fundamental rights as laid down in the Charter of Fundamental Rights of the European Union[58]. Only positive impacts can be expected, first on the right to environmental protection thanks to the reduction of GHG emissions, and second on the right to consumer protection, resulting from the enhanced reparability of the products, the expected reduction of the energy bill, and the better consumer information consequence of the improvement of the energy label.

6.5.6. Geographical distribution

PO2+PO3 and PO2+PO3+PO4 will eliminate the least performant and cheapest tumble dryers from the market. Member States with lower household incomes are more vulnerable to the increase of prices accompanying those policy options than Member States with higher household incomes like France or Germany. This also happens to a lesser extent with PO2 and PO2+PO4. The penetration rate of tumble dryers in Southern Member States is low and it will be Eastern Member States, which account for 20% of EU tumble dryer ownership as displayed in Figure A12-2 in Annex 12, that are likely to be more negatively affected by the rising purchase costs. Section 6.5.1 shows that the increase of the prices will be offset by energy savings, which in principle would make the more ambitious policy options also interesting in Easter Member States. Nevertheless, the calculations have been carried out with average EU prices, but the reality is that energy costs greatly vary between Member States, and according to the prices for the second semester of 2021 taken from Eurostat[59], Eastern European Member States generally have lower electricity prices than those in Western Europe.

To identify the electricity price threshold that would make the different policy options interesting at national level, the same methodology as in section 6.5.1 can be used, keeping the same parameters but imposing a 12 or 14-year payback (depending on the average lifetime of a tumble dryer under the different policy options). This gives the threshold for the electricity price that would allow to offset the purchase costs within the lifetime of the appliance via savings in the operating costs. This shows that the minimum energy price to offset purchase costs in 14 years for PO2+PO3+PO4 is 0,14 €/Kwh, which would leave Croatia, Bulgaria and Hungary as the only Member States where this policy option would not bring benefits because they have lower electricity prices. These three countries have a very minor share (around 2%) of the EU market of tumble dryers. For PO2+PO3, the threshold would be 0,11 €/Kwh and only Hungary would not benefit from that option because of its very low electricity prices.

It can be concluded therefore that, in general, the impact of the policy options would not bring benefits with historic electricity prices in a limited geographical area which accounts for a very small share of the market. In the current context of much higher electricity prices analysed in section 8.4, it is possible that the abovementioned countries would also benefit under the most ambitious policy options.

[58] Charter of Fundamental Rights of the European Union (OJ L 326, 26.10.2012, p.391).
[59] File:Electricity prices for household consumers, second half 2021 .png - Statistics Explained (europa.eu)

7. HOW DO THE OPTIONS COMPARE?

The options identified are individually assessed against the two first specific objectives set out in Section 4. Each option is assessed for its effectiveness in achieving those objectives, its efficiency in doing so and its coherence. The third objective is addressed by measures that will be included in any of the policy options, and therefore it does not need to be addressed.

7.1. Effectiveness

Effectiveness refers to the extent to which the proposed policy options combinations would achieve the objectives set out in Section 4. Effectiveness has been assessed from two angles: cumulative energy savings (Effect1), which will also lead to a proportionate reduction of GHG emissions and acidification, and resource efficiency (Effect2) in terms of cumulative reduction of fuel use (coal/lignite, petroleum products, natural gas and biomass/waste) and cumulative reduction of material use for the production of tumble dryers. Besides the impacts considered in Effect2, further impacts related to resource efficiency/circular economy will be assessed in Annex 7. The values used for the calculation of the efficiencies are cumulative, and the periods of analysis are from 2025 (year of implementation of the policy option) to 2030 and from 2025 to 2040.

	Cumulative energy saved - Effect1					Cumulative material saved - Effect 2				
	In-use (Twh)		In-use and embedded (Twh)			Fuel saved (ktons)		Material saved for manufacturing (ktons)		
	2025-2030	2025--2040	2025-2030	2025-2040	Score	2025-2030	2025-2040	2025-2030	2025-2040	Score
PO2	0,96	3,60	0,90	3,51	+	222,8	818,5	-106,9	-472,1	Neutral
PO2+PO3	2,11	12,89	1,93	12,49	++	488,9	2.915,6	-345,4	-2.133,6	Neutral
PO2+PO4	0,96	3,60	0,90	6,39	+	222,8	818,5	-105,7	14.952,6	+
PO2+PO3+PO4	2,11	12,57	1,93	15,01	+++	488,8	2.843,1	-343,0	13.105,4	+

Table 7-1. Effectiveness of the different measures

Table 7-1 shows that all measures lead to a reduction of energy consumption (Effect1), with PO2+PO3 achieving the greatest savings in terms of in-use energy. PO2+PO3+PO4 is the option that delivers the largest total energy savings due to the extension of the lifetime of the appliances and the subsequent reduction of manufacturing activity.

As far as Effect2, PO2+PO3 yields the biggest mass of fuel saved due to the reduction of in-use energy consumption, and PO2+PO4 is the option that saves more material because enhanced reparability reduces the level of production.

Considering jointly both Effect1 and Effect2, PO2+PO3+PO4 appears as the best option, as it is the best in Effect 1 (considering total energy) and also Effect2 (considering the sum of fuel and material savings). For Effect2, PO2 and PO2+PO3 have been scored as neutral because they decrease the mass of fuel consumed but increase the material used by a similar amount.

Trade-offs

Table 7-1 shows a trade-off between the extension of the lifetime of the appliances and the increase of in-use energy consumption. By 2040 PO2+PO3 saves 0,32 Twh more in-use energy than PO2+PO3+PO4. This is because PO4 extends the lifetime of tumble dryers by two years, from 12 to 14, resulting in less efficient appliances remaining in use for a longer time. Nevertheless, the embedded energy saved by the PO4 component largely offsets the excess of in-use energy consumed, and the net balance is 2,5 Twh in favour of PO2+PO3+PO4. The trade-off happens as from 2034, when tumble dryers start to fail. There is also a similar trade-off between PO2 and PO2+PO4, although in this case the replacement of conventional tumble dryers by heat pump tumble dryers is significantly lower, which means that the in-use energy consumption difference

between PO2 and PO2+PO4 due to the extension of the lifetime of the appliances is almost insignificant.

7.2. Efficiency

This section assesses the benefit generated per monetary unit spent by the consumer for each policy option with respect to BAU, or in other words, the efficiency of consumer expenditure in generating the benefits addressed in section 7.1, namely in-use energy savings (Effic1), embedded energy savings (Effic2), fuel savings (Effic3) and material savings (Effic4).

The analysis performed in section 6.1 already shows that the investment carried out by the consumer in newer and more energy-efficient equipment following the application of PO2 and PO3, reduces in-use energy, GHG emissions and fuel consumption. Circular economy measures included in PO4 in turn, reduce consumer expenditure in new equipment, leading to a subsequent decrease in the use of embedded energy, material for manufacturing and GHG emissions, but at the same time have a negative influence on in-use energy consumption. In other words, the variation of in-use energy consumption is due to an investment made by the consumer on both the purchase of new equipment (which decreases in-use energy consumption) and on repair costs (which increases it), whereas the reduction of embedded energy is solely the consequence of an investment on repair costs. It is convenient therefore to assess in-use energy savings and embedded energy savings separately, as they are the consequence of different investments from the consumer. On the same basis, fuel and manufacturing material consumption will also be assessed separately. Consequently, consumer expenditure for calculating Effic1 and Effic3 will add both the purchase of new equipment and the repair costs, whereas for Effic2 and Effic4 only repair costs will be used.

The values used for the calculation of the efficiencies are cumulative rather than annual, because of the time gap between the year of the consumer investment and the return on that investment, which takes place over several years until the end of the lifetime of the appliance. A yearly value would not catch the effect of an investment made on that year, which will take place over the next years. It seems therefore more appropriate to use accumulated values over a period of analysis rather than a yearly value, as the former will give a more representative figure. The periods of analysis are from 2025 (year of implementation of the policy option) to 2030, and from 2025 to 2040.

	Effic1			**Effic2**			**Effic3**			**Effic4**		
	In-use energy savings/costs (purchase+repair) (Kwh/€)		**Score**	**Embedded energy savings/repair costs (Kwh/€)**		**Score**	**Fuel saved/costs (purchase+repair) (ton/M€)**		**Score**	**Material saved/repair costs (ton/M€)**		**Score**
	2030	**2040**		**2030**	**2040**		**2030**	**2040**		**2030**	**2040**	
PO2	1,93	1,64	+	-1,65	-0,09	-	64,1	20,3	+	-1.737	-11.011	-
PO2+PO3	1,95	3,81	+	-0,54	-0,39	-	37,5	13,1	+	-5.609	-49.763	-
PO2+PO4	1,93	-6,74	++	-0,11	1,00	+	82,9	-83,3	+	-803	135.604	++
PO2+PO3+PO4	1,95	13,30	++	-1,71	0,88	+	42,3	47,1	++	-2.615	132.489	++

Table 7-2. Efficiency of the different measures

The first aspect to be noted from table 7-2 is that PO2+PO4 features negative Effic1 and Effic3 values for 2040 (shown in red). This does not mean that PO2+PO4 is inefficient, but happens because the total consumer expenditure for PO2+PO4 is smaller than for BAU, since the economic saving due to lengthening the lifetime of the appliance offsets the increase of purchase and repair costs. The consumer investment with respect to BAU is therefore negative compared to BAU and turns the efficiency value also negative, meaning that PO2+PO4 is able to save in-use energy and fuel even with a consumer expenditure that is smaller than BAU. The negative net investment only happens in the long term, and therefore only for 2040 and not for 2030, as circular economy measures are noticed from 2034. Apart from this particular case, PO2+PO3+PO4 is the most

efficient policy option as regards Effic1 and Effic3, as each € invested by the consumer will save 13,3 Kwh of in-use energy and 47,1 tons of fuel in 2040.

As far as Effic2 and Effic4, the only policy options producing positive efficiencies in the long term are PO2+PO4 and PO2+PO3+PO4, as they are the only ones that produce embedded energy savings and therefore allow to reduce the material used during the production phase. PO2 and PO2+PO3 are not efficient at reducing material use or embedded energy because the complexity of manufacturing heat pump tumble dryers increases the embedded energy used throughout the whole period of analysis.

7.3. Coherence with other EU policies and legislative framework

7.3.1. Coherence with the overarching objectives of EU policies

The proposed measures are fully consistent with energy and environmental EU policy goals such as set out in the Green Deal. In particular the options contribute explicitly to *the EU's climate ambition for 2030 and 2050* and *mobilising industry for a clean and circular economy* as discussed in that Communication. The options have also been assessed to ensure as far as possible a simplification of requirements in line with Better Regulation goals. Nevertheless, some of the measures are more consistent than others with the Green Deal objectives. In particular, PO2 and PO3 will produce an increase of embedded energy that will result in total energy consumption going up in the first three to five years of application of the measures, with a corresponding increase of GHG emissions. This is later offset by means of better energy efficiency. Since the circular economy measure contributes to reduce embedded energy, it can be concluded that including PO4 will enhance coherence with the Green Deal objectives. PO4 will also enhance coherence with the European Climate Law, since circular economy has been recognised as one of the essential elements to achieve climate neutrality.

7.3.2. Coherence with Article 15(5) of the Ecodesign Framework Directive

All policy options fulfil the criteria set out in Article 15(5) of the Ecodesign Framework Directive, as shown in table 7-4.

Significant impacts as stipulated in Article 15 of the Ecodesign Directive	BAU	PO2	PO2+PO3	PO2+PO4	PO2+PO3+PO4
No significant negative impacts on the functionality of the product from the perspective of the user	✓	✓	✓	✓	✓
Health, safety, and environment shall not be adversely affected	✓	✓	✓	✓	✓
No significant negative impact on consumers in particular as regards affordability and life-cycle costs	✓	✓	✓	✓	✓
No significant negative impacts on industry's competitiveness	✓	✓	✓	✓	✓
Setting of an ecodesign requirement shall not have the consequence of imposing proprietary technology on manufacturers	✓	✓	✓	✓	✓
Impose no excessive administrative burden on manufacturers	✓	✓	✓	✓	✓

Table 7-4. Evaluation of policy options in terms of their impacts compared to the baseline (BAU)

All policy options are coherent with Article 15(5) of the Ecodesign Framework Directive, although with different degrees. In particular, PO2 and PO3 lead to an initial increase of purchase prices and consequent negative impact on consumers, which is later offset by means of better energy efficiency. PO4 does not increase prices and is therefore somewhat more coherent with Article 15(5) than the other two. However, PO4 on its own would lead to very limited benefits, being necessary to combine it with more effective measures.

7.4. Coherence with the proportionality and subsidiarity principles

Subsidiarity has already been assessed in section 3. All policy options have implications for the design and characteristics of the products and are therefore better addressed at EU rather than national level in order to ensure the correct functioning of the internal market.

As far as proportionality, as indicated in section 5, the most ambitious policy options PO2+PO3 and PO2+PO3+PO4 will accelerate the current market tendency towards a growing share of heat pump tumble dryers. Heat pump tumble dryers have been marketed for over a decade and can be considered as a mature technology, although some moderate future improvements on energy efficiency are expected to take place. The EEI proposed will only remove a part of the current market of heat pump tumble dryers besides heating element and air vented appliances, but will still leave 63% of the current models in the market. This implies that PO3 is proportional to the objectives, yielding energy savings whilst not setting targets that are difficult to achieve and that could significantly raise production costs causing affordability problems or too long payback times. Nevertheless, both PO2 and PO3 have short-term negative impacts on user expenditure and energy consumption, as indicated in sections 7.2 and 7.3.

There will be additional but very limited one-off administrative costs because there is no modification of the certification procedure. These have been calculated in section 6.5.2.2.

Concerning national authorities, there is no modification of the current market surveillance framework and therefore no additional burden.

8. PREFERRED OPTION

8.1. Preferred policy option

Based on the analysis presented in sections 6 and 7, PO2+PO3+PO4 is the preferred policy option from the point of view of effectiveness. This option provides the largest energy savings and GHG emissions reduction potential. PO2+PO3+PO4 is also the best option regarding total material (fuel + manufacturing) saved.

From the point of view of efficiency, PO2+PO3+PO4 the best option, with the highest in-use energy and fuel savings per monetary unit invested, and the second highest savings of primary energy and material used for production after PO2+PO4.

When efficiency and effectiveness are jointly considered, it looks clear that PO2+PO3+PO4 is the most balanced option, as it is as efficient as PO2+PO4, and is significantly more effective.

Summary of benefits for the preferred policy option PO2+PO3+PO4	**2030**	**2040**
Cumulative in-use energy savings (Twh)	2,1	12,6
Cumulative total energy savings (Twh), including both in-use and embedded energy	1,9	15,0
Cumulative fuel saved (ktons)	489	2.843
Cumulative material saved (ktons)	-343	13.105
Cumulative GHG emissions savings (Mton CO2eq)	-0,3	1,7
Cumulative additional manufacturer turnover, including repair turnover (M€)	455	1.077
Cumulative decreased consumer expenditure (M€)	-464	2.826
Cumulative direct employment (jobs)	4.195	-11.716
Cumulative net (direct+indirect) employment (jobs)	6.003	9.303
Cumulative water saved (Mlitres)	-270	1.673
Cumulative waste saved (tons)	35.295	186.058
Payback period (years)	12,1	

Table 8-1. Summary of cumulative values for the preferred policy option (PO2+PO3+PO4)

A more detailed summary of the results for the preferred policy option is displayed in Annex 3, including a monetisation of GHG emissions.

8.2. REFIT (simplification and improved efficiency)

A number of the measures proposed in this impact assessment will simplify the application of the legislation and reduce the legislative burden, in particular:

- The simplification of the layout of the energy label by aligning it with the washer dryers label will reduce labelling costs for manufacturers and increase understanding for consumers.
- The availability of documentation at the manufacturers' website concerning dismantling and recycling will make it easier for EoL processers to comply with the WEEE Directive.
- Legislative redundancies as regards low power modes are avoided by removing tumble dryers from the scope of the Horizontal Standby Regulation and adding the corresponding requirements in the TD Ecodesign Regulation.
- The number of test runs for the calculation of the condensation efficiency under the TD Ecodesign Regulation will be aligned to the number of test runs set out in the harmonized standard EN 61121:2013[60], thus removing ambiguity and legal uncertainty.

8.3. Application of the 'one in, one out' approach

As explained in section 6.5.2.2, there will be around 1 M€ administrative costs linked to the modification of the energy efficiency classes, which obliges manufacturers to draft an updated DoC for each model and makes suppliers to have to update the corresponding data in EPREL. No potential cost savings in administrative activities performed by businesses were identified that could be included in the current revision of the legislative framework on tumble dryers.

8.4. Overview in the context of current energy prices

Energy prices are sharply increasing due to the current situation of political instability. This increase, which was not foreseen when the review study was carried out, could encourage the introduction of more stringent ecodesign measures, because higher prices would push up operating costs and would call for even more demanding policies on energy efficiency. For instance, a stricter EEI lower than the proposed 85 could have been set for PO3.

It is not possible at this stage to make a fully-fledged evaluation of impacts considering current energy prices because the available scenarios have not yet been updated with those prices in time to be included in this report. It is however possible to carry out an analysis of the impact on affordability (payback period) of introducing additional ecodesign measures. Affordability is a key parameter as it measures directly the impact of the measures on the consumer. Other impacts such as lowering energy consumption or GHG emissions can only improve with the inclusion of stricter ecodesign measures, but this does not mean that the latter are affordable or economically viable.

The analysis in this section considers a stricter PO3 ($PO3_{str}$) with an EEI $\leq$ 71, so that only tumble dryers belonging to the A to C classes of the energy label instead of A to D can be placed on the market. The affordability analysis is carried out for PO2+$PO3_{str}$+PO4 with both REF2020 prices and the more recent EU electricity prices (2 semester 2021) for household consumers made available by Eurostat[61]. The latter are assumed to remain constant over the period of study, from

[60] In the current regulation the first test run is eliminated for calculation of the condensation efficiency.
[61] Electricity price statistics - Statistics Explained (europa.eu)

2025 to 2050. Electricity prices will in reality vary with time, but it is not possible to predict that variation reliably. Using fixed energy prices can give an idea of whether stricter ecodesign requirements could be justifiable. The result of the affordability analysis is given in table 8-1.

	Electricity prices	Increase in average unit price vs BAU, sale		Average lifetime (years)	Discounted increase repair costs (€)	Discounted in-use energy savings (€)	Discounted payback period (years)
		%	€				
PO2+PO3+PO4	REF2020	5%	526	14	1,7	3,7	**12,1**
PO2+PO3$_{str}$+PO4	REF2020	17%	585	14	1,7	5,3	**No payback**
PO2+PO3$_{str}$+PO4	2 semester 2021 (0,2369 €/kWh)	17%	585	14	1,7	5,7	**No payback**

Table 8-1. Affordability under higher energy prices

The inclusion of stricter ecodesign measures with REF2020 and with 2021 electricity prices is not cost-efficient, because the purchase price of tumble dryers will increase by 17% and the energy savings will not offset the increase in purchase cost. Figure 8-1 shows that with the higher electricity prices the Least Life Cycle Cost is in class D, which will be excluded from the market by setting the minimum EEI = 71. The purchase price dominates the life cycle cost for the high efficiency appliances, and above class D it is no longer cost-efficient to set stricter requirements, with the energy prices considered.

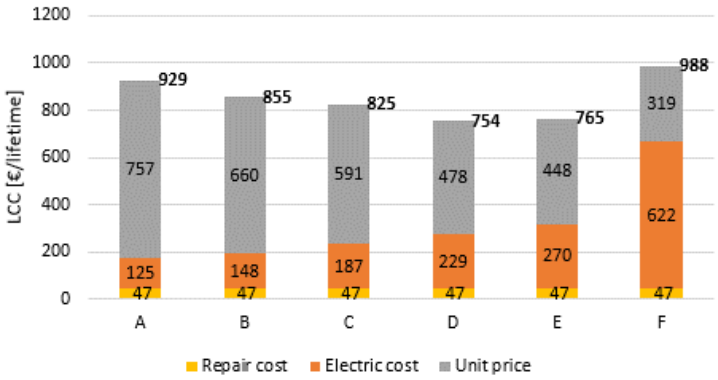


Figure 8-1. Costs distribution by energy label class with 1st semester 2021 prices

Nevertheless, as energy prices increase payback periods get significantly shorter and at a price of 0,340 €/Kwh payback period it is 13 years. Thus for prices consistently above 0,340 €/Kwh, more stringent ecodesign limits are interesting from a consumer point of view. This context of high prices needs to be nevertheless sustained in order to reduce the payback time below the average lifetime of tumble dryers.

9. HOW WILL ACTUAL IMPACTS BE MONITORED AND EVALUATED?

The main tool to verify compliance with ecodesign and energy labelling requirements is the market surveillance carried out by national authorities. Further, homogeneity of market surveillance actions at EU level is guaranteed by the Administrative Cooperation Groups, under which MSAs and the Commission meet several times per year with the objective to ensure efficient, comprehensive and consistent market surveillance across the Union.

In addition, surveillance activities are supported by the Commission through EEPLIANT3, which is a Horizon 2020-funded project with a budget of €6,85 million[62], involving a consortium of MSAs. The project runs from June 2019 until November 2023 and includes a work package on tumble dryers[63]. In the frame of EPPLIANT3, MSAs carry out market surveillance activities on their national markets and make recommendations that can be used later by national authorities in their regular enforcement activities. Those activities are focused on document inspections, analysis of the content of web-shops, verification of EPREL data and testing of appliances.

The summary of the results on tumble dryers is the following: on the document inspection chapter, the rate of non-compliance has been high, either due to missing documents or errors in the information provided. However, after MSAs enquiry a significant amount of information was returned, which proves the positive effect of enforcement on documentation quality. Regarding online sites inspections, the results have not been positive due to the general lack of knowledge about the legislative requirements on websites. EPREL inspections show acceptable results, being the percentage of tumble dryers not present in EPREL remarkably reduced through enforcement, from 24% to 11%. On the negative side, the percentage of missing entries for each model is still high. Finally, as far as testing, the percentage of dryers with severe problems is low. However, it must be noticed that the time for discussions between MSAs and economic operators is sometimes very long.

All in all, EPPLIANT3 has identified specific parameters with different levels of compliance. MSAs will need to take the EPPLIANT3 conclusions into account in order to optimise efforts by focussing on those specific issues where level of compliance is deficient. This will also apply to the enforcement of the upcoming TD Ecodesign and Energy Labelling Regulations.

Finally, the revised TD Ecodesign and Energy Labelling Regulations will include a review clause that will mandate, inter alia, the evaluation of the effectiveness and regulatory efficiency and relevance of the revised regulations 5 years after their entry into force. The results of this evaluation, which will involve the acquisition of independent market data as well as extensive stakeholder consultation, should be presented to stakeholders and Member States in the Ecodesign and Energy Labelling Consultation Forums. In addition, the European Commission will be regularly analysing the latest updated data of the database EPREL, in order to monitor the evolution of the market as regards of energy efficiency, condensation efficiency, refrigerants used in the appliances and level of noise. This will allow the Commission to early detect possible technical improvements that can be assessed in future revisions.

[62] https://cordis.europa.eu/project/id/832558
[63] https://eepliant.eu/index.php/new-products/tumble-dryers

ANNEX 1: PROCEDURAL INFORMATION

1. LEAD DG, DECIDE PLANNING/CWP REFERENCES

DG ENER, PLAN/2019/5480 (Ecodesign) and PLAN/2019/5479 (Energy Labelling), Commission Ecodesign and Energy Labelling Working Plan 2022-2024 (C/2022/2026 final).

2. ORGANISATION AND TIMING

A draft of the impact assessment report was submitted on 21 March 2022 to the members of the Ecodesign and Energy Labelling Interservice Steering Group, involving the following DGs: SG, AGRI, BUDG, CLIMA, CNECT, COMM, COMP, DEFIS, EAC, ECFIN, ECHO, EMPL, ENV, ESTAT, FISMA, GROW, JRC, JUST, IDEA, MOVE, REFORM, REGIO, RTD, SANTE, SJ, TAXUD. A meeting took place on 4 April. There were no major comments, DG CLIMA asked to include in the impact assessment report that the latest proposal for a revised F-gas Regulation would ban the use of F-gases from 2025 in heat pump tumble dryers, In addition, DG CLIMA also asked to better reflect what is already prescribed in Article 12 of the F-gas Regulation to avoid contradictions or duplications.

3. CONSULTATION OF THE RSB

The RSB was consulted on 25 May 2022. Prior to that, an upstream meeting took place on 21 March 2022, in which the Board provided several recommendations to DG ENER, highlighting in particular the need to ensure coherence with related initiatives, the importance to consider scenarios in line with the "fit-for-55" package, the relevance of the cost-benefit analysis as basis for the adoption of choices on the policy options and the inclusion of sensitivity analysis on crucial parameters.

4. EVIDENCE, SOURCES AND QUALITY

The departing point for this impact assessment is the review study carried out by the consultant Viegand Maagoe, finished and handed over to the Commission in May 2019. The review study provides some of the input data used for the baseline scenario and the modelling of the different policy options related to sales, stock and product prices, all of them split by product type. Sales data have been purchased to GfK, a reputed company in the field of data provision and analysis for the consumer goods industry. Stock data have been derived from sales and average lifetime for each product family. Average lifetime has been extensively discussed with APPLiA, the main industry interlocutor at EU level.

The EPREL database established by the Commission, which gathers information about all the models placed on the market since 2010, has been consulted for confirmation of data on energy efficiency and condensation efficiency per model.

Electricity prices and carbon intensity have been obtained from REF2020, in order to ensure coherence and consistency of the impact assessment baseline scenario with existing energy policies. The REF2020 electricity prices have nevertheless proved to be unrealistic in the context of the current energy crisis with escalating energy prices. Unfortunately, at the time of finalisation of this impact assessment no scenario incorporating the influence of the new situation on the energy market was available. For this reason, section 8.4 has been included in this IA, featuring a preliminary analysis on how higher energy prices could influence the configuration of the policy options and the choice of the preferred option. The electricity price taken for this analysis is the EU average electricity price of the second semester of 2021, which is still below current prices but it is

the latest officially available for the whole EU. In the absence of information about the future evolution of electricity prices, it has been assumed that the electricity price considered in the analysis is fixed throughout the whole period. This assumption, even being a simplification, does not prevent drawing valid conclusions about the policy option to be chosen in a context of sharp rise of energy prices.

ANNEX 2: STAKEHOLDER CONSULTATION (SYNOPSIS REPORT)

1. REVIEW STUDY AND STAKEHOLDER CONSULTATIONS

The review study was launched in October 2017 and was completed in May 2019. It covered a technical, environmental, and economic analysis performed according to the MEErP methodology, in order to assess the need to update the requirements for household tumble dryers and to identify the corresponding policy options.

The study was developed in an open process, taking into account inputs from relevant stakeholders including manufacturers and their associations, environmental NGOs, consumer organisations and Member States. During the study, two open stakeholder consultation meetings were organised at the Commission premises in Brussels on 26 June 2018 and 4 December 2018. Drafts of the study were discussed at these meetings, reviewed and validated with input from stakeholders. A review of different consumer surveys was also presented as part of the results, including one carried out by APPLiA in 2018. Annexes VII and X of the review study present a thorough overview of the contributions from stakeholders to both meetings.

In addition, a dedicated open public website was set up on which the interim results and other relevant material were published. The purpose of the website was to facilitate the monitoring of the status of the study, get related information and be able to track the meetings taking place.

2. FIRST CONSULTATION FORUM ON 18 SEPTEMBER 2019

The Consultation Forum consists of a balanced representation of Member State representatives, industry associations and NGO's, in line with Article 18 of the Ecodesign Framework Directive.

A Consultation Forum specifically on tumble dryers was held on 18 September 2019. The Commission services prepared two Working Documents with separate ecodesign and energy labelling requirements, which were based on the policy options described in section 7.2.2 of the review study report (Table 60 of that study). The documents were circulated to the members of the Consultation Forum and were also provided to the secretariat of the Environment, Public Health and Food Safety (ENVI) Committee of the European Parliament for information. The minutes, working documents and stakeholder comments received in writing before and after the Consultation Forum meeting were posted on CIRCABC[64]. 11 documents with comments were received from the following stakeholders and Member States: APPLiA (providing three documents with comments), ANEC, ECOS (on behalf of ECOS, Coolproducts, EEB, Topten, Right to Repair, Rreuse), Denmark, Germany, France, Sweden, Austria, Bulgaria and the Netherlands.

3. STAKEHOLDERS COMMENTS DURING AND AFTER THE CONSULTATION FORUM

The comments on the two Working Documents are summarized as follows:

- Scope: it was discussed if small capacity dryers should be left out of the scope of the TD Ecodesign Regulation. Two NGOs and three Member States proposed to include them with different rules in order to make it possible for them to continue being marketed, consisting of either less stringent EEI limits or a correction factor. Two Member States proposed to also include in those specific rules household tumble dryers that can be powered by batteries. Further, APPLiA and three Member States proposed to explicitly state that the regulation would

[64] ecodesign - Library (europa.eu)

not apply to products covered by the Machinery Directive. The option assessed in this IA is to include small tumble dryers in the scope, although exempting them from the new EEI limits.

- Name of the standard programme: APPLiA and some Member States supported to rename the standard cotton programme used during testing as 'Eco', to align tumble dryers with washing machines and to make clear to consumers what the best programme is in terms of energy efficiency. APPLiA also emphasizes that there should be some flexibility for manufacturers. For instance they should be allowed to additionally indicate the drying target and/or the type of textile: e,g, "cotton eco cupboard dry" and/or "cotton eco". ECOS does not support the use of "eco" for the standard programme, but instead the use of the term "standard/normal", since it would ensure that more consumers instinctively use that programme. In addition, NGOs and two MS representatives called to regulate other programmes or at least to regulate the most commonly used programme rather than the most efficient. This point needs to be discussed further with stakeholders before the legislative proposal is formalised, noting in the meantime that the final decision does not have any relevance for the assessment in this IA.

- Cycle time duration: two NGOs and five Member States proposed to regulate the duration of the standard programme to avoid extensively long cycle times. Moreover, two Member States proposed to limit the duration of the wrinkle guard function. A Member State proposed to include these requirements in a review clause for future updates of the legislation. As result of the discussion, a unique cap to the duration of all programmes is proposed in this IA.

- Energy Efficiency Index (EEI) requirements: the NGOs and five Member States proposed to set very stringent EEI requirements. Among them, five NGO's proposed EEI $\leq$ 80, with two NGOs proposing a second tier with EEI $\leq$ 60. The proposal in this IA is EEI $\leq$ 85, since this is the value that would phase out condenser and air vented dryers and even some of the least efficient heat pump tumble dryers. For further energy savings, incentive to manufacturers by means of the rescaling of the label is considered to be sufficient.

- EEI intervals in the energy label: nearly all stakeholders proposed to revise the current intervals, first to cater for the revision of the minimum EEI level and second, to make the distribution among all classes more even whilst making them consistent with a verification tolerance of 6%. Two Member States and APPLiA commented that the proposed intervals do not incentivise manufacturers to produce class A tumble dryers as the class is very hard to reach. The options considered provide a good balance between expected energy efficiency gains and the amount of effort that is necessary to achieve those gains.

- Calculation of Standard Energy Consumption per cycle (SEc): two NGOs proposed to align the calculation of SEc with the ecodesign and energy labelling legislation on washing machines and APPLiA commented that current correction factors are too high and would prevent tumble dryers reaching class A.

- Condensation efficiency: two Member States proposed to set two stages to reach a more demanding minimum condensation efficiency, with 90% applicable from 2024. In addition, one Member State proposed to remove the split between full and half loads for the calculation of the average condensation efficiency ('C'), since the conditioning cycle that is run in every test is able to level differences between both. APPLiA proposed to rescale the condensation efficiency classes by making them wider than the proposal. The IA keeps a minimum condensation efficiency of 80%, as it is considered that higher values could significantly increase the average price of tumble dryers, and because the benefit of efficiencies higher than 80% would only be

noticed in very reduced enclosures with a very high humidity. Condensation efficiency classes have been removed.

- Spare parts availability requirements: NGOs and five Member States proposed to extend the list of available spare parts so that they are aligned to the lists provided in the washing machine ecodesign Regulation. In addition, the NGOs and three Member States proposed that spare parts should not only be available to professional repairers but also to end-users. APPLiA proposed to reduce spare parts availability time to 7 years to avoid unnecessary waste and stock, while two NGOs proposed to extend it to 12 years, which is the average lifetime of tumble dryers. Three Member States proposed to keep the European Commission proposal of 10 years. Two NGOs proposed to specify the definition of professional repairers in a way that it does not exclude credible repair actors. APPLiA also advised against the inclusion of the heat pump system in the list of spare parts, because the high costs and complexity of the replacement would mean the repair would not be cost-efficient. With this background, it is proposed that the TD Ecodesign Regulation include a list of spare parts that is similar to that in the Ecodesign Regulation on washing machines, since many of the components are of similar nature, and completing this list with components that are specific from tumble dryers. For safety reasons, the list should differentiate between parts available for consumers and parts restricted to professional repairers. Finally, the availability period should be 10 years, which is considered a reasonable mid-point.

- Disassembly requirements: two NGOs proposed to include disassembly requirements for the materials and components listed in Annex VII of the WEEE Directive, in particular as regards the heat pump. Materials and components shall be easily disassembled with relevant information disclosed by manufacturers. APPLiA pointed out that replacing the heat pump is too costly and complex and that the environmental savings are not worth the replacement. The TD Ecodesign Regulation should not mandate tumble dryer design requirements to make disassembly easier because this measure could have a negative impact on costs. Nevertheless, enhanced information for disassembly and recyclability of the product will need to be available by manufacturers on a public website.

- Filter maintenance requirements: two NGOs and one Member State proposed to include a requirement for the appliances to automatically signal when the filter needs to be cleaned and to provide guidance information on proper cleaning to avoid micro-plastic release. An alternative is to signal in a visible part of the appliance the need to periodically change the filters.

- Low power mode requirements: APPLiA, two NGOs and three Member States proposed to include low power mode requirements in the TD Ecodesign regulation and remove them from the Horizontal Standby Regulation, in line with other ecodesign regulations such as washing machines.

- Requirements on the use of high GWP refrigerants: two NGOs proposed the implementation of a rating system penalising the EEI when using high GWP refrigerants, as well as adding information to consumers about the type and GWP of the refrigerant. One Member State supported the inclusion of information about the refrigerant.

- Requirements to restrict the use of halogenated flame retardants: two NGOs proposed to integrate restrictions on these chemicals for the largest components of the tumble dryer, as it has been done for the electronic display regulation. Product marking could be used as a complimentary tool. Nevertheless, the use of halogenated flame retardants is a crosscutting

issue that should be addressed from horizontal legislation, and it has been considered that the TD Ecodesign Regulation is not the appropriate framework to address it.

- Implementation dates: five Member States and two NGOs proposed to align implementation dates for both ecodesign and energy labelling regulations, otherwise it would create confusion and market surveillance issues. APPLiA and one Member State proposed to implement ecodesign requirements in two tiers, the first applying 18 months after the entry into force of the legislation and the second 36 months after the first tier.

- Verification tolerances: APPLiA and a Member State representative proposed to add tolerances for full and half-load, and for networked and information display standby modes. However, verification tolerances are consistent with the accuracy of the current testing methodologies as confirmed via a Round Robin Test carried out by several manufacturers in different testing facilities.

- Pictograms in the energy label: five Member States proposed to remove from the energy label the pictogram indicating the type of tumble dryer, since gas dryers have been phased out and all of them are electric. One Member State proposed to add a pictogram indicating the GWP of the refrigerant and two NGOs proposed to add pictograms on guarantee period and spare parts availability.

4. PUBLICATION OF THE CALL FOR EVIDENCE AND STAKEHOLDERS FEEDBACK

For this IA, a derogation has been granted and therefore no 12-weeks Open Public Consultation has been carried out. A Call for Evidence has nevertheless been published on the Have your Say Portal[65] on 21 January 2022, and the feedback period finished on 18 February 2022. A total of 9 comments were received from the following stakeholders: EIA, AIG, APPLiA, ZERO, ECOS, ANEC/BEUC, VDZ, EMMA/ENPA and Principia Association. The feedback to the call for evidence does not deviate from the opinions summarised in Section 3 of this Annex. The comments provided are the following:

- Energy efficiency threshold: ANEC/BEUC pledges for setting out a stricter EEI.

- High global warming potential refrigerants: EIA, ZERO and ECOS ask for the consideration of measures restricting the use of refrigerants with high global warming potential.

- Rescaled label: APPLiA states that it is not necessary that A class in the new label is empty at the beginning, and that intervals between classes should be narrowed down. ANEC/BEUC does support that A class should be left empty in order to incentivise technical development.

- Transitional periods: APPLiA asks not to phase out tumble dryers in current classes F and G before 2026. ANEC/BEUC does not support a two-tier approach and asks the measures to be implemented in one time and as soon as possible.

- Availability of spare parts: generally supported by industries and environmental NGO's, although with different levels of ambition.

- Noise: ANEC/BEUC asks for an introduction of noise limits.

[65] Ecodesign and energy labelling requirements for household tumble dryers [review] (europa.eu)

5. SECOND CONSULTATION FORUM ON 10 MARCH 2022

A second Consultation Forum took place on 10 March 2022, with the main objective of updating stakeholders on the progress of the IA and gathering their opinion about some specific issues that still needed discussion. The issues related to the application of energy labelling and ecodesign requirements in two stages, the list of spare parts, the denomination of the programme under test and the incorporation of low power modes in the TD Ecodesign Regulation. Stakeholders generally agreed to the approach proposed by the Commission to each of the topics. NGO's and a number of Member States asked for the merging of the two stages for the application of the requirements in a single one, 18 months after the entry into force of the amendments to the TD Ecodesign and Energy Labelling Regulations.

ANNEX 3: WHO IS AFFECTED AND HOW?

1. PRACTICAL IMPLICATIONS OF THE INITIATIVE

The TD Ecodesign and Energy Labelling Regulations will apply to the manufacturers established in the EU and to importers of equipment produced by other manufacturers which are not established in the EU, which will need to comply with the requirements summarised in table A3-1.

Who	What	When
Manufacturers, importers and authorised representatives	New rescaled energy label and new EEI calculation method	2025
Manufacturers, importers and authorised representatives	Minimum energy efficiency requirements	2026
Manufacturers, importers and authorised representatives	Resource efficiency: - Minimum spare part availability of 10 years - Repair and maintenance information - End of life information - Cleaning of filters	2025
Manufacturers, importers and authorised representatives	Information on the type of refrigerant	2025

Table A3-1. Summary of Energy Labelling and Ecodesign requirements

2. SUMMARY OF COSTS AND BENEFITS

I. Overview of Benefits (total for all provisions) – Preferred Option		
Description	***Amount***	***Comments***
Direct benefits		
Cumulative energy savings (see table 8-1)	1,9 TWh in the period 2025-2030 and 15,0 TWh in the period 2025-2040	
Cumulative GHG-emissions savings (see table 8-1)	-0,3 Mt CO_2eq in the period 2025-2030 and 1,7 Mt CO_2eq in the period 2025-2040	
Cumulative economic savings due to the reduction of externalities	-30 M€ in the period 2025-2030 and 170 M€ in the period 2025-2040	CO2 emission savings have been monetised at a price of 100€/tonCO_2eq, which is a common value used by the Commission in its IAs.
Cumulative fuel savings (see table 8-1)	489 ktons in the period 2025-2030 and 2.843 ktons in the period 2025-2040	
Cumulative material savings for manufacturing (see table 8-1)	-343 ktons in the period 2025-2030 and 13.105 ktons in the period 2025-2040	
Cumulative water savings (see Annex 7)	-270 Mlitres in the period 2025-2030 and 1.673 Mlitres in the period 2025-2040	
Cumulative waste savings (see Annex 7)	35.295 tons in the period 2025-2030 and 186.058 tons in the period 2025-2040	
Cumulative consumer expenditure savings (see table 8-1)	-464 M€ in the period 2025-2030 and 2.826 M€ in the period 2025-2040	
Cumulative increased employment in the society (see table 8-1)	6.003 jobs in the period 2025-2030 and 9.303 in the period 2025-2040	This is the sum of direct and indirect jobs. PO2+PO3+PO4 causes a loss of employment of 11.716 direct jobs.

*Administrative cost savings related to the 'one in, one out' approach**		
	0	No savings of administrative costs resulting from the preferred option

Table A3-2. Overview of Benefits (total for all provisions) as compared to the baseline– Preferred Option

II. Overview of costs – Preferred option							
		Citizens/Consumers		**Businesses**		**Administrations**	
		One-off	**Recurrent**	**One-off**	**Recurrent**	**One-off**	**Recurrent**
Costs of increasing the production of heat pump tumble dryers	Direct adjustment costs				147 M€/year in 2030 and 152 M€/year in 2040		
	Direct administrative costs						
	Direct regulatory fees and charges						
	Direct enforcement costs						
	Indirect costs						
Costs of making available spare parts	Direct adjustment costs				81 M€/year in the period in 2030 and 84 M€/year in 2040		
	Direct administrative costs						
	Direct regulatory fees and charges						
	Direct enforcement costs						
	Indirect costs						
Costs of drafting a new DoC for existing models and updating the information in EPREL	Direct adjustment costs						
	Direct administrative costs			1 M€			
	Direct regulatory fees and charges						
	Direct enforcement costs						
	Indirect costs						
Costs related to the 'one in, one out' approach							
Total	Direct adjustment costs						

	Indirect adjustment costs						
	Administrative costs (for offsetting)			1 M€			

Table A3-3. Overview of costs (total for all provisions) as compared to the baseline– Preferred Option

RELEVANT SUSTAINABLE DEVELOPMENT GOALS

III. Overview of relevant Sustainable Development Goals – Preferred Option		
Relevant SDG	**Expected progress towards the Goal**	**Comments**
SDG no. 7 - affordable and clean energy, target 7.3 (double the improvement of energy efficiency)	Expected increase of the share of tumble dryers placed on the market and enhanced reparability will save 2,1 TWh of in-use energy by 2030 and 12,6 TWh by 2040, contributing to positively reduce the energy bill in households	
SDG no. 8 – Decent work and economic growth; target 8.4 (improve resource efficiency in consumption and production)	The preferred policy option will save both fuel used for powering the tumble dryers (2,8 Mtons by 2040) and material used during the manufacturing process (13 Mtons by 2040).	The fuel saving will mostly correspond to the reduction of in-use energy consumption by PO2 and PO3, whereas material savings will be due to PO4.
SDG no. 10 – Reduced inequalities; target 10.1 (reduced income inequalities)	Energy efficiency measures will yield direct cumulative consumer savings of 2.826 M€ by 2040, thus increasing by the same amount consumers' available income.	
SDG no. 11 – Sustainable cities and communities; target 11.6 (reduce the environmental impact of cities)	There is no data about the distribution of local tumble dryers between cities and rural areas, but it is expected that the vast majority of energy savings take place in densely populated areas, thus helping to reduce the environmental impact in cities.	
SDG no. 12 – Responsible consumption and production; targets 12.2 (sustainable management and use of natural resources), 12.5 (substantially reduce waste generation)	Availability of spare parts will save 2,4 Twh of accumulated embedded energy by 2040 (target 12.2) and will generate material savings up to an amount of 13 Mtons in 2040 (target 12.5) thus reducing waste at EoL.	
SDG no. 13 – Climate action; target 13.1 (strengthen resilience and adaptative capacity to climate-related hazards)	Expected accumulated GHG emissions decrease by 1,7 MTCO2-eq. by 2040, contributing to the deceleration of climate change.	The decrease in energy GHG emissions is faster during the first years, and levels off over time due to the progressive decarbonisation of the energy grid

Table A3-4. Overview of relevant Sustainable Development Goals – Preferred Option

As indicated in section 7.1, there is a trade-off between SG12 and SG13, because the extension of the lifetime of the appliances increases the in-use energy consumption. Even if there are some minor losses of in-use energy savings due to the extension of the lifetime of the appliances, the analysis carried out in section 7.1 shows that the embedded energy saved by PO4 largely offsets the excess of in-use energy consumed, being the net balance of +0,6 Twh/year in favour of PO2+PO3+PO4.

ANNEX 4: ANALYTICAL METHODS

1. MODEL DESCRIPTION

The impact assessment is based on an analytical model specifically developed by the external consultant Viegand Maagøe A/S for impact assessments. The model is built in Excel using a 1-year time step, and covers the years 2000 – 2050. A shorter timeframe can be used, but for product with long (10 years+) lifetimes, the effects of the ecodesign measures are not immediate and thus a longer timeframe is needed to properly evaluate the effects. This also applies to tumble dryers as they have a product lifetime of around 12 years.

The analytical model is divided into three different connected sub-models, each covering a different aspect of the analysis. The models are:

A **Data input model** containing raw data for sales, power consumption, energy efficiencies and other input data gathered from the sources available for the project. Data are sorted, scaled, and normalised depending on the data quality, source, and scope.

A Stock model, calculating the stock of tumble dryers from the sales and product lifetime data coming from the data input model. The available data is extrapolated to cover the whole period 2000 – 2050.

Finally, **a scenario model** calculates all relevant impacts for the study including, but not limited to:

- total user expenditure, retail, wholesalers and manufactures turnover;
- total energy consumption in the use phase, production and end-of-life of the products;
- GHG emissions for the use phase, production and end-of-life;
- jobs;
- material consumption.

The scenario model will first define a BAU scenario and subsequent policy options will then be linked to this BAU scenario. All policy options and their effects will be calculated in a single excel workbook available to the Commission.

Labour costs, GDP and inflation rates are taken from Eurostat. Learning rates and multivariable regressions are used to project product prices following new ecodesign regulations. Energy prices and energy intensity follow REF2020.

Life cycle costs and embedded emissions corresponding to the manufacturing, distribution and recycling of tumble dryers have been calculated according to the MEErP methodology. The effect of CO2e emissions related to refrigerants being released to the ambient during EoL processing are calculated separately to the regular GHG emissions.

2. DATA SOURCES, SCALING AND PROJECTIONS

2.1. Sales data

The primary data source for this study is market data supplied by the market research institute GfK, which is broken down by type of tumble dryer and includes the following information: sales

numbers, unit prices, energy label, rated capacity, power consumption, condensation efficiency, cycle time, noise levels, programme duration and the inclusion of smart technology on specific models.

The raw data have been supplied for the years 2017, 2018 and 2019 and have been combined with data from the review study from 2000 – 2016 to have a strong dataset for 2000 to 2019 and to ensure consistency with the review study.

GfK also supplies a coverage ratio per country, this is the share of each national market covered by the data[66]. The coverage ratios are weighted with the population and GDP of each country and compared to the full population and GDP for EU-27 in order to evaluate the data coverage of the full EU-27 market. The sales data is then scaled with the weighted coverage ratios[67] in order to give a best possible estimation for the real sales values for the EU-27 market, including Estonia, Latvia and Lithuania which was not covered in the market analysis by GfK.

Country\Period	Jan 17-Dec 17	Jan 18-Dec 18	Jan 19-Dec 19
Austria	91	91	91
Belgium	87	87	87
Croatia	75	75	75
Czech Republic	93	93	93
Denmark	95	95	95
Finland	90	90	90
France	94	83	90
Germany	95	94	94
Great Britain	95	95	95
Greece	92	92	92
Hungary	91	91	91
Ireland	85	85	85
Italy	93	93	93
Luxembourg	65	65	65
Netherlands	90	90	90
Poland	88	88	88
Portugal	94	95	95
Romania	90	90	90
Russia	65	65	65
Slovakia	92	92	92
Slovenia	85	85	85
Spain	80	80	80
Sweden	90	90	90
Switzerland	22	22	22
Ukraine	91	91	91
Estonia	not audited	not audited	not audited
Latvia	not audited	not audited	not audited
Lithuania	not audited	not audited	not audited

Table A4-1. Coverage ratio of the supplied GfK data. Numbers are percentage of the market covered in the analysis

[66] E.g., a 90% coverage ratio means that 90% of the sold models in the given country is covered in the supplied data set.
[67] The following scaling factors have been found: For GDP: 115%, for population: 131%. Average: 123%.

The sales data are then projected from 2019 to 2050. High model accuracy cannot be expected for projections longer than 10 – 15 years, but a long timeframe is needed to evaluate the long-term effects of the legislation.

For the non-heat pump tumble dryer models, the projections are based on an exponential smoothing algorithm based on the years 2017 – 2019, weighted with the projected growth of households in the EU from Eurostat. For heat pump models, an s-curve[68] function is used to model the initial growth of this dryer type. The function was first fitted to the known data from 2013 to 2019, and then continued towards 2050. The s-curve function gives a moderate growth projection compared to using a linear or exponential function that would otherwise have excessively increased the penetration rate of heat-pump tumble dryers.

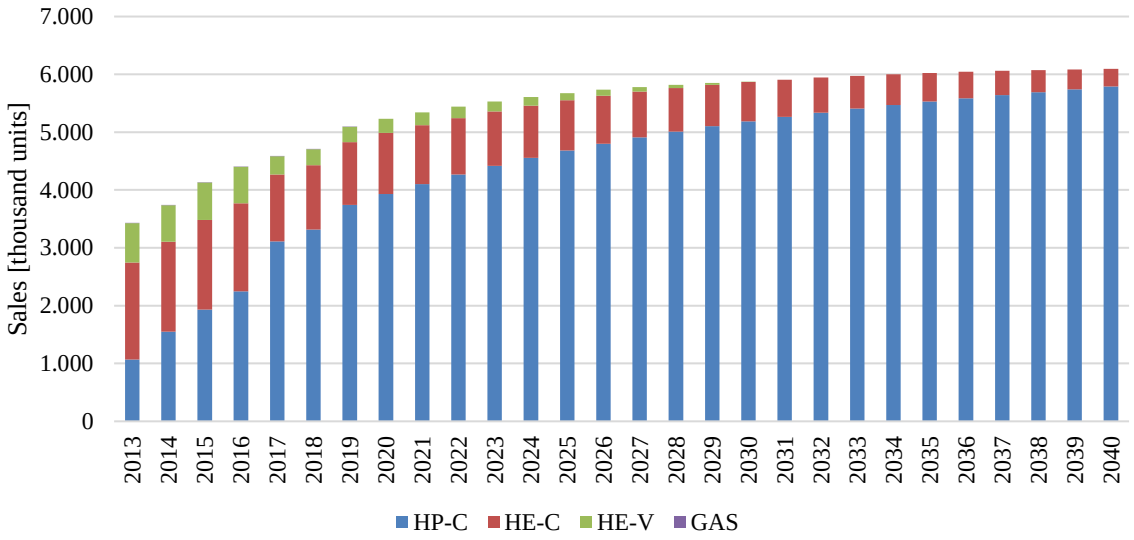


Figure A4-1. BAU - EU27 sales numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

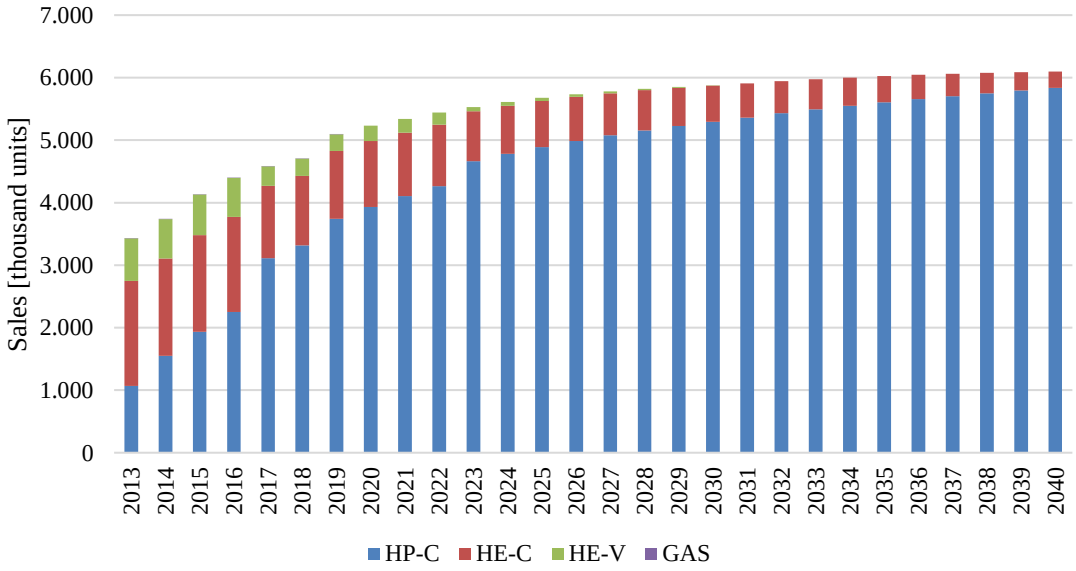


Figure A4-2. PO2 - EU27 sales numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

[68] A logistic growth function – also called a sigmoid function.

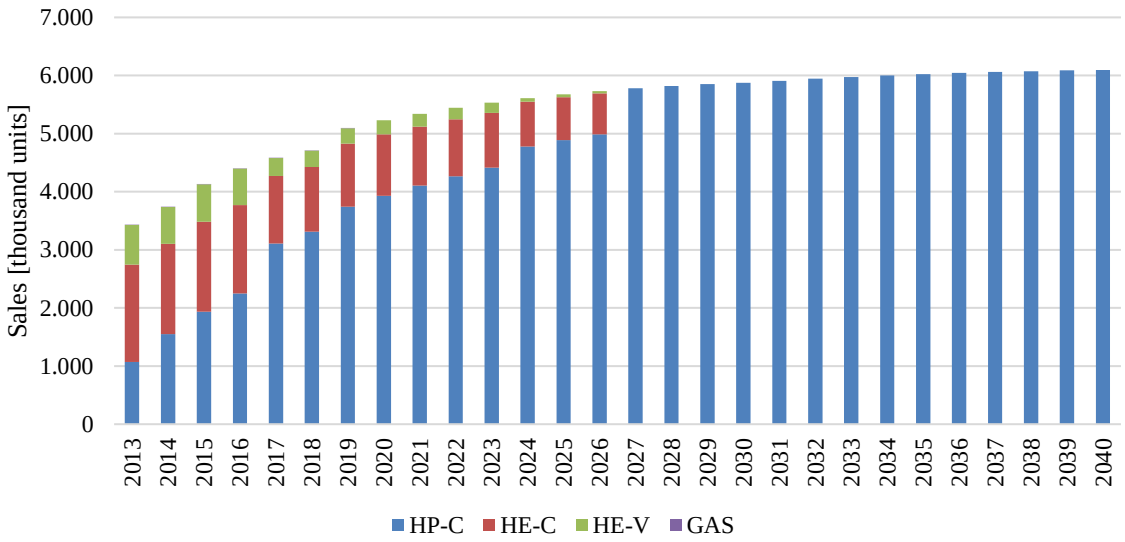


Figure A4-3. PO2+PO3 - EU27 sales numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented.

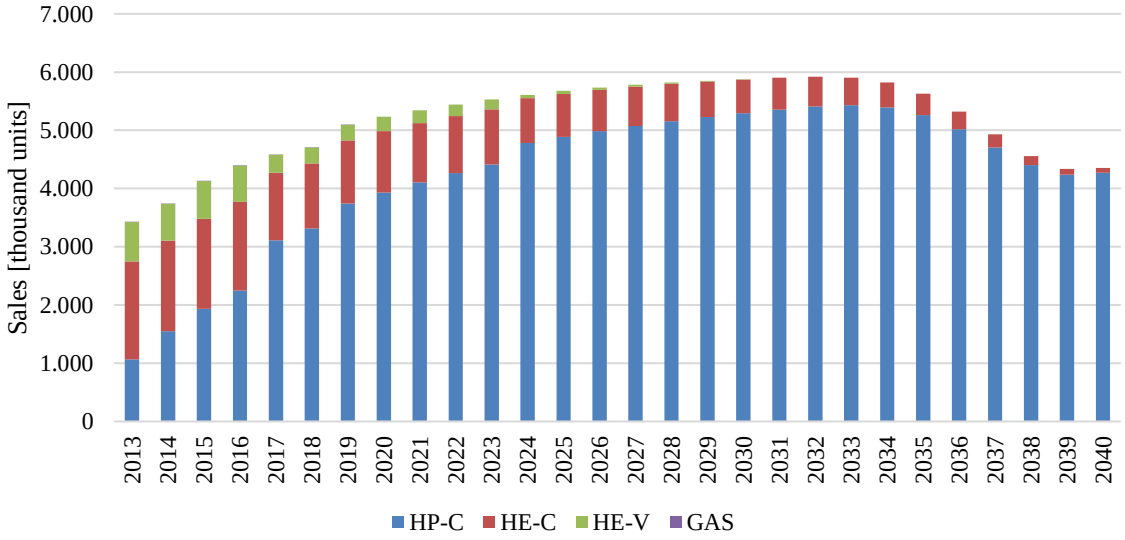


Figure A4-4. PO2+PO4 - EU27 sales numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers. HE-C = Heating element condensers. HE-V = Heating element air vented

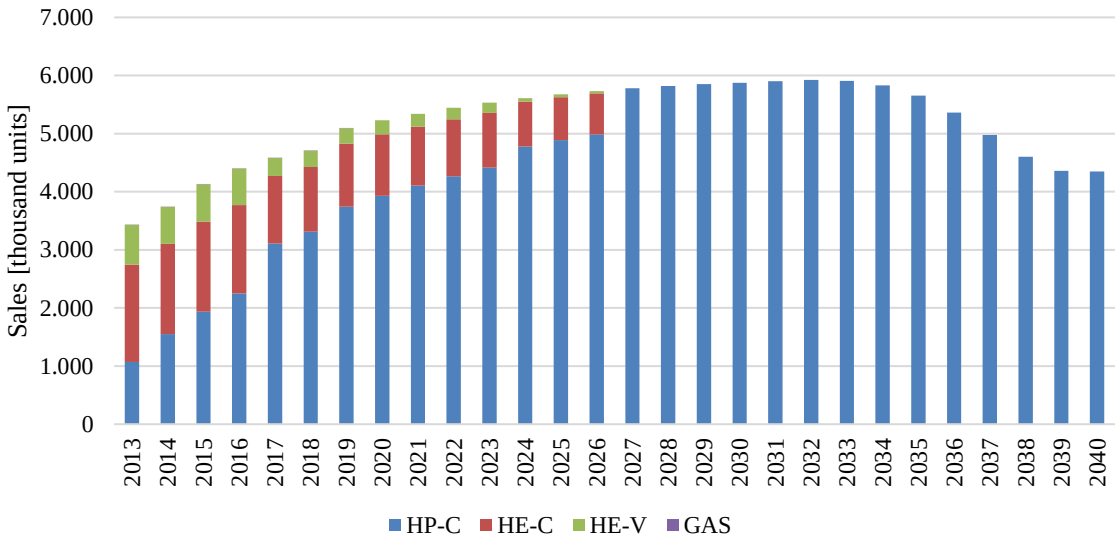


Figure A4-5. PO2+PO3+PO4 - EU27 sales numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

Figure A4-6 shows the percentage of heat pump dryers sold in the different scenarios. For PO2+PO3, the stringent ecodesign limit will remove all non-heat pump dryers between 2026 and 2027.

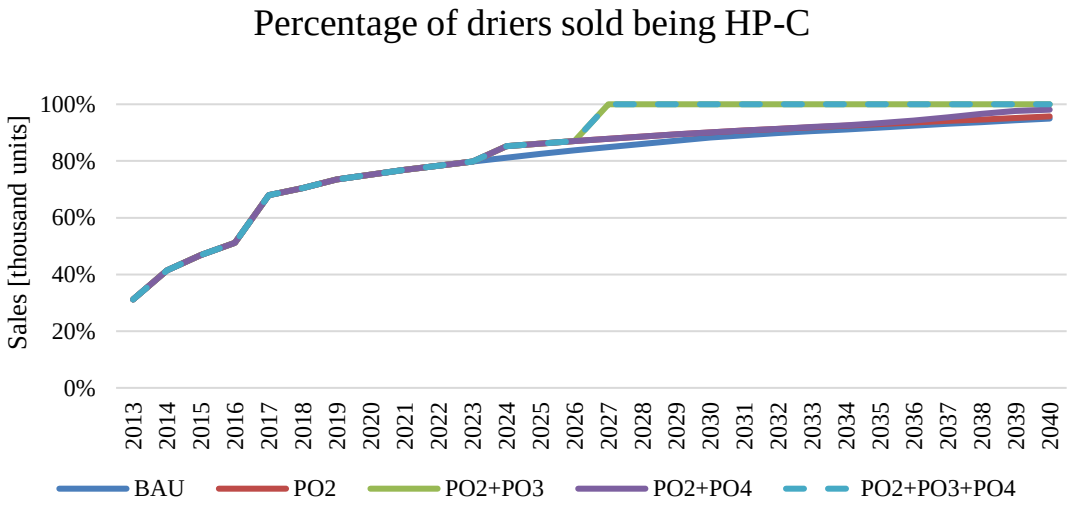


Figure A4-6. EU27 percentage of sold heat pump tumble dryers. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

2.2. Stock

The stock of tumble dryers in the EU has been estimated on the basis of sales and expected product lifetimes from the review study. This is done as follows: for each given year, the sales of a tumble dryer type on that year are combined with a Gaussian lifetime distribution to calculate the share of those sales that will be in stock for the remaining years until the end of the product average lifetime. The use of a Gaussian distribution entails that 50% of the units of a given model are still in use when the average model lifetime has been reached. The total stock in each year is derived by summing the stock estimated for the sales of different years.

The total stock is the same in all policy options because none of them will affect the need of consumers to have a tumble dryer. Nevertheless, the share for the different models of tumble dryers varies among scenarios.

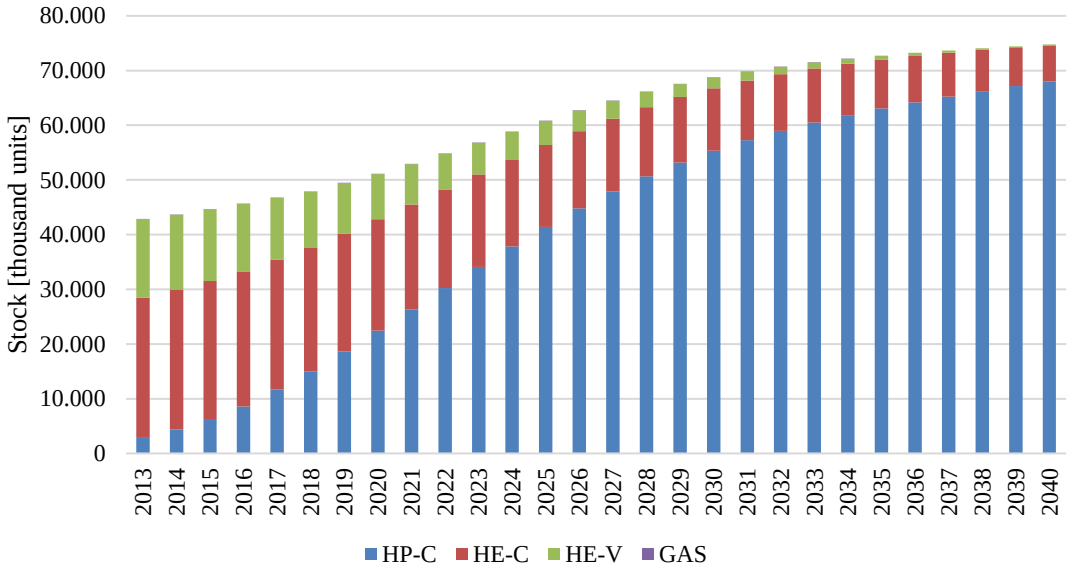


Figure A4-7. BAU - EU27 stock numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

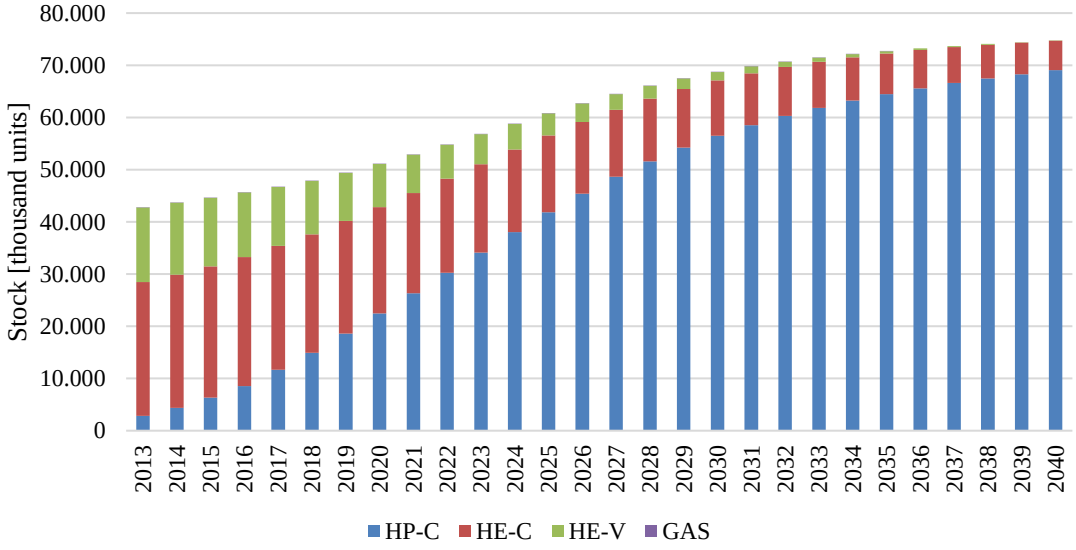


Figure A4-8. PO2 - EU27 stock numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

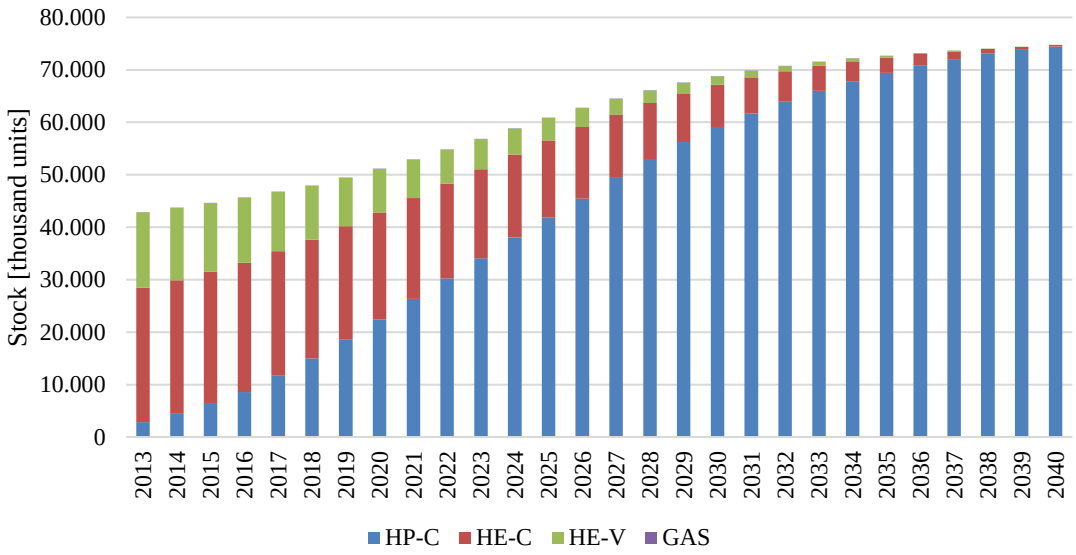


Figure A4-9. PO2+PO3 - EU27 stock numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

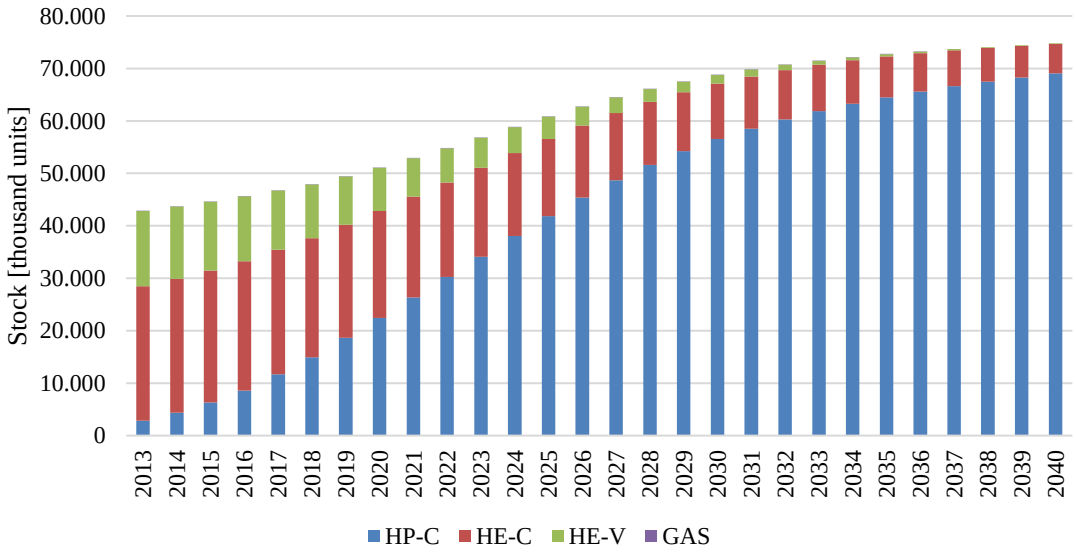


Figure A4-10. PO2+PO4 - EU27 stock numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

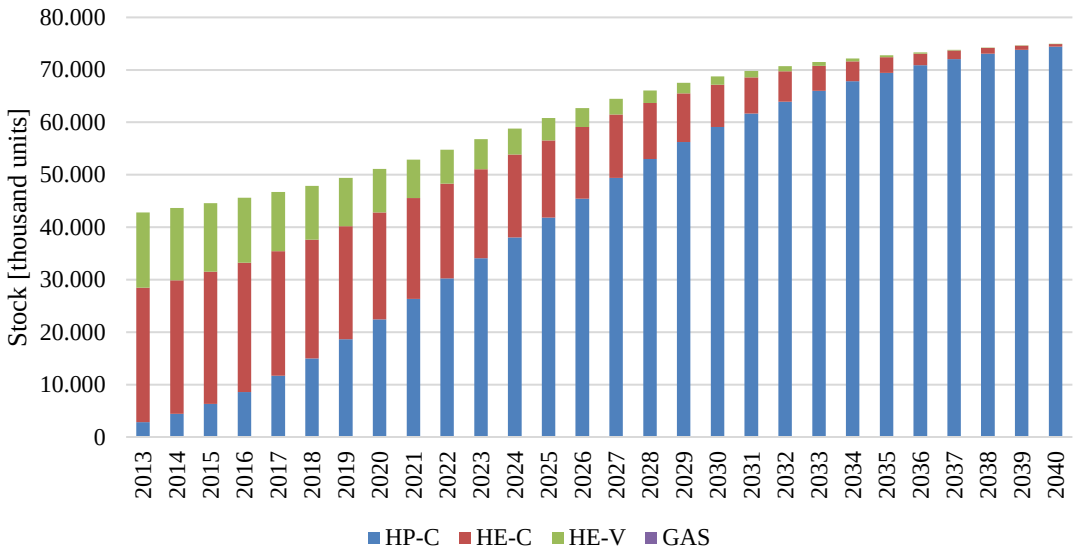


Figure A4-11. PO2+PO3+PO4 - EU27 stock numbers from 2013 to 2040. Source: GfK (2013 – 2019), Viegand Maagøe (2020 – 2040). HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented

2.3. Price versus performance

Purchase price and level of efficiency are correlated according to a regression formula, so that higher energy efficiency (lower energy consumption) corresponds to higher prices. Nevertheless, the GfK dataset is not disaggregated by model, and it is therefore not possible from this data to obtain an average price for each tumble dryer type, which is necessary to carry out the correlation between purchase price and level of efficiency. Instead, the 2019 APPLiA model database has been used to get those prices, allowing a relation to be established between efficiency and price according to the regression displayed in figure A4-12. For condenser tumble dryers (both heating element and heat pump), a power regression is used. For air-vented, a linear regression fits better.

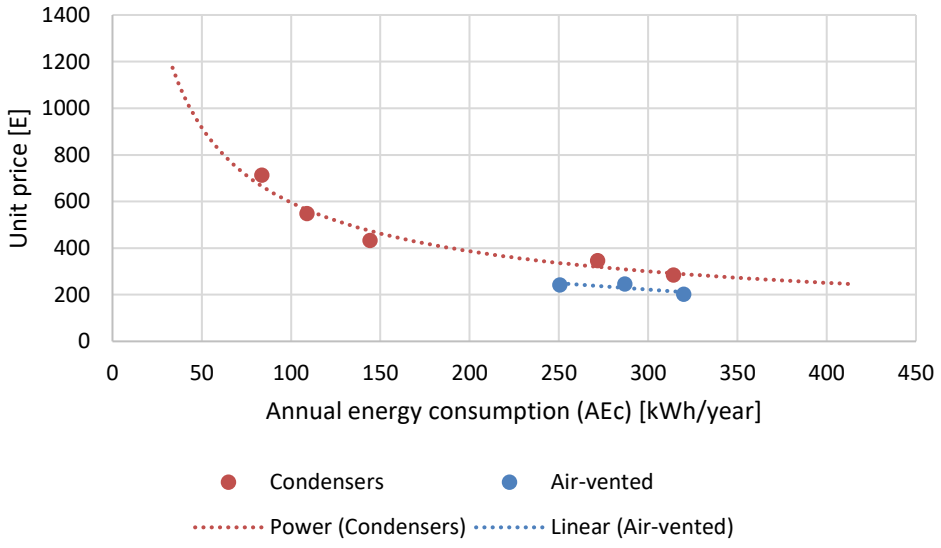


Figure A4-12. Price vs annual energy consumption. Source, GfK, APPLiA

The evolution of unit prices over time has been modelled through the combination of a learning rate approach with an exponential smoothing algorithm. Data for 2013 to 2019 are used as a reference to fit the model, which is then projected towards 2040. The evolution of unit price for the current energy labelling regulation is shown in Figure A4-13.

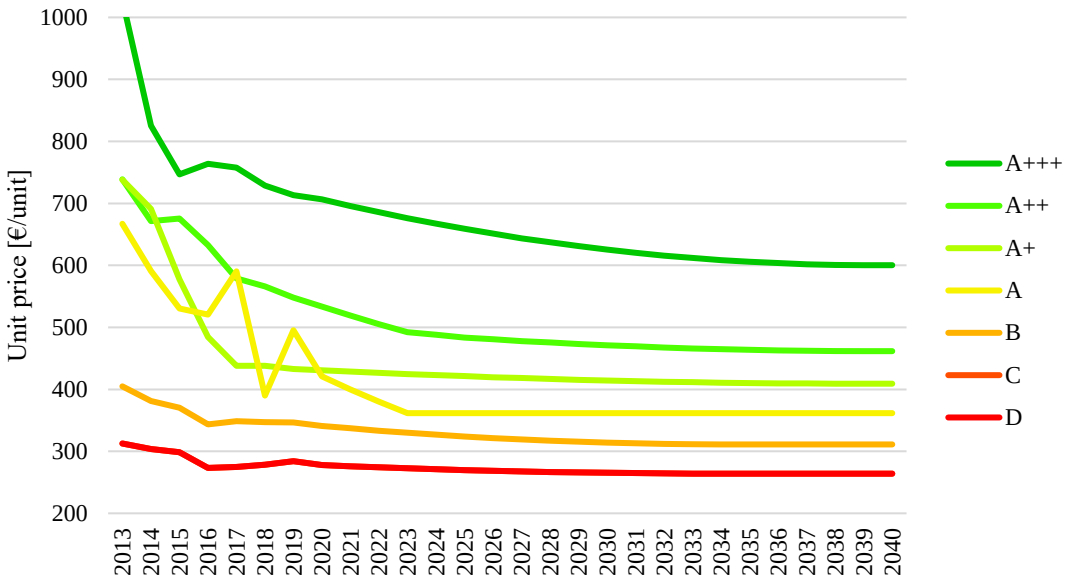


Figure A4-13. Projected unit prices based on current energy labels from 2013 to 2040. Source: GfK and Viegand Maagøe

For scenarios with rescaling of the energy labelling (proposed regulation), the current energy classes are converted to the new energy label classes based on the energy consumption per cycle (Ec).

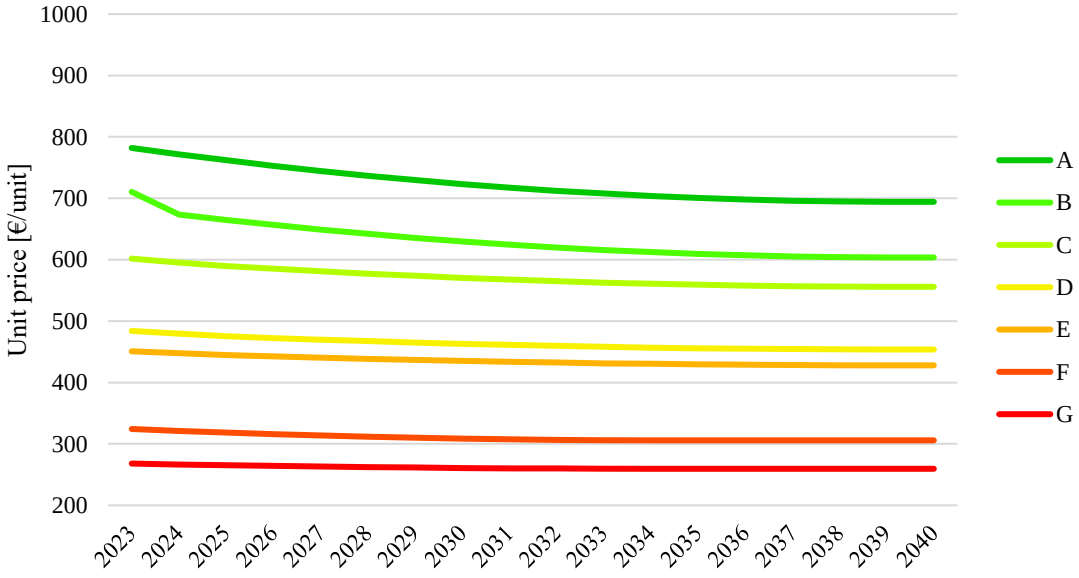


Figure A4-14. Projected unit prices for rescaled energy label from 2013 to 2040. Source: GfK and Viegand Maagøe

The development of the average sales weighted unit price per policy option from 2013 to 2040 is shown in Figure A4-15. The average unit price in PO2+PO3 increases when the more demanding energy efficiency requirements enter into force, PO2 increases the prices slightly due to some cheap vented dryers being omitted from the market due to the rescaling.

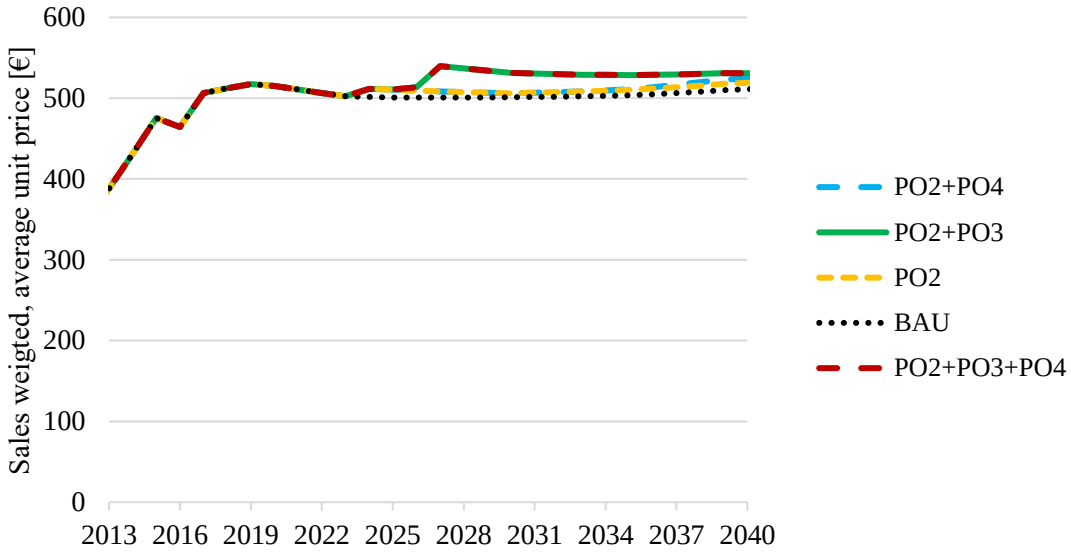


Figure A4-15. Sales weighted unit prices

2.4. Rated capacity

The rated capacity of tumble dryers has been steadily increasing since 2010 and this tendency is expected to continue until 2030. From 2030 to 2040 no further increase in capacity is foreseen. The capacity tends to be higher for condenser dryers than for air vented dryers. The assumed development in the rated capacity is shown in Figure A4-16. The values until 2019 are based on market data from GfK.

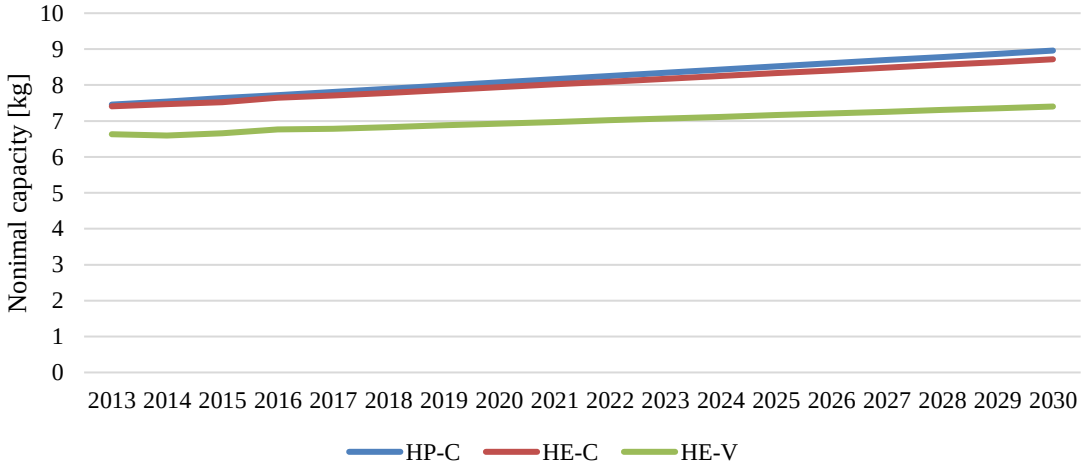


Figure A4-16. Average nominal capacity per tumble dryer type from 2013 to 2030. Values from 2030 to 2040 assumed constant in the model. Source: GfK (2013-2019), Viegand Maagøe (2020 - 2030)

2.5. Electricity costs and emission factors

Electricity prices (household prices) and $CO2_e$ emission factors for electricity consumption come from REF2020 The MIX scenario (which is in line with the fit-for-55 objectives) is evaluated in the sensitivity analysis in Annex 8.

Electricity	***2010***	***2015***	***2020***	***2025***	***2030***	***2035***	***2040***	***2045***	***2050***
Households	169	176	193	201	210	208	208	209	224

Table A4-2. Electricity prices from PRIMES in €/MWh (REF2020)

Electricity	*2010*	*2015*	*2020*	*2025*	*2030*	*2035*	*2040*	*2045*	*2050*
Intensity	318	305	214	179	143	109	75	61	61

Table A4-3. Emission factors/intensity for electricity consumption in $gCO2_e/kWh$ (REF2020)

2.6. Employment

It is assumed that the impact of the measures on employment is proportional to the turnover, corrected negatively by an increase in labour productivity. For the manufacturing industry there are data available to approximately verify this assumption, i,e, from Eurostat detailed enterprise statistics Eurostat[69] for the years 2011-2016 and from APPLiA for 2017-2018. The graph below shows turnover versus number of employees. Between 2013 and 2015 the number of employees does not follow turnover trend[70], but for the period 2011-2013 and 2015-2018 the assumption seems correct. In the view of the available data, the impact assessment considers that every 1% increase in turnover is equivalent to 0,55% increase in direct jobs, which is rounded up to 0,6%. The swift to heat pump tumble dryers, which are more labour intensive than the other types of tumble dryers due to the complexity of the heat pump system (consisting of multiple metal sheets welded with very little space between each other) may have pushed the ratio of 0,55% somewhat higher, as according to information from the industry, a heat pump tumble dryer is 1,25% more labour intensive than a conventional tumble dryer. The effect on employment is supposed to be nevertheless small, as heat pump systems hold already a 70% of the market share, and it is estimated that an additional 30% (100% market share of heat pump tumble dryers) could increase the ratio to around 0,59%.

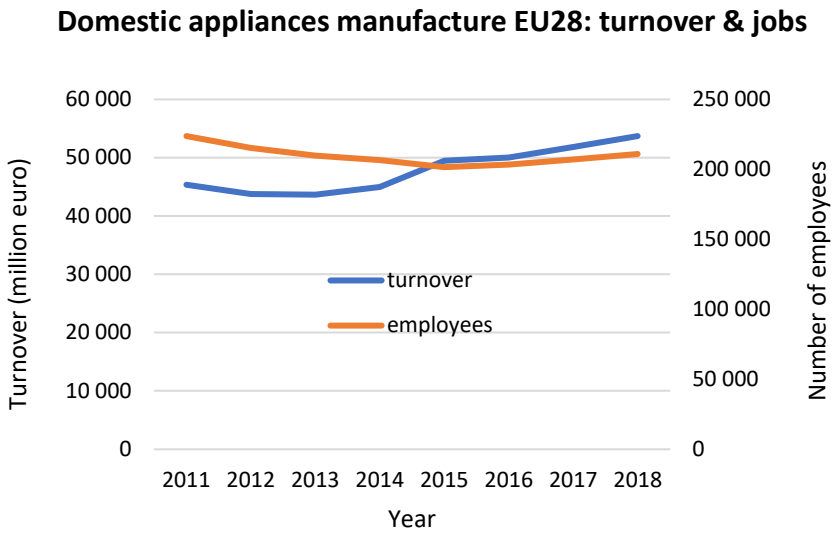


Figure A4-17. Turnover and jobs domestic appliance manufacturer 2011-2018. Source: Eurostat 2021, APPLiA 2020

2.7. Mark-up factors

Based on feedback from the sector, the split of the purchase price for tumble dryers is 47% manufacturing, 39% retail and 14% taxes[71], percentages that are also used as employment factors.

[69] Eurostat, Annual detailed enterprise statistics for industry (NACE Rev. 2, B-E) Year 2016 [sbs_na_ind_r2]
[70] This can be caused by a statistical correction (e.g. redefinition of manufacturing previously belonging to another sector) or an actual introduction of new manufacturing activities.
[71] VAT 21% but considering also a share B-2-B sales at zero VAT.

2.8. Material input

Material input used in the calculations to estimate embedded (production, distribution and recycling) GHG emissions are based on the values from the review study. Calculations are carried out with the EcoReport Tool.

Material Type	Materials (examples)	Air-vented – Heat element (g)	Condenser – Heat element (g)	Condenser – heat pump (g)
Bulk Plastics	PP, PP GF, ABS, PA GF[72]	9.300	12.800	13.900
TecPlastics	Elastomer	900	679	1.200
Ferrous	Sheet metal steel	18.700	23.473	18.500
Non-ferrous	Aluminium, copper	150	1.364	3.500
Coating		0	0	0
Electronics	Various	5.600	6.040	13.350
Misc,		2.800	2.800	6.800
Total		37.450	47.156	57.550

Table A4-4. Bill of Materials for tumble dryers

2.9. Performance data

The following three major assumptions related to the calculated energy consumption per cycle are included in the model:

- Average cycles per year: 107.
- Average load per cycle (in kg, of laundry used): 4,4.
- Initial moisture contents: 56% for cotton, 42% for synthetics[73] (for reference, the testing standard indicates an initial moisture contents of 60% and 45% for cotton and synthetics respectively)

The parameters are used for calculation of the yearly energy consumptions in the scenario analysis[74]. For the initial moisture content, a factor is used to adjust the energy consumption per cycle in relation to the test standard to obtain a realistic value that takes into account the more efficient spinning in washing machines.

The above parameters have been investigated in detail for the review study and the values provided in that study have been kept as no new reliable sources of information have been identified during this impact assessment.

[72] Nomenclature used in the EcoReport tool. PP=Polypropylene, PP GF=Polypropylene Glass Reinforced, PA=Polyamide, PA GF=Polyamide Glass Reinforced.
[73] This results in a reduction of 6.8% of energy consumption per kg of laundry compared to the energy consumption stated on the label from the testing standard.
[74] Yearly energy consumption is estimated as energy consumption per kg*kg per cycle*cycles per year.

ANNEX 5: EEI CALCULATION AND RESCALING OF THE ENERGY LABEL

1. CALCULATION OF THE NEW EEI

The formula to calculate the EEI is the following:

$$EEI = \frac{Etc}{SEc} \times 100$$

with:

- **EEI** = Energy Efficiency Index
- **E_{tc}** = Weighted energy consumption of the active mode per cycle [kWh]
- **SEc** = Standard energy consumption per cycle [kWh]

Following the conclusions from the review study[75], new formulas are proposed for **E_{tc}** and **SEc (AE_c and SAE_c in the current regulation)** in order to better reflect real-life use of tumble dryers. Furthermore, it is proposed to align Ec with the new regulations for washing machines and dishwashers by displaying on the label the energy consumption per 100 cycles instead of per year. This will require that the Energy Efficiency Index (EEI) is also calculated per 100 cycles.

1.1. New calculation method for E_{tc} and T_t

Currently, the calculations of the weighted[76] energy consumption per cycle (E_{tc}) and the weighted cycle time (T_t), are based on 3 cycles with full load, and 4 cycles with half-load, which is equivalent to an average load per cycle of ~71% of the rated capacity of the tumble dryer.

In order to better reflect the average load reflected in the APPLiA consumer study (corresponding to 62% of the rated capacity), the following new formula for calculating the weighted energy consumption is proposed:

$$E_{tc} = 0{,}24 \times E_{dry} + 0{,}76 \times E_{dry½} \quad [77]$$

with:

- **E_{dry}** being the average energy consumption at full load for the standard cotton program [kWh]
- **$E_{dry½}$** being the average energy consumption at partial (half) load for the standard cotton program [kWh]

[75] Review study on household tumble dryers. Final Report. June 2019.
[76] "Weighted" refers in this case to the weighing between the energy consumption at full load, and the energy consumption at half/partial load.

No change of the current test method is proposed, but only to the weighing between full and partial load.

The same approach is used to estimate the average cycle time:

$$T_t = 0{,}24 \times T_{dry} + 0{,}76 \times T_{dry½}$$

with:

- **T_{dry}** being the average cycle time at full load for the standard cotton program [kWh]
- **$T_{dry½}$** being the average energy consumption at partial (half) load for the standard cotton program [kWh]

Similarly, for the condensation efficiency:

$$C_t = 0{,}24 \times C_{dry} + 0{,}76 \times TC_{dry½}$$

with:

- **C_{dry}** being the average condensation efficiency at full load for the standard cotton program [-]
- **$C_{dry½}$** being the average condensation efficiency at partial load for the standard cotton program [-]

For gas-fired dryers, the energy consumption per cycle at full and half load ($E_{dry}, E_{dry½}$) is defined as:

$$E_{dry} = \frac{Eg_{dry}}{f_g} + Eg_{dry,a}$$

$$E_{dry½} = \frac{Eg_{dry½}}{f_g} + Eg_{dry½,a}$$

with

- **Eg_{dry}** being the gas consumption at full load for the standard cotton program [kWh]
- **$Eg_{dry½}$** being the gas consumption at partial load for the standard cotton program [kWh]
- **$Eg_{dry,a}$** being the auxiliary electricity consumption at full load of the standard cotton program [kWh]
- **$Eg_{dry,a}$** being the auxiliary electricity consumption at partial load of the standard cotton program [kWh]

- $\mathbf{f_g}$ being a conversion factor between primary energy and electricity. Currently, this factor is 2,5, but it needs to be changed to 2,1[78] to better reflect the average EU electricity generation efficiency.

1.2. New calculation method for SEc

The new SEc is based on the average weighted energy consumption per cycle of the three tumble dryer types (condenser heat pump, condensing heating element and air vented) per rated capacity, to which the sales distribution in 2019 are applied in order to give a sales-weighted value per rated capacity at 7, 8, and 9 kg[79]. A power regression ($y = a \times x^b$) is then applied to the resulting values, which has resulted in the corresponding power regression coefficients: a = 0,46 and b = 0,63.

The proposed SEc for condensing dryers will then be:

$$SEc = 0{,}46 \times c^{0{,}63}$$

with "c" being the rated capacity.

Rated capacity [kg]	Weighted energy consumption per cycle (SE_{tc}) [kWh/cycle]			
	HP-C	HE-C	HE-V	Sales weighted average
7	1,18	2,76	2,77	1,57
8	1,17	3,07	3,17	1,65
9	1,33	3,38	3,56	1,85
Sales distribution (2019)	75%	20%	5%	

Table A5-1. Weighted energy consumption per cycle (E_{tc}) per rated capacity and type and the estimated sales distribution in 2019. Sources: APPLiA, GfK. HP-C = Condensing heat pump dryer, HE-C = Condensing heating element dryer, HE-V = air-vented heating element dryer.

Figure A5-1 shows that the proposed SEc formula has a much lower slope (the a-coefficient) than the current one, due to the market mostly consisting of heat pump dryers that feature a generally lower energy consumption per cycle. The b-coefficient (the exponent) is also lower (0,63 compared to 0,8) and makes the curve flatter than the current formula. This will prevent manufacturers to make artificially larger tumble dryers in order to reduce the EEI of their models.

[78] Based on information from the Commission and "Evaluation of primary energy factor calculation options for electricity, Anke Esser (FhG-ISI), Frank Sensfuss (FhG-ISI), 2016"
[79] Not enough data points were available for 10 kg dryers.

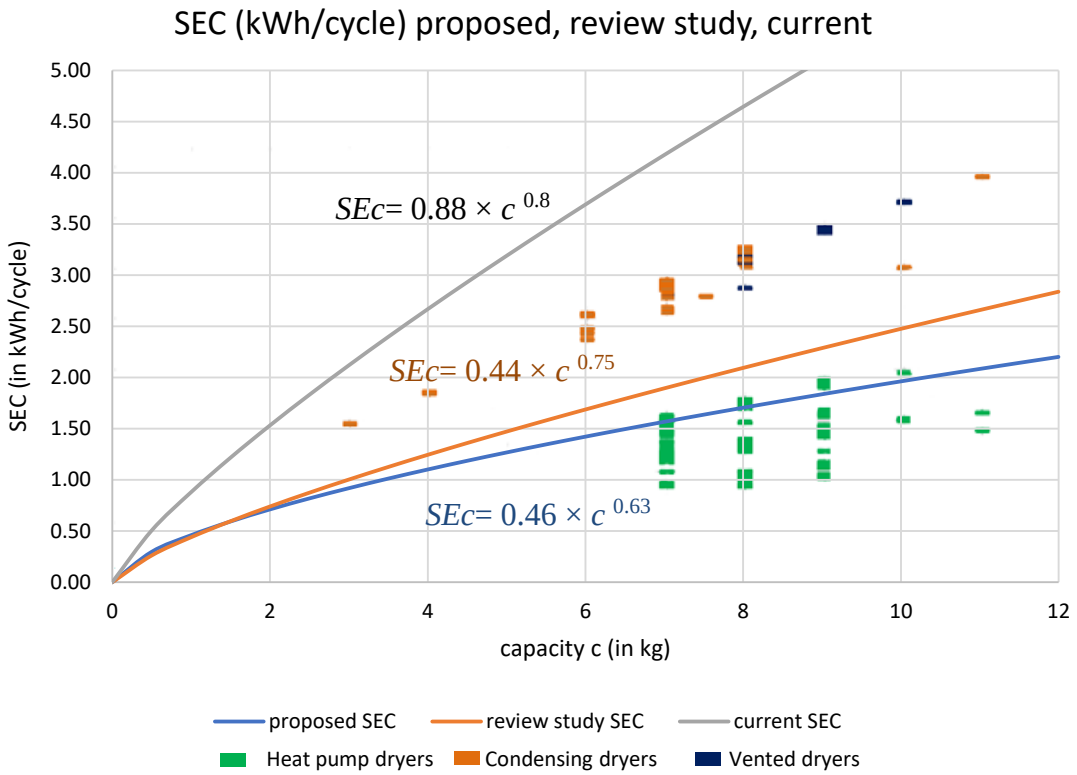


Figure A5-1. Different SEc curves and available data points

For air-vented dryers, a correction factor is used to cater for their impact on secondary energy systems. In the current regulation, the correction factor corresponds to a ~5% decrease in the SEc value per hour of cycle time for a 7kg dryer with a 123-minute cycle time. The proposed calculation method, instead, imposes this percentage reduction directly, without lowering the correction factor for dryers with longer cycle time. The reduction is increased from the 10% per cycle in the current ecodesign regulation to 17%[80], based on the conclusions from the review study Task 3. As result, the proposed SEc for air-vented dryers is:

$$SEc = 0{,}46 \times c^{0{,}63} \times \left(1 - \frac{Tt}{60} \times 0{,}083\right)$$

with:

- **c** being the rated capacity [kg]
- $\mathbf{T_t}$ being the weighted cycle time [minutes],

2. ENERGY LABEL DISTRIBUTION

The current energy label distribution is displayed on Figure A5-2, which shows a large gap in energy efficiency between heating element dryers (blue dots) and heat pump dryers (red dots). This means that a large part of the current energy scale is not utilised.

[80] 17% per cycle with a cycle time at 123 minutes is equivalent to 17%/ (123/60) = 8.3% per hour.

The A-class is almost completely empty, and the C and D classes have both been removed from the market due to the ecodesign limits under the current legislation. In 2025 it is projected that heat pump models account for 83% of the total market (in sales) and that more than 90% of these will have an A++ or A+++ label[81].

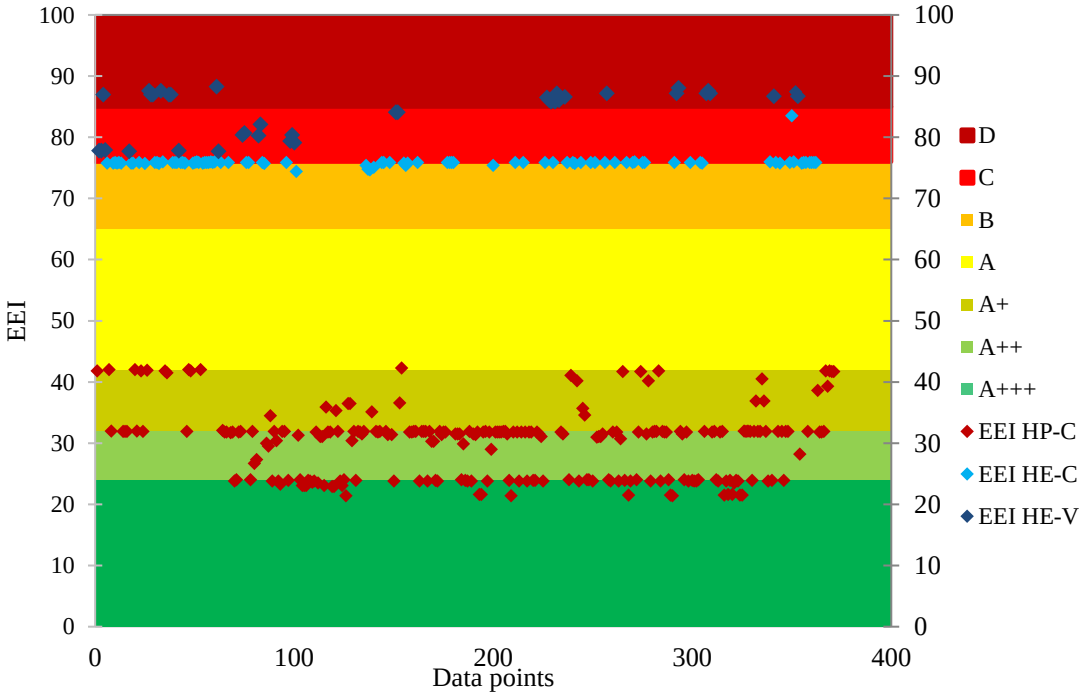


Figure A5-2. Energy label distribution, current regulation. Each dot represents a tumble dryer model. Source: APPLiA

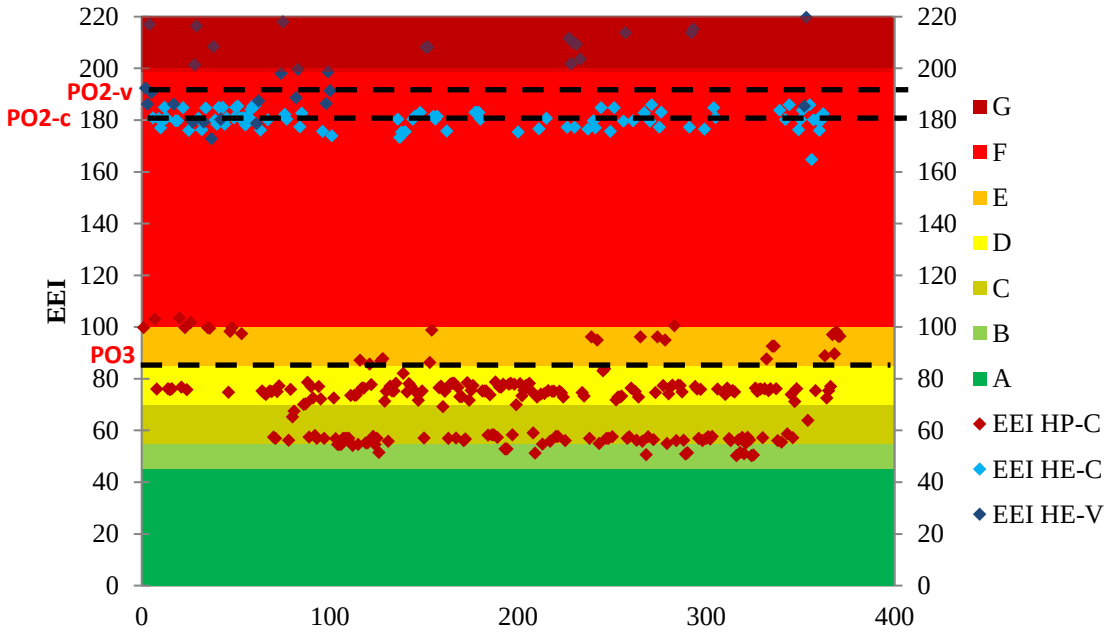
Following the 2017 framework Energy Labelling Regulation mandating an A-G energy scale with an empty A-class, a rescaling of the energy label is needed. Figure A5-3 and Table A5-2 show the proposed intervals with the new EEI calculation method explained in section 1 of this annex. The intervals have been derived as follows:

- The A-class interval limit is calculated by subtracting a 10% of the EEI for the best-in-class model. This gives manufactures a goal that is not easy but possible to achieve and secures the pull-effect of the energy label[82].

- Similar sized B to E intervals making it easier for consumers to evaluate the performance between models with different labels, increasing the incentive for manufacturers to improve the energy performance of their models.

- F and G classes including all the worst performing models, which will disappear in tier 2 with the application of PO3.

[81] Based on GfK sales data.
[82] A more stringent limit at best-in-class subtracted 20% was originally proposed but manufactures stated that this level was not achievable in the next 10 years. The EEI limit was thus increased.



FigureA5-3. Energy label distribution, proposed intervals with the proposed EEI calculation method. The dotted line represents the ecodesign limits proposed for PO3, PO2-v indicates the EEI limit for vented dryers (EEI=193) and PO2-c indicates the EEI limit for heating element dryers (EEI=180). The PO3 limit is 85. Source: APPLiA, Viegand Maagøe

EEI Interval
A ≤ 45
46 < B ≤ 55
56 < C ≤ 70
71 < D ≤ 85
86 < E ≤ 100
100 < F ≤ 200
201 ≤ G

TableA5-2: Proposed EEI intervals

As displayed in Table A5-3, class B of the proposed TD Energy Label Regulation will correspond to class A+++ in the current Regulation. The most efficient appliances in the current energy class A++ will be placed in the proposed C class while the less efficient in class A++will be placed in class D along with the most efficient share of class A+. There are no models located in the class A of the future label, which is therefore empty, 6% of the models fall in class B, 17% in class C, and 33% in class D.

Current class		Proposed class	Current classes, distribution	Proposed classes, distribution
		A	0%	0%
A+++	→	B	17%	6%
A++ (Top)	→	C	29%	17%
A++ (Bottom)	→	D		33%
A+ (Top)	→	D	17%	

Current class		Proposed class	Current classes, distribution	Proposed classes, distribution
A+ (Bottom)	→	E		7%
A	→	E	1%	
A	→	F		37%
B	→	F	19%	
C	→	G	11%	
D	→	G	6%	

Table A5-3. Correlation between current and proposed energy class distribution (based on models in APPLiA database and not on sales data)

3. PROJECTION OF THE ENERGY LABELLING DISTRIBUTION

To model the future evolution of the distribution of tumble dryers across the different classes of the energy label, historic performance data and energy label distributions from GfK from 2013 – 2019 have been used. We first obtain an average specific energy consumption in kWh per kg-dried laundry, based on the average EEI per label class, the average nominal capacity drawn from GfK data and the average load and cycles per year (4,4 kg and 107 cycles/year from the review study as declared by APPLiA). This specific energy consumption is then projected towards 2040 to estimate the average improvement in efficiency. For heating element dryers, the projection is based on a decaying exponential smoothing algorithm, for the heat pump dryers an s-curve approach is used.

An algorithm is used to distribute the energy labels so that the average specific energy consumption matches with the one first calculated. The distribution is made on a narrow Gaussian distribution. Figure A5-4 shows the specific energy consumption used for the BAU scenario as an example.

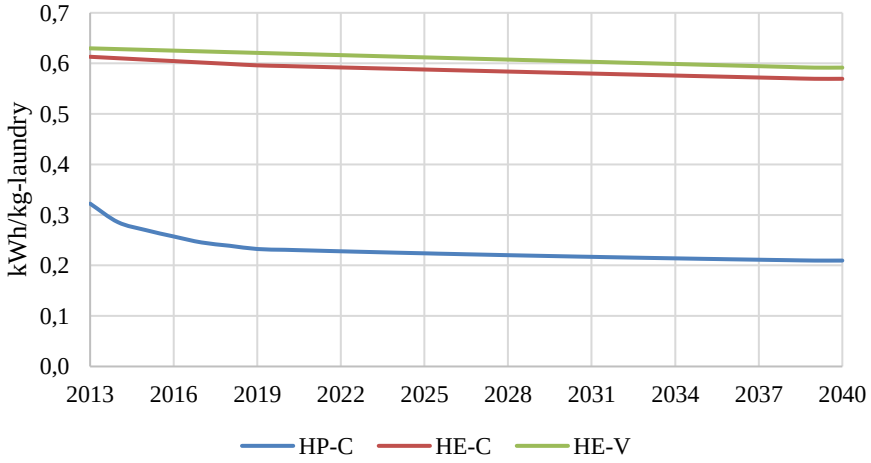


Figure A5-4. Specific energy consumption from 2013 to 2040 in the BAU scenario. HP-C = Heat pump condensers, HE-C = Heating element condensers, HE-V = Heating element air vented. Source: Viegand Maagøe based on GfK and APPLiA data

Tables A5-5 to A5-8 show the corresponding energy class distributions for all the policy options. Note that all options except BAU follow the rescaled energy class distributions and hence cannot be 1:1 compared to the BAU distribution.

BAU

		2015	2020	2024	2026	2030	2035	2040
Heat pump condenser	A+++	8%	32%	35%	36%	38%	42%	45%
	A++	59%	58%	57%	56%	55%	52%	50%
	A+	31%	10%	9%	8%	7%	6%	5%
	A	2%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element condenser	A+++	0%	0%	0%	0%	0%	0%	0%
	A++	0%	0%	0%	0%	0%	0%	0%
	A+	0%	0%	0%	0%	0%	0%	0%
	A	0%	0%	0%	0%	0%	0%	0%
	B	81%	99%	100%	100%	100%	100%	100%
	C	19%	1%	0%	0%	0%	0%	0%
	D	0,2%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element air vented	A+++	0%	0%	0%	0%	0%	0%	0%
	A++	0%	0%	0%	0%	0%	0%	0%
	A+	0%	0%	0%	0%	0%	0%	0%
	A	0%	0%	0%	0%	0%	0%	0%
	B	15%	7%	7%	7%	8%	11%	14%
	C	80%	93%	93%	93%	92%	89%	86%
	D	5%	0%	0%	0%	0%	0%	0%

Table A5-4. Energy label distribution for BAU

PO2								
		2015	2020	2024	2026	2030	2035	2040
Heat pump condenser	A	8%	32%	35%	2%	3%	3%	4%
	B	59%	58%	57%	10%	10%	12%	14%
	C	31%	10%	9%	32%	33%	35%	37%
	D	2%	0%	0%	40%	39%	37%	35%
	E	0%	0%	0%	16%	15%	12%	10%
	F	0%	0%	0%	0%	0%	0%	0%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element condenser	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	81%	99%	100%	0%	0%	0%	0%
	F	19%	1%	0%	100%	100%	100%	100%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element air vented	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	15%	7%	7%	0%	0%	0%	0%
	F	80%	93%	93%	100%	100%	100%	100%
	G	5%	0%	0%	0%	0%	0%	0%

TableA5-5. Energy label distribution for PO2

PO2+PO3								
		2015	2020	2024	2026	2030	2035	2040
at pu m p co nd	A	8%	32%	35%	2%	3%	3%	4%

	B	59%	58%	57%	10%	10%	12%	14%
	C	31%	10%	9%	32%	33%	35%	37%
	D	2%	0%	0%	55%	54%	49%	45%
	E	0%	0%	0%	0%	0%	0%	0%
	F	0%	0%	0%	0%	0%	0%	0%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element condenser	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	81%	99%	100%	0%	0%	0%	0%
	F	19%	1%	0%	100%	100%	100%	100%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element air vented	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	15%	7%	7%	0%	0%	0%	0%
	F	80%	93%	93%	100%	100%	100%	100%
	G	5%	0%	0%	0%	0%	0%	0%

Table A5-6. Energy label distribution for PO2+PO3,

PO2+PO4								
		2015	2020	2024	2026	2030	2035	2040
Heat pump condenser	A	8%	32%	35%	2%	3%	3%	4%
	B	59%	58%	57%	10%	10%	12%	14%
	C	31%	10%	9%	32%	33%	35%	37%
	D	2%	0%	0%	55%	54%	49%	45%
	E	0%	0%	0%	0%	0%	0%	0%
	F	0%	0%	0%	0%	0%	0%	0%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element condenser	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	81%	99%	100%	0%	0%	0%	0%
	F	19%	1%	0%	100%	100%	100%	100%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element air vented	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	15%	7%	7%	0%	0%	0%	0%
	F	80%	93%	93%	100%	100%	100%	100%
	G	5%	0%	0%	0%	0%	0%	0%

TableA5-7. Energy label distribution for PO2+PO4

PO2+PO3+PO4								
		2015	2020	2024	2026	2030	2035	2040
Heat pump condenser	A	8%	32%	35%	2%	3%	3%	4%
	B	59%	58%	57%	10%	10%	12%	14%
	C	31%	10%	9%	32%	33%	35%	37%

	D	2%	0%	0%	55%	54%	49%	45%
	E	0%	0%	0%	0%	0%	0%	0%
	F	0%	0%	0%	0%	0%	0%	0%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element condenser	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	81%	99%	100%	0%	0%	0%	0%
	F	19%	1%	0%	100%	100%	100%	100%
	G	0%	0%	0%	0%	0%	0%	0%
		2015	2020	2024	2026	2030	2035	2040
Heating element air vented	A	0%	0%	0%	0%	0%	0%	0%
	B	0%	0%	0%	0%	0%	0%	0%
	C	0%	0%	0%	0%	0%	0%	0%
	D	0%	0%	0%	0%	0%	0%	0%
	E	15%	7%	7%	0%	0%	0%	0%
	F	80%	93%	93%	100%	100%	100%	100%
	G	5%	0%	0%	0%	0%	0%	0%

TableA5-8. Energy label distribution for PO2+PO3+PO4

ANNEX 6: KEY ASSUMPTIONS FOR POLICY OPTIONS

Due to the lack of available data, some assumptions according to stakeholders input during the review study, notably from the industry, have been taken in order to carry out the calculation of the impacts.

1. FOR BAU

- Heat pump models will continue to increase their market share each year at the same pace as until 2019.
- The current trends observed in terms of energy efficiency gain of tumble dryers will continue.
- The average lifetime of dryers will stay in the current 12 years.
- Refrigerants with high GWP will be gradually phased out due to the limits imposed by the F-gas Regulation (annual quotas on the placing on the EU market of HFCs by producers and importers). The proposal for a new F-gas Regulation ensures a phase-out of f-gases with a GWP > 150 from 2025 onwards.
- 70% of the refrigerant of each appliance is assumed to be recovered during EoL, based the Review study. Based on stakeholders' input, 80% of tumble dryers models on the market in 2020 are assumed to have high GWP refrigerants such as R134a[83], with an average of 0,3 kg of refrigerant per model.

2. FOR PO2

- Moderate pull effect towards more energy efficient products from the rescaling of the energy label and adaption of energy efficiency requirements to new calculation method.
- Average lifetime of dryers remains 12 years.

3. FOR PO3

- Average lifetime of dryers remains 12 years.

4. FOR PO4

- The average lifetime of dryers increases to 14 years. Information from repair and service businesses estimates that the lifetime of tumble dryers could be increased with 1,5 years if the proposed resource efficiency requirements are implemented. It is assumed that the lifetime could be extended with an additional half a year by the proposed information requirements regarding cleaning of filters, to a total of 2 years.
- According to APPLiA the average repair cost per unit is around 150 €. There are very different views on how often a tumble dryer is repaired, varying from "it is often not repaired" according to APPLiA to "twice during the lifetime" according to R.U.S.Z. The review study suggests that the machines on average will be repaired 0,4 times during their lifetime[84] (5 €/year x 12 years/150 € per repair). To extend the lifetime, it

[83] A GWP100 of 1,430 kg CO2eq/kg is used in the calculations.
[84] 5 €/year in 12 years is used in the review study.

is assumed that more repairs are required and that the repair and maintenance cost per year will increase to 8 €/year. With this assumption a tumble dryer will on average be repaired 0,75 times during its lifetime of 14 years.

ANNEX 7: ADDITIONAL IMPACTS RELATED TO RESOURCE EFFICIENCY

From the EcoReport tool, two impacts are considered to be relevant from the point of view of resource efficiency, on top of the analysis already made as regards fuel and material consumption: water use and generation of waste.

1. WATER CONSUMPTION

According to the EcoReport tool, water is fundamentally consumed during the production phase. Cumulative water consumption savings for each product subgroup have been calculated by multiplying the water consumption per unit and product subgroup provided in the EcoReport Tool, by the cumulative sales for the periods considered (2025-2030 and 2025-2040). Adding the subtotals for each product subgroup gives the total water consumption per policy option, which is then subtracted from water consumption under BAU, giving the cumulative water savings for each policy option.

	Cumulative water consumption (Mlitres)		**Total water saved (Mlitres)**	
	2025-2030	**2025-2040**	**2025-2030**	**2025-2040**
BAU	9.700	26.735		
PO2	9.842	26.997	-143	-263
PO2+PO3	9.970	27.252	-270	-517
PO2+PO4	9.842	24.706	-143	2.029
PO2+PO3+PO4	9.970	25.062	-270	1.673

Table A7-1. Cumulative water savings per policy option

Table A7-1 shows that only the policy options including circular economy measures save water, following the reduction of the production that takes place. PO2+PO4 is the best policy option in this respect, since sales of heat pump tumble dryers, which is the type which production consumes more water per unit manufactured, is lower than for PO2+PO3+PO4.

2. WASTE

Waste is produced in all the phases of the product lifecycle, namely: manufacturing, use and EoL. The use phase is more important than it could look like at a first glance, since obtaining the resources to generate electricity generates big amounts of waste. Energy efficiency has therefore a very significant role in reducing waste.

Cumulative waste savings for each product subgroup have been calculated by multiplying the waste consumption per unit and product subgroup provided in the EcoReport Tool, by the cumulative sales for the periods considered (2025-2030 and 2025-2040). Adding the subtotals for each product subgroup gives the total waste consumption per policy option, which is then subtracted from waste consumption under BAU, giving the cumulative waste savings for each policy option.

	Cumulative waste (tonnes)		Cumulative waste saved (tonnes)	
	2025-2030	**2025-2040**	**2025-2030**	**2025-2040**
BAU	640.096	1.722.077		
PO2	621.993	1.687.652	18.103	34.425
PO2+PO3	604.800	1.653.159	35.295	68.918
PO2+PO4	621.993	1.573.523	18.103	148.553
PO2+PO3+PO4	604.800	1.536.018	35.295	186.058

Table A7-2. Cumulative waste savings per policy option

Table A7-2 shows that all policy options yield waste savings, being PO2+PO3+PO4 the best option. This is because PO2+PO3+PO4 reduces waste by means of both increasing energy efficiency and slowing down the pace of production of new devices.

ANNEX 8: SENSITIVITY ANALYSIS

During the performance of the IA, some assumptions have been made which are not sufficiently supported by available data. Notably in the field of circular economy measures, there is no sufficiently reliable data available on how PO4 would extend the lifetime of tumble dryers, and by how much repair costs would increase with respect to BAU. On the basis of the information provided by repair businesses and APPLIA, the IA assumes an increase from 12 to 14 years on the lifetime of the tumble dryers, and an increase from 5€ to 8€ per year on repair expenditure. Nevertheless, there is no statistical data supporting that information. Due to the high degree of uncertainty related to the assumptions taken, a sensitivity analysis is carried out in this Annex to observe how a variation on those two assumptions would change the results of the preferred option (PO2+PO3+PO4).

In addition, as it has been indicated in section 8.4, the current energy prices are extremely volatile and clearly overcome those used in this IA, which are taken from the REF2020 scenario. This Annex contains a sensitivity analysis with the electricity prices included in the MIX scenario. Although the MIX scenario energy prices are higher than those in REF2020, they are still low compared to the current ones, although a sensitivity analysis based on the former could give a first idea on how policies would need to be adapted in a situation of uncertain energy supply.

A number of other assumptions have been made throughout the study, although these have been based on reliable data or model outputs and therefore are not considered to be critical.

All sensitivity analyses will be evaluated as regards of:

- user expenditure, which includes purchase price, electricity consumption and repair costs;
- total energy consumption, which includes embedded energy;
- net greenhouse gas emissions, which includes use, production and EoL.

1. SENSITIVITY ANALYSIS REGARDING THE VARIATION IN REPAIR COSTS AND PRODUCT LIFETIME

1.1. Baseline values

Table A8-1 shows the baseline values for the different policy options,

Baseline value	Energy consumption [TWh/year]		GHG emissions [mtCO2/year]		User expenditure [Bln. €/year]	
	2030	2040	2030	2040	2030	2040
BAU	11,21	10,45	2,52	1,95	5,26	5,26
PO2+PO3+PO4	10,58	8,72	2,53	1,52	5,33	4,43

Table A8-1. Baseline values for energy consumption, GHG emissions and user expenditure in 2030 and 2040

1.2. Influence of an increase in repair costs

Table A8-2 shows that in 2040, when the full effect of the increased lifetime has taken place, an added repair cost of up to 12 €/year will just be enough to produce a little

increase of the user expenditure compared to BAU in 2040. There is therefore a wide margin to increase repair costs and still the preferred option will be worth from the point of view of user expenditure with higher repair costs.

Repair cost that are added to 8 €/year	User expenditure [bln. €/year]	
	2030	2040
PO2+PO3+PO4 baseline: 8 €/year	5,33	4,43
4	5,48	4,74
8	5,57	5,01
12	**5,67**	**5,29**
16	5,77	5,57
20	5,86	5,84

Table A8-2. Sensitivity analysis on the effect on total user expenditure of the added repair cost per year for the preferred option PO2+PO3+PO4.

1.3. Increase of the lifetime

The increase of the lifetime of a tumble dryer reduces the number of appliances being produced per year and consequently also reduces the energy consumption and GHG emissions related to the production of those appliances. Increasing the lifetime of a tumble dryer will also result in a slight increase of in-use energy consumption since the longer used old appliance is assumed to be less efficient than a newer model. The net effect of lengthening the lifetime of the tumble dryer is overall beneficial and results in a net reduction of the energy consumed, GHG emissions and user expenditure. In addition, table A8-3 also shows that if the circular economy measures did not have any effect on the extension of the lifetime (increase of lifetime = 0), the balance of the preferred option would still be positive with respect to BAU for the three impacts considered, due to the combined effect of PO2 and PO3.

Increased lifetime of tumble dryers [years]	Total energy consumption [TWh/year]		Total GHG emissions [mtCO2/year]		User expenditure [Bln. €/year]	
	2030	2040	2030	2040	2030	2040
0 (PO2+PO3)	10,58	9,32	2,53	1,91	5,26	5,12
PO2+PO3+PO4 baseline: 2	**10,58**	**8,72**	**2,53**	**1,52**	**5,33**	**4,43**
4	10,58	8,57	2,53	1,24	5,33	3,94
6	10,58	8,44	2,53	1,09	5,33	3,68

Table A8-3. Sensitivity analysis of the impact on the preferred option of increased lifetime for tumble dryers

2. SENSITIVITY ANALYSIS ON THE USE OF THE MIX SCENARIO INSTEAD OF REF2020

While the calculations and scenario modelling in the IA were carried out using energy prices and carbon intensity from REF2020, a sensitivity assessment using the MIX scenario (which is in line with the fit-for-55 objectives with lower carbon intensity and higher energy prices) is made to cater for future fluctuations of those parameters.

	2010	***2015***	***2020***	***2025***	***2030***	***2035***	***2040***	***2045***	***2050***
Households electricity price	169	176	192	204	211	217	225	230	225

Table A8-4. Electricity prices from PRIMES in €/MWh (MIX scenario)

	2010	*2015*	*2020*	*2025*	*2030*	*2035*	*2040*	*2045*	*2050*
Carbon intensity	318	305	213	149	100	37	0	0	0

Table A8-5. Carbon intensity for electricity consumption in gCO2e/kWh (MIX scenario)

The results for both REF2020 and MIX are shown in tables A8-6 and A8-7, respectively. Following the lower carbon intensity in the MIX scenario (0 in 2040), the greenhouse gas emissions in 2040 are only due to refrigerant losses, production and EoL, and therefore GHG emissions reduction is larger for MIX than for REF2020. User expenditure would be somewhat bigger for MIX than for REF2020 due to the higher electricity costs, although the former still encountering important savings.

Reference scenario (BAU)	**Total energy consumption [TWh/year]**		**Total GHG emissions [mtCO2/year]**		**User expenditure [Bln. €/year]**	
	2030	2040	2030	2040	2030	2040
BAU	11,21	10,45	2,52	1,95	5,26	5,26
PO2+PO3+PO4	10,58	8,72	2,53	1,52	5,33	4,43

Table A8-6. Key values for BAU and preferred policy option for energy consumption, GHG emissions and user expenditures in 2030 and 2040 for REF2020 scenario

MIX Scenario	**Total energy consumption [TWh/year]**		**Total GHG emissions [mtCO2/year]**		**User expenditure [Bln. €/year]**	
	2030	2040	2030	2040	2030	2040
BAU	11,21	10,45	2,13	1,34	5,28	5,41
PO2+PO3+PO4	10,58	8,72	2,13	0,92	5,42	4,62

Table A8-7. Key values for BAU and preferred policy option for energy consumption, GHG emissions and user expenditures in 2030 and 2040 for MIX scenario

ANNEX 9: LIST OF SPARE PARTS

Manufacturers, importers or authorised representatives must make available spare parts for a minimum period of 10 years after the placing of the last unit of a model on the market. The requirement will cover availability of spare parts to professional repairers but also to the consumers for some specific spare parts that can be replaced without professional assistance.

A list of essential spare parts to be included in the TD Ecodesign Regulation is compiled from information available and comments from stakeholders. Furthermore, supplementary interviews with a number of repairers have been performed to further develop the list of critical spare parts. The resulting list is the following:

1) For professional repairers and end-users:

- Door, door handles, locking and hinge
- Lint filters
- Plastic peripherals
- Gaskets and seals
- Switches and knobs
- Condensate pump, condensate tank
- Air filter

2) Only for professional repairers:

- Motor and motor brushes
- Transmission between motor and drum
- Pumps
- Fan, fan wheel
- Drum and bearings
- Piping and related equipment including hoses, valves, and filters
- Cables and plugs
- Printed circuit boards
- Electronic displays
- Pressure switches
- Thermostats and sensors (including humidity sensor)
- Software and firmware updates including reset software,
- Shock absorbers and springs
- Heating elements
- Electric fuses
- Tension pulley, support roller

ANNEX 10: SME TEST

The SME test is carried out according to Tool#23 of the Better Regulation Toolbox.

1. IDENTIFICATION OF THE AFFECTED BUSINESSES

In the absence of specific data on SMEs in the sector of tumble dryers, the number of tumble dryer manufacturers, repairers and the corresponding share of SMEs has been worked out from data from the Eurostat Structural Business Statistics. Eurostat provides turnover and number of SMEs for the entire appliance sector (NACE 275). The share of SMEs that are manufacturers of tumble dryers has been estimated for 2020 by multiplying the ratio *turnover of the tumble dryers industry under BAU (data from this IA)/turnover for the entire appliance sector (data from Eurostat),* by the *number of SMEs for the appliance sector* also from Eurostat, assuming therefore that the number of SMEs is proportional to the abovementioned ratio. The ratio number of SMEs/turnover for the whole appliance sector is 2.983 (SMEs)/54.336 (M€) = 0,054 in 2020, which multiplied by 2.840 M€ of turnover in 2020 for tumble dryers manufacturers, gives 153 SMEs that are affected by the measures. Assuming that the share SMEs/total companies remains the same for tumble dryers as for the entire appliance sector, from Eurostat results that 153 SMEs would be equivalent to 96% of the manufacturers, thus being 160 the total number of manufacturers in the tumble dryer sector. A search carried out in EPREL database confirms that the number of tumble dryer manufacturers is in the order of 160.

In addition, the repair sector is also affected by the measures included in this IA. Repair of tumble dryers is included in NACE-S95 (repair of personal and household goods). As it has been done for manufacturers, the number of companies dedicated to the repair of tumble dryers can be determined by multiplying the ratio *SMEs/turnover* for the whole sector by the *turnover calculated for the repair sector* in this IA. This gives 140.480 (SMEs)/48.880 (M€) = 2,87, which multiplied by 260 M€ that has been estimated as turnover for the repair sector in 2020, gives 747 companies, with an average of two workers per company. If it is assumed that the share SMEs/total companies remains the same for tumble dryers repair sector as for the entire repair sector, it results that all repairers in the tumble dryer sector are SMEs.

2. THRESHOLD QUESTION: TO WHAT EXTENT IS THE INITIATIVE RELEVANT FOR SMES?

Although SMEs are in the scope of the initiative, the latter is not specifically targeting SMEs. The initiative will apply to the whole sector of tumble dryers, with no distinction between big companies and SMEs.

As already addressed in sections 6 and 7 of the main report, the measures proposed will not have negative impacts on the companies affected. SMEs will not be negatively affected because, as indicated in sections 5.2.3 and 8.4, measures that could entail a significant effort of adaptation and therefore an increase of the costs have been discarded. Mainly, the setting of more stringent ecodesing limits has been ruled out on the basis of the least life cycle cost criterion developed in section 8.4. There are therefore no negative effects, on the contrary, both tumble dryer manufacturers and repairers will see their turnover to grow thanks to the combination of energy efficiency and resource efficiency measures.

In addition, after a thorough revision of the feedback provided by stakeholders during the process of legislative review, and in particular the relevant industry representatives, no particular concerns have been noted about SMEs.

Finally, the SME filter list[85], which screens all the initiatives published in the "have your say" portal after 1 January 2022, does not mention ecodesign for tumble dryers as one of the initiatives that could be relevant for SMEs.

Given the background above, it can be concluded that the initiative does not entail any negative impact on SMEs and subsequently there is no need to carry out the next steps of the SME test as laid down in the Better Regulation Toolbox.

[85] SME envoys network (europa.eu)

ANNEX 11: REFRIGERANTS IN TUMBLE DRYERS

The refrigerants equipping heat pump tumble dryers range from F-gases such as R134a to organic gases such us R290 (Propane). The type of refrigerant is chosen based on the intended temperatures and pressures.

R134a is an F-gas and is therefore currently undergoing phase-out because of its high GWP. F-gases phase-out has reduced their availability on the European market and consequently increased their price.

1. F-GAS REGULATION

Because F-gases are potent greenhouse gases, they have been regulated in the EU since 2014 and their emissions have started to decline after the peak reached in 2014. The F-gas Regulation (EU) No 517/2014 implements an EU-wide phase-down of HFCs, which started in 2015, with the aim of cutting emissions by two thirds by 2030 in the EU compared with 2014 levels.

The HFC phase-down under the F-gas Regulation is being implemented by annual quantitative limits (quotas) on the placing on the EU market of HFCs by producers and importers, as displayed in Figure 23.

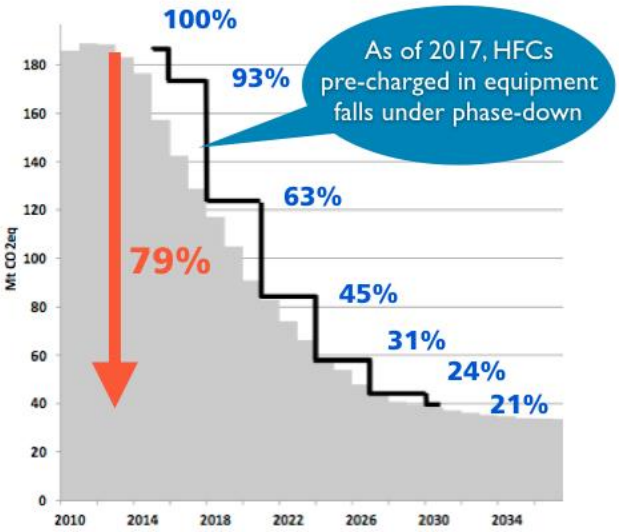


Figure A11-1. Quota system stepwise phase-down of HFCs[86], Phase down from 2017 includes HFCs pre-charged in equipment as is the case for heat pump tumble dryers

In 2019, EU-wide placing on the market of HFCs was 2% below the 2019 overall market limit set by the quota system (1 % in 2017 and 2018).

[86] Global Market Trends for Natural Refrigerants in commercial refrigeration. Danish Technological Institute and shecco Base. Refrigeration in the supermarket sector. November 2017.

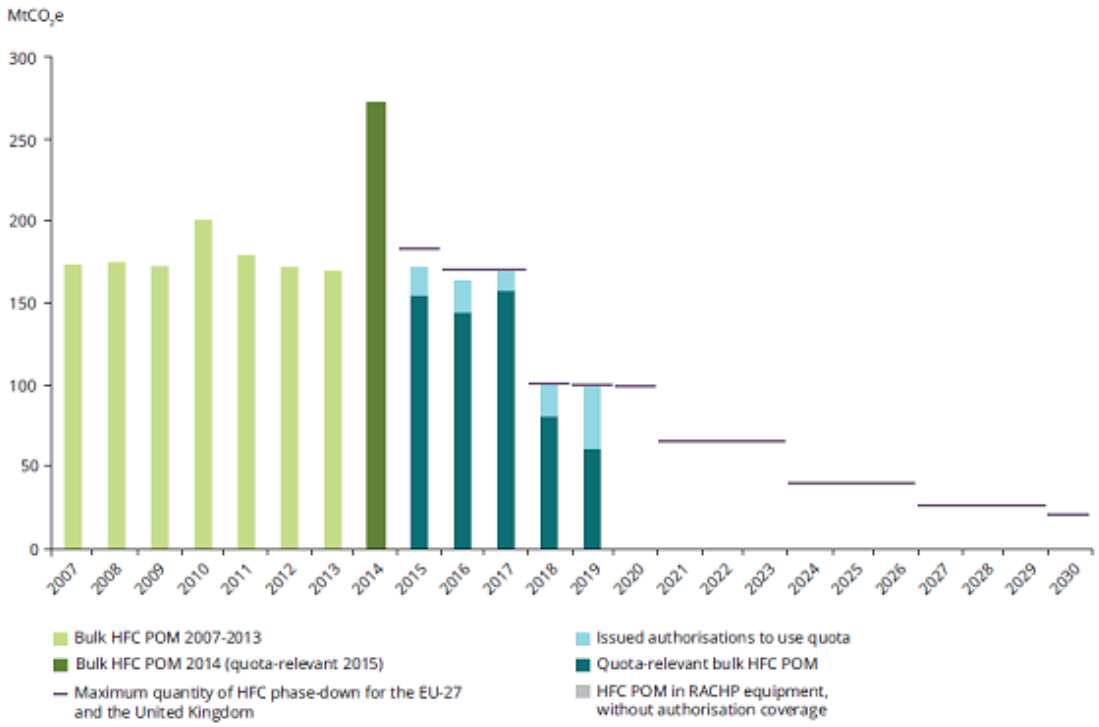


Figure A11-2. Progress of the HFC phase-down under EU regulation[87]

The demand for HFCs has been decreasing because EU industries have been moving to gases with lower GWP's.

The Quota system phase-down includes HFCs pre-charged in equipment and therefore also R134a used in heat pump tumble dryers. However, unlike for other products such as fridges or air conditioners, the F-gas regulation does not prohibit the use of HFCs with high GWP in tumble dryers.

The F-gas Regulation has been under review and the Commission has adopted a new proposal on 5 April 2022. The proposal increases ambition of the quota system and adds new sub-sectoral prohibitions, including a prohibition to use F-gases with a GWP higher than 150 in self-contained heat pump equipment from 2025.

2. PRICE DEVELOPMENT OF REFRIGERANTS FOR HEAT PUMPS

The price of conventional HFCs on the EU market has been strongly affected by the phase-down measure. In the first two years of the implementation of the F-gas Regulation (2015 and 2016), there was no significant impact on the prices of commonly used high GWP HFC refrigerants. However, from mid-2017 onwards prices for R404A, R410A, R134a as well as R407C rose significantly until reaching a peak in early 2018 of 6 to 13 times higher than the original price.

In the same period natural alternatives such as CO2 (R744), ammonia (R717) and propane (R290) with very low GWPs have seen modest price decreases compared to

[87] Fluorinated greenhouse gases 2020. EEA Report No 15/2020. European Environment Agency.

2015. There is no constraint on their availability on the EU market, and they are available at low cost, comparable to the price level of high GWP HFCs in 2015[88].

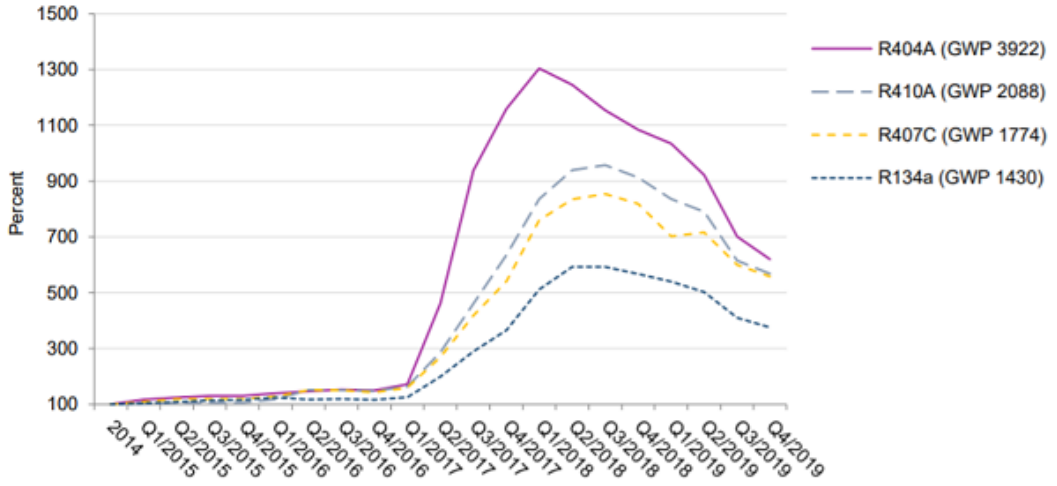


Figure A11-3. Average purchase prices of the most commonly used HFCs at service company level (price index, 2014 = 100 % (baseline)). Source C(2020) 8842 final



Figure A11-4. Development of average purchase prices of natural alternatives at service company level (price index, Q2/2017= 100 % (baseline)). Source C(2020) 8842 final

3. TRANSITION TO HEAT PUMPS WITH LOW GWP REFRIGERANTS

The most relevant alternative to R134a in tumble dryer heat pumps seems to be propane (R290), which transition is already under way. Some major European manufacturers of tumble dryers have announced the change to R290 in all their tumble dryers:

- B/S/H has already completely changed their full portfolio to R290[89],
- Miele has informed in 2019 that they will do the same[90],

[88] Report from the Commission on availability of hydrofluorocarbons on the Union market. C(2020) 8842 final

[89] Comment provided by BSH Group (BSH Group includes various global brands such as Siemens, Bosch, and NEFF and a number of local brands). …

[90] https://www.miele.de/en/m/faster-more-convenient-and-greener-the-new-washing-machines-and-tumble-dryers-from-miele-5036.htm

- Electrolux has announced that in Europe their new highly efficient range of heat pump dryers will use a hydrocarbon in their compressors[91].

Table A11-1 displays a non-exhaustive list of models using propane as refrigerant that have already been placed on the market.

Model	Capacity	Energy label	Price	Refrigerant (charge)
Miele TSB 143 WP	7 kg	A++	695 EUR	R290
BEKO DE744RX1	7 kg	A++	849 EUR	R290
AEG T7DBZ41570	7 kg	A++	695 EUR	R290
Bosch WTR85V80[92],[93]	7 kg	A++	708 EUR	R290 (0,120 kg)
Sharp KD-GHB7S7GW2-DE[94]	7 kg	A++	408 EUR	R290
Constructa CWK33R400[95]	8 kg	A+++	890 EUR	R290
Miele TWB140 WP [96]	7 kg	A++	770 EUR	R290
LG Eco Hybrid™ FDV909W[97]	9 kg	A+++	£949	R290 (0,145 kg)
Siemens WT47w5w0iQ700[98]	8 kg	A+++	609 EUR	R290
Grundig GTN 38250 TGCW	8 kg	A++		
Bosch WTR854A0 Serie 6[99]	7 kg	A+++	539 EUR	R290 (0,149 kg)
Siemens WT47W5W0iQ700[100]	8 kg	A+++	609 EUR	R290 (0,149)
AEG T9ECOWP[101]	8 kg	A+++	867 EUR	R290

Table A11-1. Examples of tumble dryers where propane (R290) is used as refrigerant. Information collected from various homepage and user manuals in 2021[102]

According to the review study and information from stakeholders, the energy efficiency of R290 heat pumps compares favourably to conventional refrigerants and allows for a reduction of the charges, namely the amount of refrigerant in the heat pump circuit.

Due to safety measures to address flammability of propane, the production cost of the tumble dryer might be slightly higher, but the latter could be offset by the lower refrigerant price and lower volume charge.

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[91] Electrolux Sustainability Report 2020

[92] https://www.euronics.de/haus-und-haushalt/waschen-trocknen-buegeln/trockner/wtr85v80-waermepumpentrockner-weiss-a-4061856066745.

[93] https://assetscdn.loadbee.com/catalogue/stdzsexzpn4nf9e3/media3.bsh-group.com/Documents/9001378516_A.pdf

[94] https://www.idealo.de/preisvergleich/OffersOfProduct/6425464_-kd-ghb7s7gw2-sharp.html

[95] https://www.constructa.com/at/produktliste/waschen-und-trocknen/waeschetrockner/CWK3R400#/Togglebox=464450815/. Constructa is a local brand of BSH group

[96] https://www.idealo.de/preisvergleich/OffersOfProduct/6863984_-twb140-wp-miele.html

[97] https://www.lg.com/uk/tumble-dryers/lg-FDV909W.

[98] https://www.alternate.de/Siemens/WT47W5W0-iQ700-W%C3%A4rmepumpen-Kondensationstrockner/html/product/1216237

[99] https://www.alternate.de/Bosch-Home/WTR854A0-Serie-6-W%C3%A4rmepumpen-Kondensationstrockner/html/product/1572493#product-details-tab

[100] https://www.alternate.de/Siemens/WT44W5W0-iQ700-W%C3%A4rmepumpen-Kondensationstrockner/html/product/1285857

[101] https://cdn.expert.de/09/c1/b7/a0e633c3c365c9443238325be4e316e96a/TechnischeInfo_1132879.pdf

[102] If no other reference is mentioned the reference is https://www.cleanthinking.de/waeschetrockner-waermepumpe-2020-testsieger-miele-beko/

Standards at international, European and national level regarding the use of flammable refrigerants appear to be an important barrier to the uptake of climate-friendly alternatives to HFCs. To overcome this barrier, the Commission issued at the end of 2017 a mandate request (M/555) to the standardization organizations to draft a European standard on the use of flammable refrigerants in refrigeration, air conditioning and heat pump equipment by February 2021. Work is currently being undertaken on the product standard IEC 60335-2-40 to expand the potential scope of applications within a boundary limit-value of 1kg for R290 (propane). The amount of propane in a heat pump tumble dryer is well below this limit. There are dryers on the market with a charge of around 150 g propane and below.

ANNEX 12: ANALYSIS OF CURRENT ECODESIGN AND ENERGY LABELLING REGULATIONS FOR HOUSEHOLD TUMBLE DRYERS

In the context of the Better Regulation policy[103], the Commission evaluates all EU activities intended to have an impact on society or economy. Many evaluations are triggered by individual clauses in legislation formulated as requiring a review. For the review of an existing ecodesign or energy labelling measure, three out of the five standard evaluation criteria foreseen by Better Regulation need to be addressed, namely whether the measure has been effective, efficient, and relevant[104].

The coherence and EU added-value criteria have already been addressed at the framework level, i,e, in 2012, when the Ecodesign Framework Directive was reviewed[105]. A joint evaluation of the Ecodesign and Energy Labelling Framework Directives[106] was carried out by the Commission in 2015. Among others it was pointed out that the ecodesign and energy labelling measures in place are effective and bring tangible and substantial energy and cost savings. The implementation of the two Directives is estimated to save 175 Mtoe primary energy per year by 2020, which corresponds to 19% savings with respect to business-as-usual energy use for those products. These policies will deliver almost half of the 20% energy efficiency target by 2020. Dependency on imports of energy would be reduced by 23% and 37% for natural gas and coal, respectively. In total, the ecodesign and energy labelling measures in place to date are estimated to save end-users 100 billion € per year in 2020 through lower utility bills (translated into roughly 500 € yearly savings in each household).

This annex presents the relevant findings of the evaluation of the Ecodesign Regulation No 932/2012 and Energy labelling Delegated Regulation No 392/2012, in addition to the evaluation carried out by the Review Study on tumble dryers 2019[107]. The evaluation addresses the effectiveness, efficiency and relevance of these regulations and builds on the information in the review study and the impact assessment.

1. EFFECTIVENESS

This section focuses on two key objectives of the current regulations, i,e,, ensuring a transition towards more energy-efficient household tumble dryers and achieving significant energy savings. Other impacts are quantified but are not analysed in depth.

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103 https://ec.europa.eu/info/law/law-making-process/overview-law-making-process/evaluating-and-improving-existing-laws/reducing-burdens-and-simplifying-law/refit-making-eu-law-simpler-and-less-costly_en

104 https://ec.europa.eu/info/law/law-making-process/planning-and-proposing-law/better-regulation-why-and-how_en

105 COM(2012) 765 final, REPORT FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT AND THE COUNCIL, Review of Directive 2009/125/EC establishing a framework for the setting of ecodesign requirements for energy-related products (recast)

106 COM(2015) 345 final, Report from the Commission to the European Parliament and the Council Review of Directive 2010/30/EU of the European Parliament and of the Council of 19 May 2010 on the indication of labelling and standard product information of the consumption of energy and other resources by energy-related products.

107 Review study on household tumble dryers. Final Report. June 2019.

1.1. Energy savings

Figure A12-1 shows the expected energy consumption without current regulations (BAU0) and with the current regulations (BAU). It appears that the regulations have saved 2 TWh in 2020 compared to no regulation, and they are expected to save additional 6 TWh in 2030. Accumulatively, savings of 9 TWh have been achieved since 2012 until now, which corresponds to 2,8 $mtCO_2$-eq.

No direct comparison with the energy consumption and savings predicted by the Impact Assessment 2012 (IA 2012)[108] have been made in this evaluation, because the assumptions regarding development of the stock and use of tumble dryers (frequency of use and load) have changed significantly. Instead, the stock and other relevant assumptions have been adjusted to make the BAU0 scenario comparable to BAU.

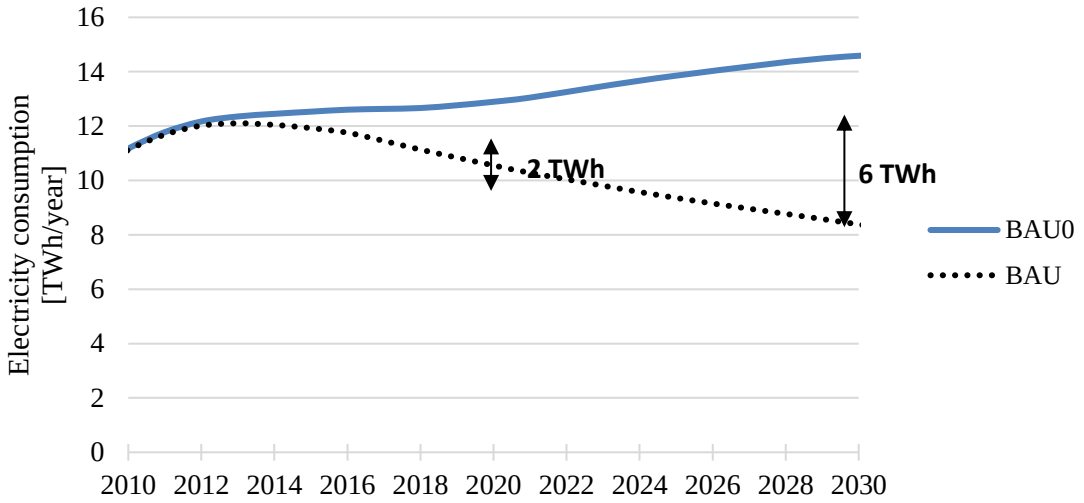


Figure A12-1. Electricity consumption of house tumble dryers 2010-2030 according to BAU (current) and BAU0 (IA 2012)

As an example, the review study from 2019 revealed that the stock estimated in the IA 2012 was more than 15 million units bigger in 2015 and about 10 million units bigger in 2030 than the corresponding stocks considered in the review study. The main reason for this is the UK leaving the EU. The UK has almost a 60% market penetration of laundry driers, which comes down to about 17 million driers[109]. Furthermore, in the aftermath of the financial crisis 2008-2009 the sales growth in Western Europe stagnated until about 2018. Only in 2019 there was an increase in sales of 10-13%[110] and dryer ownership in Germany and France is now above 40%[111] [112]. Finally, there is the increased dryer ownership in the Eastern European member states. Anecdotal data suggest for instance a

[108] COMMISSION STAFF WORKING DOCUMENT EXECUTIVE SUMMARY OF THE IMPACT ASSESSMENT Accompanying the document Commission Regulation implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for household tumble dryers.
[109] IEA ETSAP Technology brief No 9, 2012
[110] GfK data 2020.
[111] Statista, Sales of laundry driers in France 2010-2019.
[112] Statistische Bundesamt, Ausstattung von Haushalte 2020.

20% dryer ownership in Poland, starting from a low penetration <5% in 2010[113]. Nevertheless, this does not compensate for the loss of the UK from EU-figures.

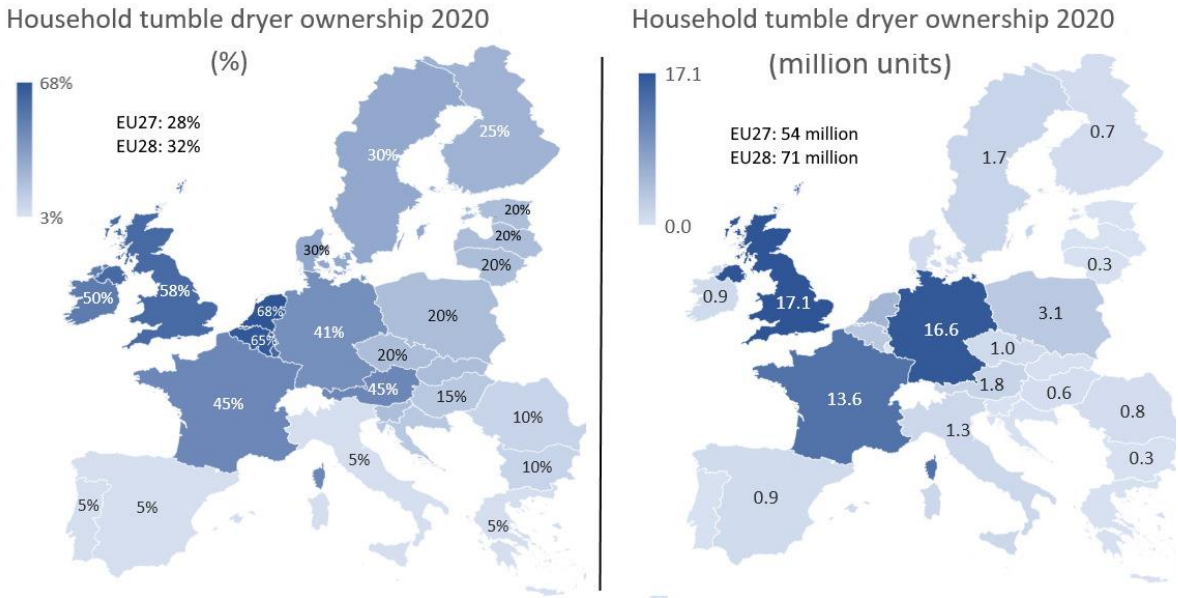


Figure A12-2. Tumble dryer ownership in % (left) and million units installed (right). Source: VHK estimate based on national statistic figures and misc, sources, 2021

A larger stock means a higher energy consumption and potentially a higher saving potential. Therefore, the saving potential indicated in IA 2012 was estimated at 8,6 TWh/year in 2030 (3,3 TWh/year in 2020), corresponding to a reduction of 25% (10,6 % in 2020). The real savings estimated during the review study were nevertheless 2 TWh/year in 2020, which would project a saving of 6 TWh/year in 2030 instead of 8,6.

Based on the above it is concluded that the current regulations have met their objectives and achieved energy savings, in spite of the numerical EU27 savings being smaller than expected because of Brexit[114] and a lower penetration rate of tumble dryers in EU households than expected.

1.2. Specific energy consumption and rated capacity

As displayed in figures A12-3 and A12-4, the current regulations have decreased the energy consumption per kg dried laundry (and thus the yearly energy consumption) more than forecasted in the IA 2012. This conclusion is somewhat misleading, because it is to some extent due to the increase of the rated capacity of the dryers, in particular in the segment of heat pump dryers.

[113] Laitala, K., et al., Use phase of apparel: A literature review for Life Cycle Assessment with focus on wool, February 2018, DOI: 10.13140/RG.2.2.25769.90729, Report number: Report No. 6 - 2017

[114] Meaning that although the UK has made considerable savings due to ecodesign and the energy label it is no longer shown in EU27 statistics.

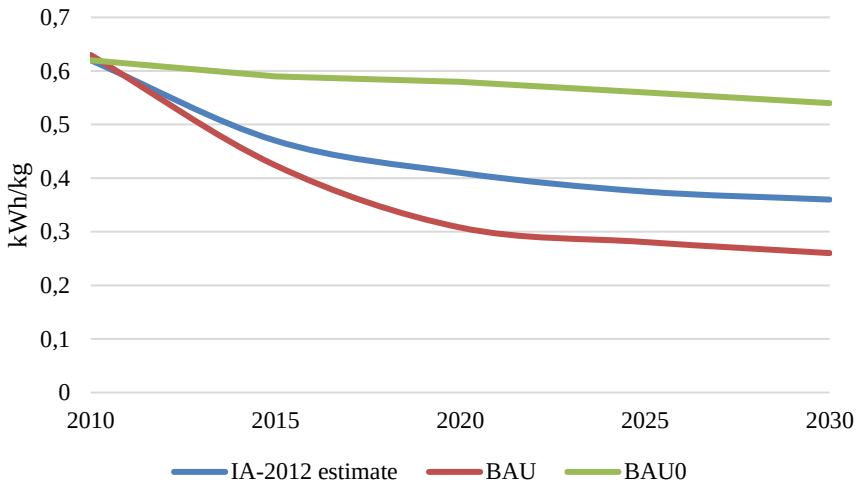


Figure A12-3. Development in specific energy consumption for condensing dryers

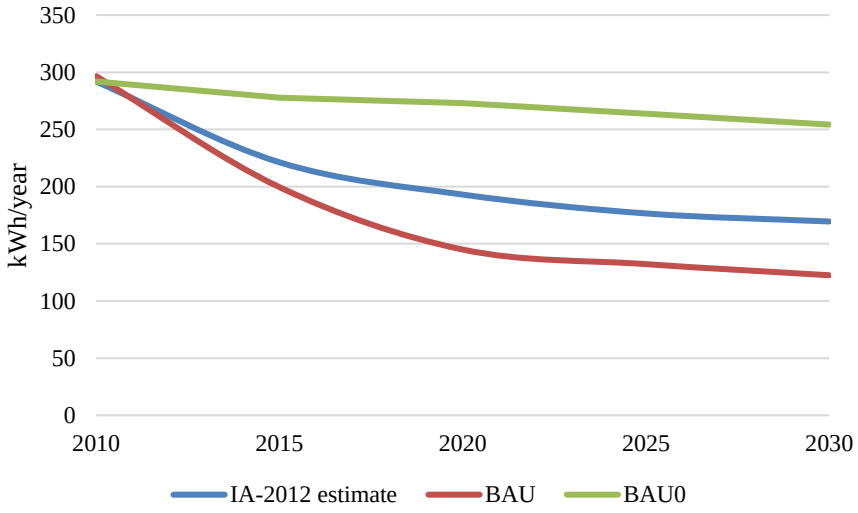


Figure A12-4. Development in yearly energy consumption for condensing dryers assuming 4,4 kg/cycle and 107 cycles/year

The rated capacity of tumble dryers has been steadily increasing since 2010 and this tendency is expected to continue. The development indicates that the current energy labelling regulations (and the equation to establish the EEI) gives the manufacturers an incentive to produce larger tumble dryers to achieve a more beneficial energy label. This will be corrected in the new proposal.

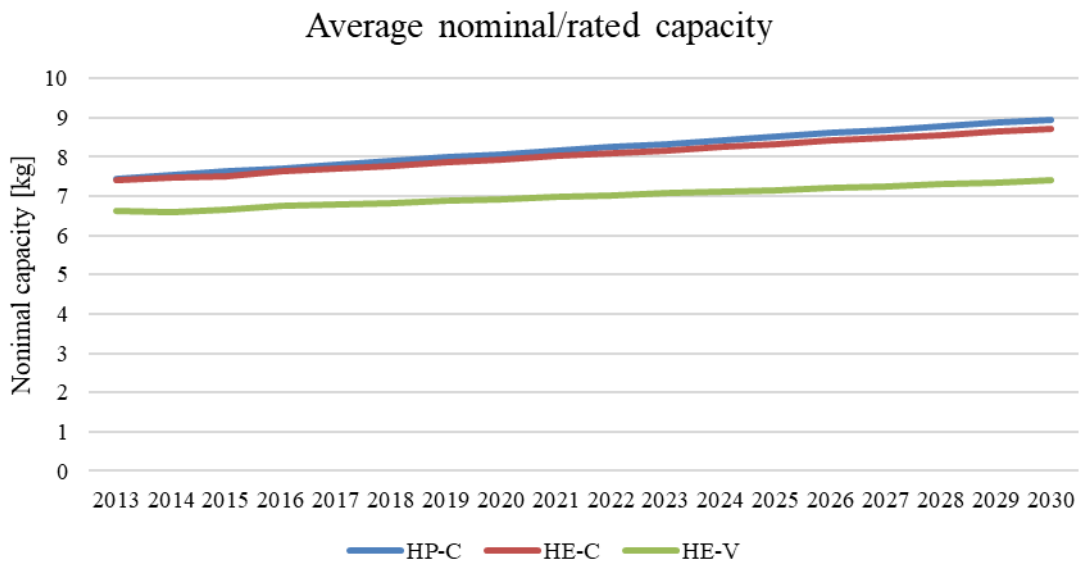


Figure A12-5. Development in rated capacity based on GfK data

1.3. Market share and price of heat pump tumble dryers

The regulations have been able to transform the market towards more energy efficient household dryers, especially condenser dryers. Only small improvements have been achieved for air-vented electric and gas-fired dryers (gas dryers have actually disappeared from the market). The energy efficiency improvement of condenser dryers is primarily due to a large increase of the market share of heat pump technology.

The IA 2012 estimated the total market share of heat pump dryers to be 3 % in 2015 and forecasted an increase up to 4 % in 2020. Nevertheless, the market share of heat pump dryers has increased much more than initially estimated. In 2015 that share was 47 % and 75 % in 2020. This increase may be to a large extent due to the more ambitious categorisation of energy classes in the current energy labelling regulation compared to the old one.

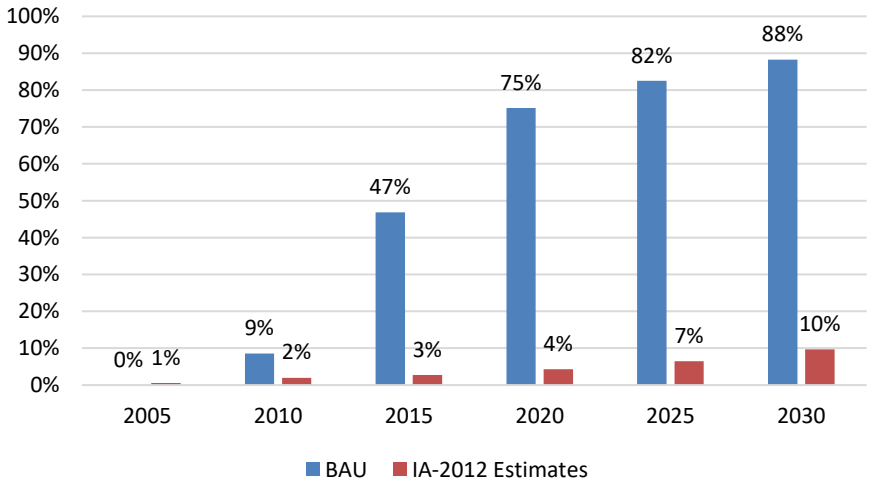


Figure A12-6. Market share of heat pump dryers. Source: data from GfK

While the market share of heat pump dryers increased rapidly, their retail price decreased. The IA 2012 also foresaw a price reduction for heat pump dryers. In the IA 2012 it was estimated that to achieve the LLCC level the consumer purchase costs for

heat pump dryers should not be above 668 EUR per unit, which was well below the purchase cost of 887 EUR per unit estimated for the heat pump BAT technology in the previous preparatory study. But a tendency of decreasing prices was seen in some countries such as the Netherlands and Germany. In the Netherlands heat pump dryers were at the time of the IA 2012 available at costs between 529 EUR and 1.524 EUR per unit and excluding the most expensive ones, the average price was around 760 EUR per unit and decreasing.

Since 2014 the retail price (consumer price including VAT) for heat pump dryers has been close to or below the LLCC level. In 2016 the price was 615 EUR per unit and in 2019 below 600. For heating element dryers, the price has been at the same level since 2015.

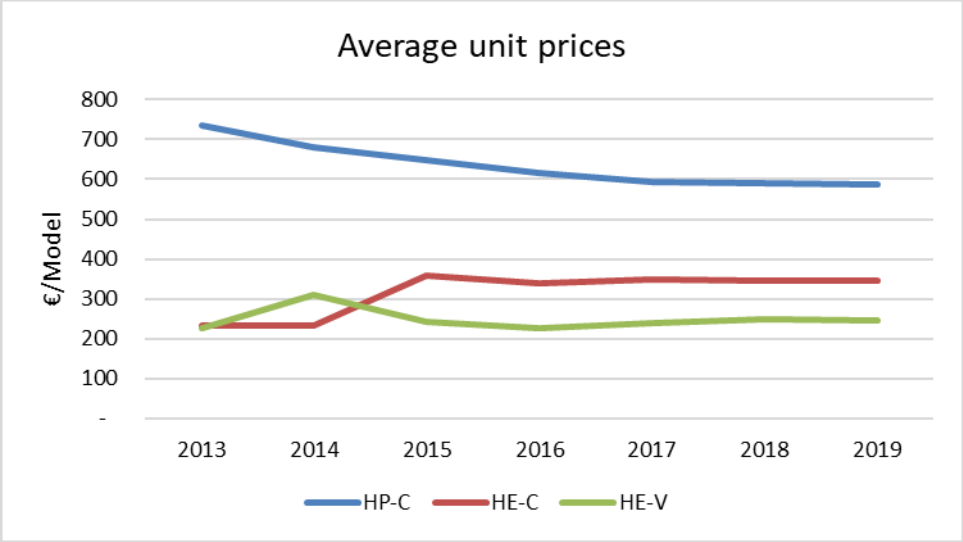


Figure A12-7. Unit retail prices in EUR for household tumble dryers. Source: Data from GfK

Based on the above it is concluded that the current regulations have been effective and achieved the expected savings and market transformation.

2. EFFICIENCY

This section describes how efficient the current regulation has been in delivering the above-mentioned benefits. Table A12-1 gives an overview of the purchase price, lifetime running cost and total lifecycle cost in 2020 of the different household tumble dryer types. All prices have been fixed in 2019 to make them comparable. The estimated electricity consumption and purchase price for BAU0 is based on data from the IA 2012, which have been forecasted to the year 2020.

To make BAU and BAU0 comparable, the updated user behaviour parameters (4,4 kg/cycle and 107 cycles/year) from the review study have been used for both BAU and BAU0.

	Purchase price (EUR/unit)	**Lifetime running cost[115] (Electricity price 2019) (EUR/unit)**	**Total lifecycle cost (EUR/unit)**

[115] Not considering repair cost

From IA 2012 (BAU0)			
Heat Pump - Condenser	527	557	1.084
Heating element - Condenser	527	768	1.296
Heating element - Ventilated	268	708	976
Gas-fired	268	381	649
Weighted Average	514	610	1.124
From IA 2020 (BAU)			
Heat Pump - Condenser	587	278	865
Heating element - Condenser	346	712	1.058
Heating element - Ventilated	246	741	988
Gas-fired	692	416	1.108
Weighted Average	518	395	912

Table A12-1. Purchase price, lifetime running cost and total life cycle cost in 2020 (in 2019 prices)

Table A12-1 shows that, on average, consumers will pay 4 EUR more for a household tumble dryer, however, during a lifetime the consumers will save 212 EUR,

The retail price of heat pump tumble dryers is higher in BAU than expected in BAU0. However, the lifetime running cost and thus the total lifecycle cost is much lower than foreseen in principle, because heat pump dryers are also more efficient than initially estimated. In addition, heating element condenser driers can be purchased at lower cost than expected in the IA 2012.

Overall, it is evaluated that the current regulation has been successful in keeping the total lifecycle cost lower than the BAU0 scenario, except for gas-fired tumble dryers, but these are considered irrelevant because they are almost non-existing in the market.

Table A12-2 shows the total revenue of the whole EU market (acquisition cost), generated from market sales and the total energy cost and consumer lifecycle cost in 2020. Furthermore, the table shows the distribution of the revenue, generated by sales, between the different sectors in the industry. The share of the revenue for the different sectors are based on share distribution used in the impact assessment for refrigerating appliances[116], listed here:

- VAT – 21 %
- Manufactures – 37,5 %
- Wholesale – 8 %
- Retail – 37,5 %

In total the industry has made an extra revenue of 110 million EUR of which 20 million EUR goes to the tax office, 40 million EUR to the manufactures, 10 million EUR to wholesale and 40 million EUR to retail. This means that the manufactures have been able to pass the extra cost of developing better performing household tumble dryers to the consumers, and both manufacturers, wholesale and retailers have benefitted from the increased turnover.

	BAU	**BAU0**	**Difference**
Acquisition cost (purchase price only) (billion EUR)	2,64	2,53	0,11

[116] https://ec.europa.eu/transparency/regdoc/rep/10102/2019/EN/SWD-2019-341-F1-EN-MAIN-PART-1.PDF

Energy cost (electricity price 2019) (billion EUR)	2,01	3,12	-1,11
Consumer lifecycle cost (billion EUR)	4,65	5,65	-1,00
VAT of acquisition cost (billion EUR)	0,46	0,44	0,02
Turnover manufactures in EU (billion EUR)	0,99	0,95	0,04
Turnover wholesale (billion EUR)	0,21	0,20	0,01
Retail turnover (billion EUR)	0,99	0,95	0,04

Table A12-2. Overview of economic impacts in 2020 of the regulation BAU compared to no regulation BAU0

3. RELEVANCE

The Review study on tumble dryers 2019[117] and the evaluation in this Impact Assessment show that the regulations support a transition towards more energy-efficient household tumble dryers effectively and at the same time delivering substantial savings for the consumers.

However, higher savings could be achieved by revising the requirements and updating the calculation methods. This forms the basis of the proposal for an updated regulation. Moreover, the current legislation only regulates the energy efficiency of the appliances. The Review study 2019 revealed that household tumble dryers can contribute to the Commission's Circular Economy Action Plan by including resource efficiency requirements.

[117] Review study on household tumble dryers. Final Report. June 2019.

ANNEX 13: ADDITIONAL CHARTS AND TABLES FOR THE DIFFERENT POLICY OPTIONS

This annex shows some additional charts that are not displayed in sections 6 and 7 due to the 40-page limit mandated for impact assessments. Those charts refer to GHG emissions (environmental impacts), retail turnover (economic impacts) and retail employment (social impacts). Since GHG emissions and retail turnover follow similar patterns as energy consumption and manufacturer turnover respectively, it has been considered that there is no need to display the corresponding figures in section 6 to understand the effects of the different policy options, and that those figures can be carried over to this Annex.

1. GREENHOUSE GAS EMISSIONS

1.1. PO2

In the short term, PO2 will slightly increase GHG emissions with respect to BAU due to the bigger carbon footprint linked to the production of heat pump tumble dryers compared to heating element and air vented tumble dryers. In the longer term, reduction of GHG emissions due to in-use energy savings will offset the initial increase, but both counteracting effects remain similar. Figure A13-1 barely shows any reduction of GHG emissions under PO2 compared to BAU.

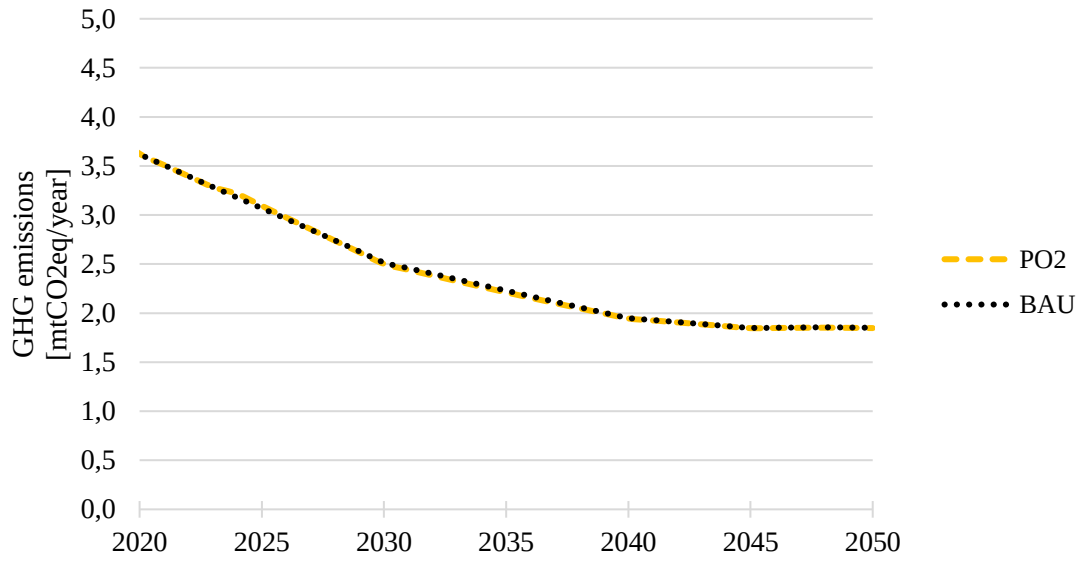


Figure A13-1. GHG emissions for PO2 and BAU

	GHG emission, in-use + embedded [mtCO2e/year]				Savings compared to BAU [mtCO2e/year]		Cumulative savings compared to BAU [mtCO2e]	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2	3,62	3,09	2,50	1,94	0,02	0,01	-0,01	0,16

Table A13-1. GHG emissions for PO2 and BAU

1.2. PO2+PO3

Alike PO2, PO2+PO3 will show in the short term an increase of GHG emissions associated to the embedded energy, although bigger than for PO2 due to a larger

production of heat pump dryers, with higher carbon footprint. This increase will be later counteracted by the decrease of GHG emissions due to in-use energy savings, although the cumulative savings in 2040 will be a little 0,1 mTCO2eq.

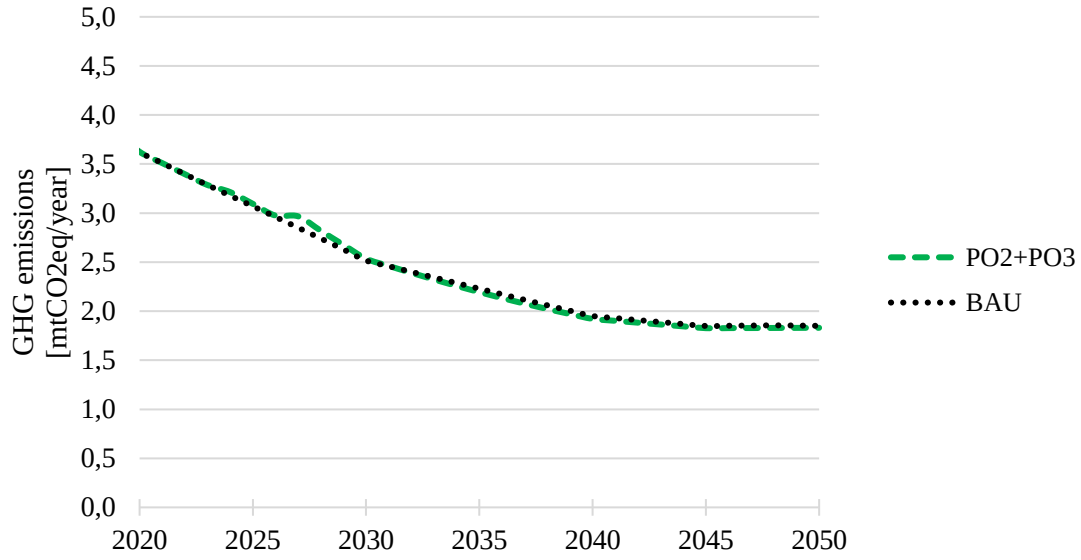


Figure A13-2. GHG emissions for PO2+PO3 and BAU

	GHG emission, in-use + embedded [mtCO2e/year]				Savings compared to BAU [mtCO2e/year]		Cumulative savings compared to BAU [mtCO2e]	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2+PO3	3,62	3,09	2,53	1,91	-0,01	0,04	-0,27	0,1

Table A13-2. GHG emissions for PO2+PO3 and BAU

1.3. PO2+PO4

Until 2033, PO2 prevails over PO4 and subsequently GHG emissions increase due to the higher embedded energy linked to PO2 and to the fact that the effects of PO4 are only realised when some of the tumble dryers bought on 2025 fail and need to be repaired. After 2033, PO4 bends the curve down and yields a net reduction of GHG emissions which is minimum in 2039, after which emissions again go up due to the increase of the manufacturing activity. Nevertheless, GHG emissions will not reach BAU because the slightly better energy efficiency of PO2 over BAU still keeps the emissions lower.

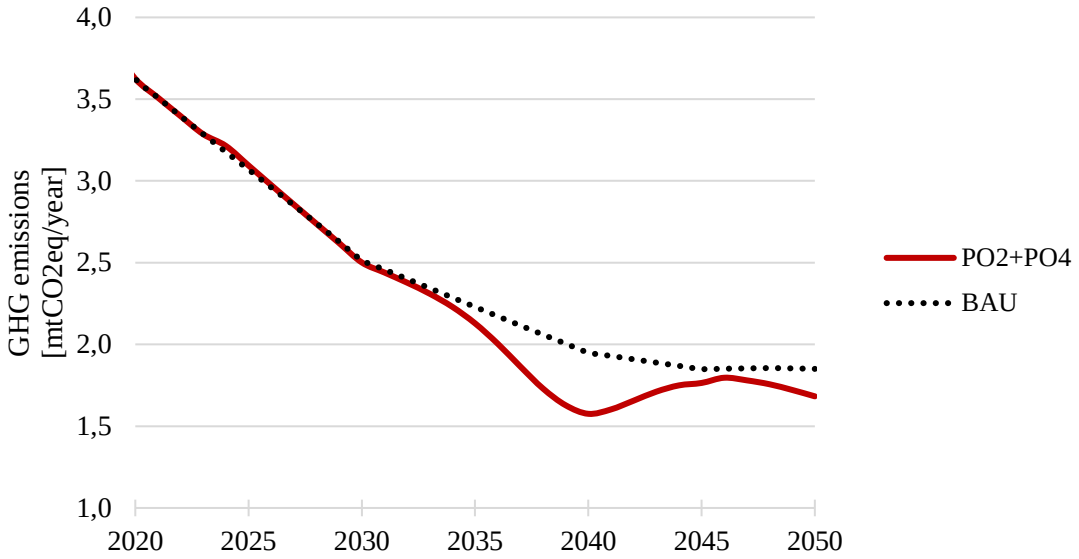


Figure A13-3. GHG emissions for PO2+PO4 and BAU

	GHG emission, in-use + embedded [mtCO2e/year]				Savings compared to BAU [mtCO2e/year]		Cumulative savings compared to BAU [mtCO2e]	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2+PO4	3,62	3,09	2,50	1,57	0,02	0,38	-0,01	1,75

Table A13-3. GHG emissions for PO2+PO4 and BAU

1.4. PO2+PO3+PO4

PO2+PO3+PO4 shows a similar behaviour as PO2+PO4, although with bigger energy losses at the beginning of the period due to the larger embedded energy consumption needed to produce heat pump tumble dryers. Those losses are offset later by means of better energy efficiency.

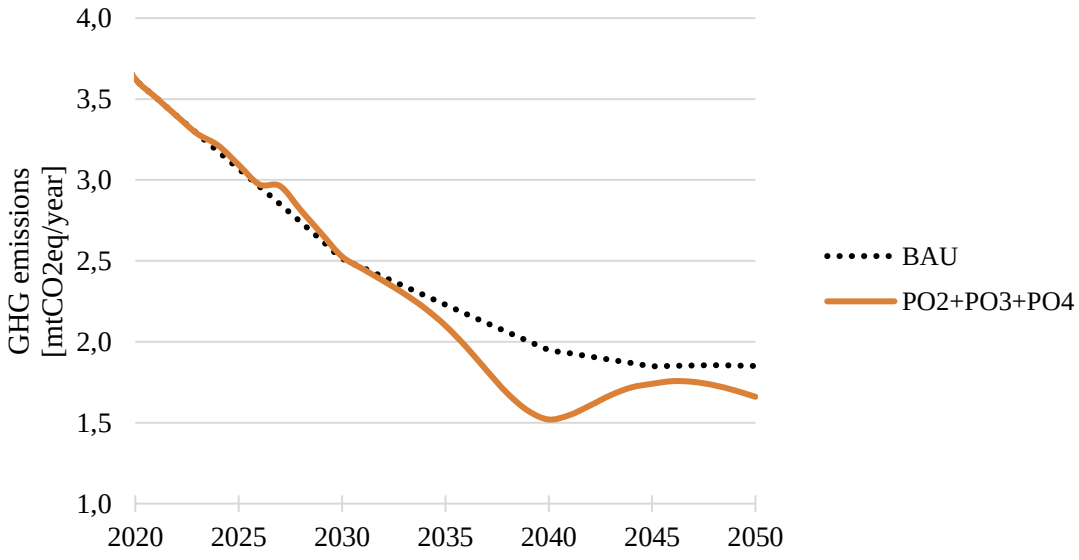


Figure A13-4. GHG emissions for PO2+PO3+PO4 and BAU

	GHG emission, in-use + embedded	Savings	Cumulative

	[mtCO2e/year]				compared to BAU [mtCO2e/year]		savings compared to BAU [mtCO2e]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	3,62	3,07	2,52	1,95			0,00	0,00
PO2+PO4	3,62	3,09	2,53	1,52	-0,01	0,43	-0,27	1,75

Table A13-4. GHG emissions for PO2+PO3+PO4 and BAU

2. RETAIL TURNOVER

2.1. PO2

Retail turnover follows the same trend as manufacturers. PO2 increases slightly the turnover with respect to BAU throughout the period of assessment due to the effect of the energy label, which increases slightly the purchase of the more expensive heat pump appliances.

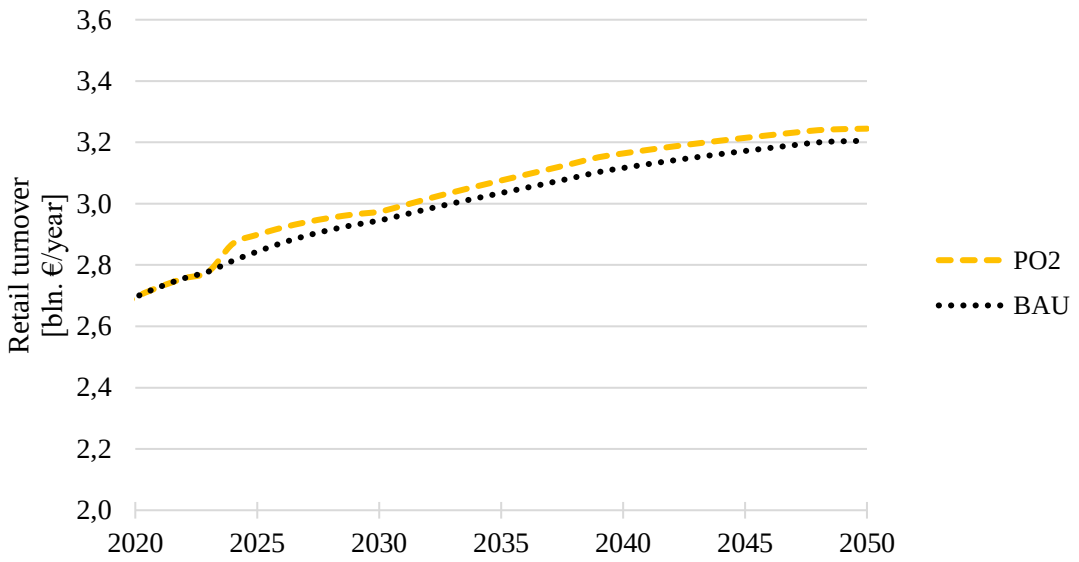


Figure A13-5. Retail turnover for PO2 and BAU

	Retail turnover [bln. €/year]				Difference compared to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	2,69	2,84	2,94	3,12				
PO2	2,69	2,88	2,96	3,16	0,01	0,04	0,19	0,52

Table A13-5. Retail turnover for PO2 and BAU

2.2. PO2+PO3

Retail turnover follows the same trend as the turnover for manufacturers. A quick increase takes place in the first years due to the accelerated production of heat pump tumble dryers, which levels off later because the sales of heat pump tumble dryers will moderate but will remain higher than BAU over the period of analysis.

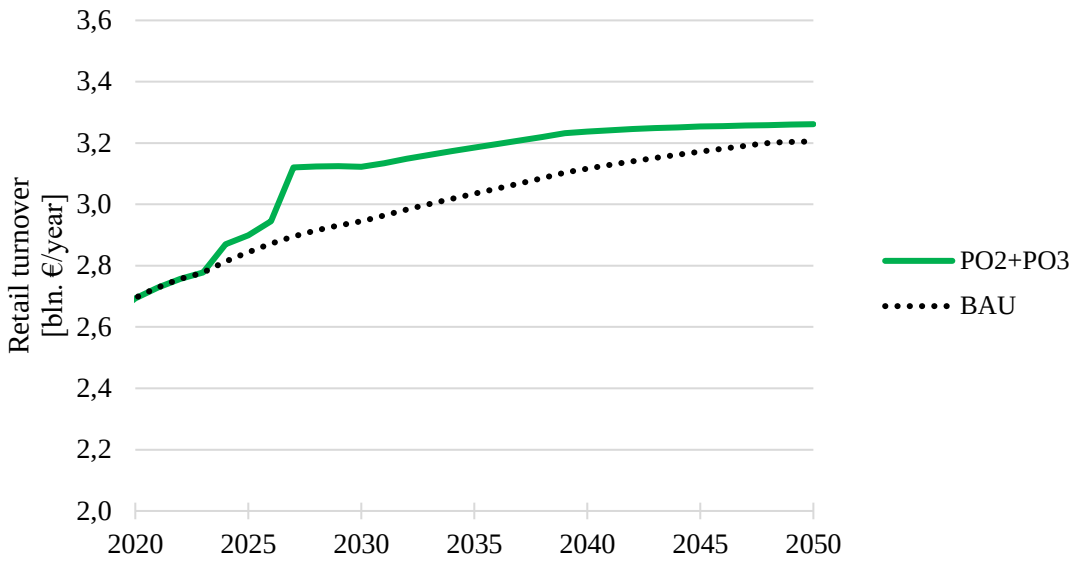


Figure A13-6. Retail turnover for PO2+PO3 and BAU

	Retail turnover [bln. €/year]				Difference compared to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	2,69	2,84	2,94	3,12				
PO2+PO3	2,69	2,88	3,09	3,22	0,15	0,11	0,80	2,08

Table A13-6. Retail turnover for PO2+PO3 and BAU

2.3. PO2+PO4

Retail turnover goes up the first years due to the rescaling of the energy label. The deceleration of sales caused by the increasing repair rate makes the turnover to decrease until 2039, when the turnover raises again as the tumble dryers which lifetime has been extended to 14 years need to be replaced. Overall, the cumulative retailer turnover decreases due to the reduction of sales. This global negative effect is not shared with industry, which loss of sales is compensated by an increase of repairs.

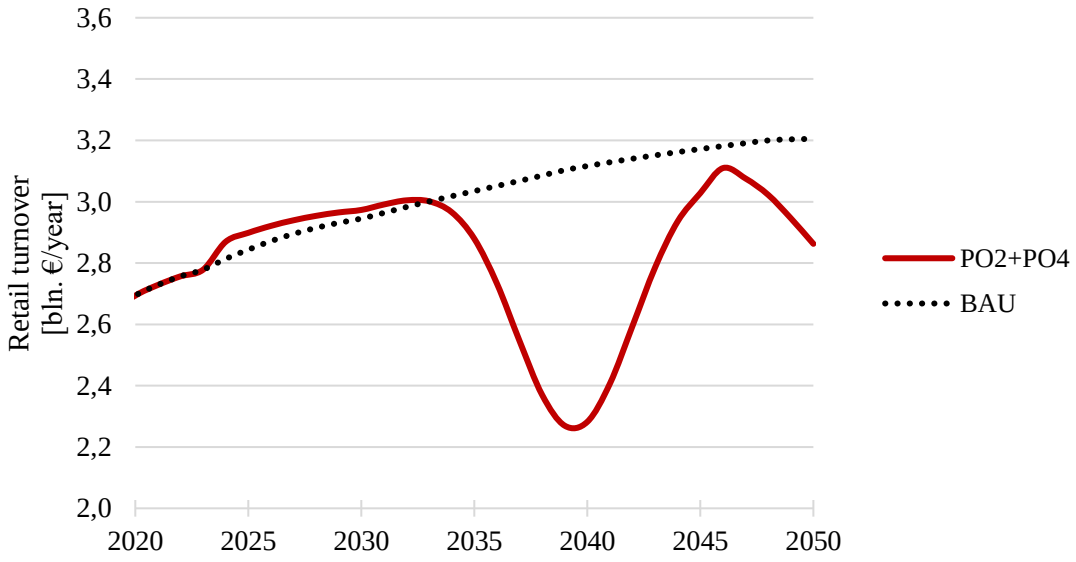


Figure A13-7. Retail turnover for PO2+PO4 and BAU

	Retail turnover [bln, €/year]				Difference compared to BAU [bln, €/year]		Cumulative difference [bln, €]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	2,69	2,84	2,94	3,12				
PO2+PO4	2,69	2,90	2,96	2,28	0,01	-0,84	0,19	-3,26

Table A13-7. Retail turnover for PO2+PO4 and BAU

2.4. PO2+PO3+PO4

For PO2+PO3+PO4, the effect is similar as for PO2+PO4, although the cumulative turnover does not fall so much as in PO2+PO4. This is due to PO3, which increases the total turnover.

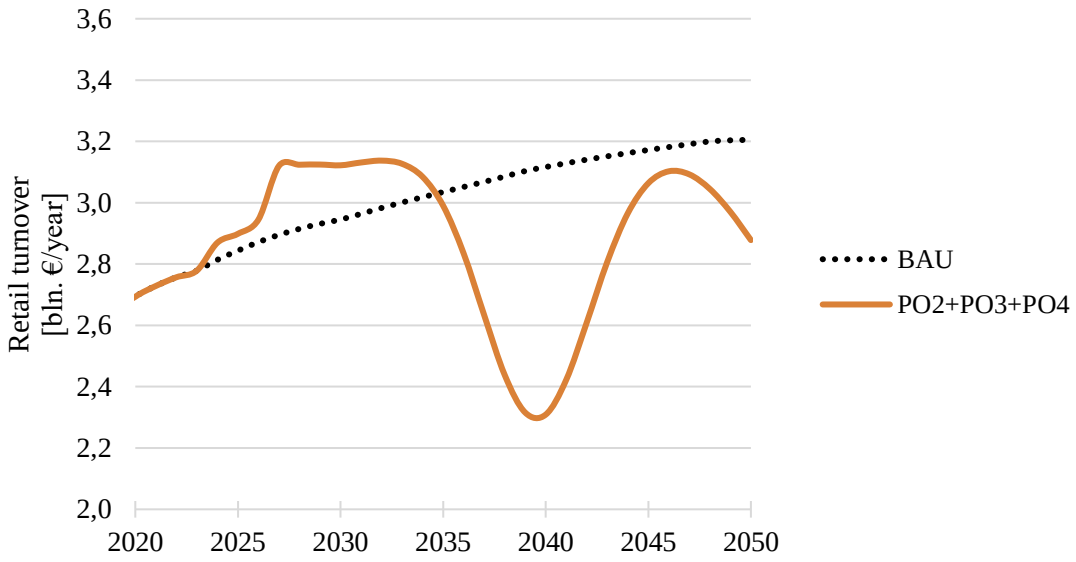


Figure A13-8. Retail turnover for PO2+PO3+PO4 and BAU

	Retail turnover [bln. €/year]				Difference compared to BAU [bln. €/year]		Cumulative difference [bln. €]	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	2,69	2,84	2,94	3,12				
PO2+PO3+PO4	2,69	2,90	3,09	2,30	0,13	-0,86	0,80	-2,32

Table A13-8. Retail turnover for PO2+PO3+PO4 and BAU

3. RETAIL EMPLOYMENT

3.1. PO2

PO2 will slightly increase retail employment, in line with the increase of the turnover commented in section 2.1 of this Annex.

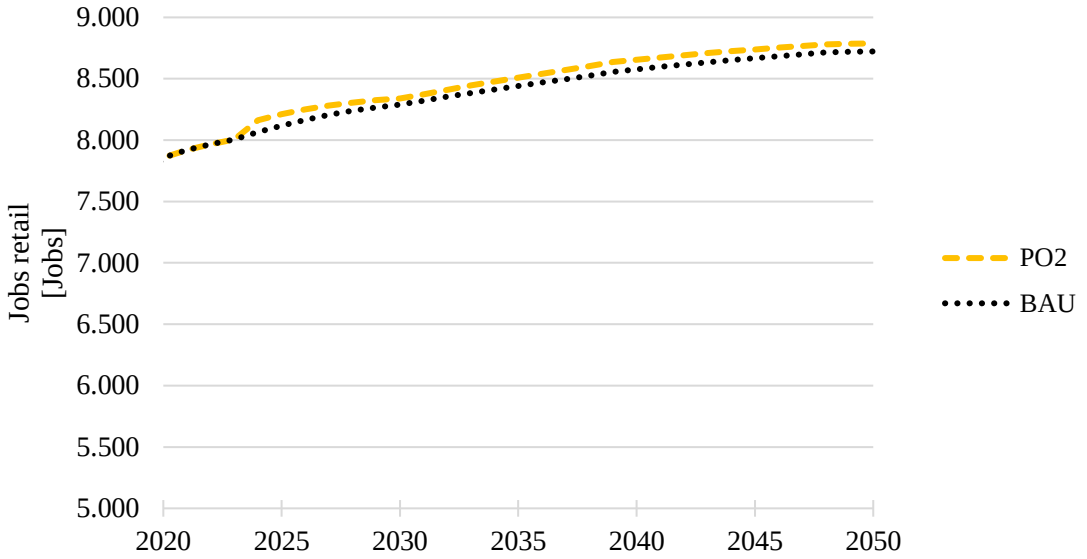


Figure A13-9. Retail employment for PO2 and BAU

	Retail employment [jobs/year]				Difference to BAU (jobs/year)		Cumulative difference (jobs)	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	7.858	8.118	8.290	8.557				
PO2	7.858	8.175	8.311	8.651	21	74	858	2.004

Table A13-9 Retail employment for PO2 and BAU

3.2. PO2+PO3

Same as for turnover, PO2+PO3 encounters an increase of employment, which is sharp during the first years due to the accelerated production of tumble dryers and smooths later as all non-heat pump tumble dryers have been phased out.

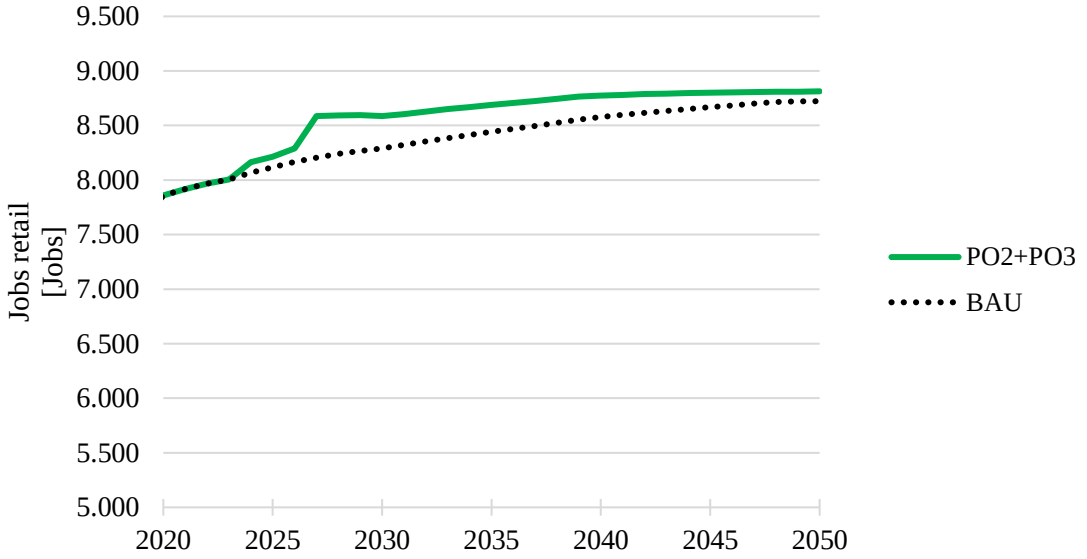


Figure A13-10. Retail employment for PO2+PO3 and BAU

	Retail employment [jobs/year]				Difference to BAU (jobs/year)		Cumulative difference (jobs)	
	2020	**2025**	**2030**	**2040**	**2030**	**2040**	**2030**	**2040**
BAU	7.858	8.175	8.539	8.757	249	180	1.755	3.751
PO2+PO3	7.858	8.118	8.311	8.651	21	74	858	2.004

Table A13-10 Retail employment for PO2+PO3 and BAU

3.3. PO2+PO4

Employment for retail sector follows a similar trajectory as for the industry assessed in section 6.3.2 of the main report, with the difference that the loss of sales caused by PO4 will not be counteracted by an increase of the repair activity (because this one has been allocated to industry), thus the drop of employment being more stressed.

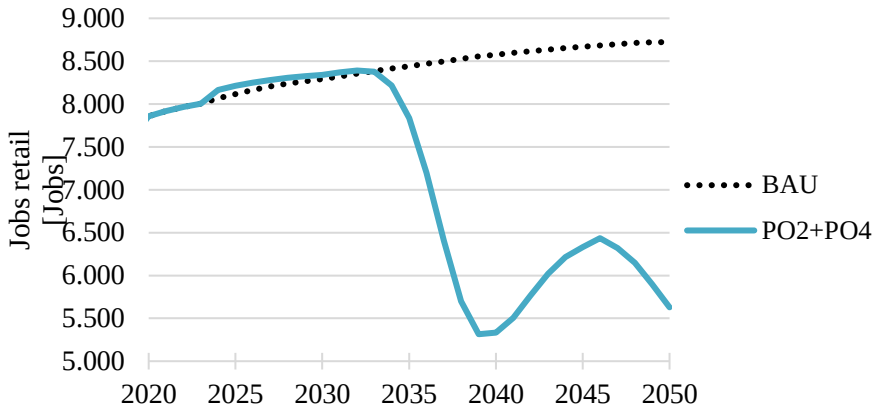


Figure A13-11. Retail employment for PO2+PO4 and BAU

	Retail employment [jobs/year]				Difference to BAU (jobs/year)		Cumulative difference (jobs)	
	2020	2025	2030	2040	2030	2040	2030	2040
BAU	7.858	8.175	8.539	8.757	249	180	1.755	3.751
PO2+PO4	7.858	8.175	8.310	5.348	20	-3.228	233	-13.161

Table A13-11 Retail employment for PO2+PO4 and BAU

3.4. PO2+PO3+PO4

For PO2+PO3+PO4, the effect is similar as for PO2+PO4, although the cumulative employment losses are somewhat smaller. This is because, even if the job losses are bigger than PO2+PO4 by the lower part of the cycle, there is also a larger creation of employment in the early years, thus being the overall cumulative effect still bigger than with PO2+PO4.

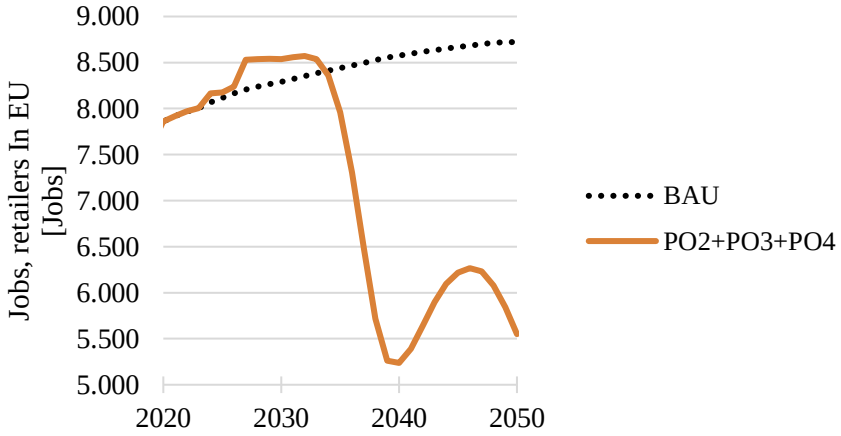


Figure A13-12. Retail employment for PO2+PO3+PO4 and BAU

	Retail employment [jobs/year]	Difference to BAU (jobs/year)	Cumulative difference (jobs)

	2020	2025	2030	2040	2030	2040	2030	2040
BAU	7.858	8.175	8.539	8.757	249	180	1.755	3.751
PO2+PO3+PO4	7.858	8.176	8.537	5.237	226	-3.415	1.276	-11.275

Table A13-12 Retail employment for PO2+PO3+PO4 and BAU

ANNEX 14: THE ECODESIGN AND ENERGY LABELLING FRAMEWORK

The Ecodesign Framework Directive and the Energy Labelling Framework Regulation are framework rules, establishing conditions for laying down product-specific requirements in regulations adopted by the Commission. The Commission's role in the implementation of delegated and implementing acts is to ensure a maximum of transparency and stakeholder participation in presenting a proposal, based on generally accepted data and information, to the European Parliament and Council for scrutiny. Figure A14-1 gives an overview of the procedures leading to a voluntary agreement, Ecodesign requirements or an Energy label.

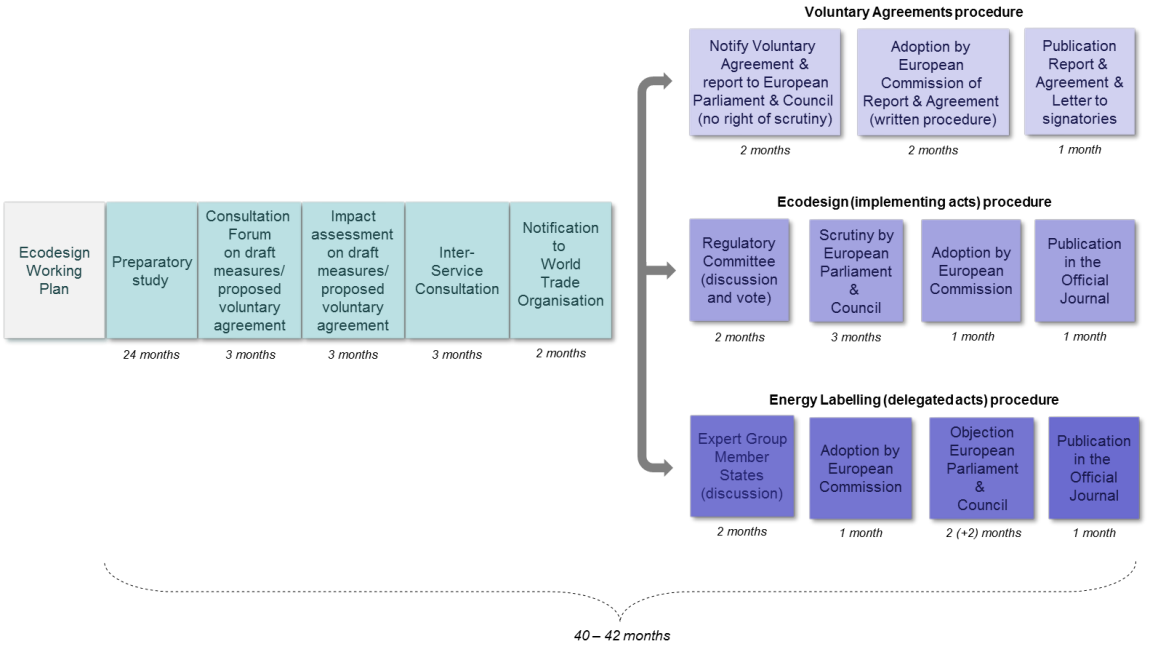


Figure A14-1. Overview of the three processes

Ecodesign and Energy labelling are complementary. Ecodesign implementing measures lay down minimum energy efficiency requirements for a product group and therefore ban worse performing products. In contrast, energy labelling provides objective comparable information to enable consumers to choose better performing products through mandatory rating. In view of the complementarity, energy labelling delegated acts are usually adopted in parallel with Ecodesign implementing acts to ensure coherence between the measures.

The process starts with establishing the priorities for Union action in this area. Priority product groups are selected based on their potential for cost-effective reduction of energy use and greenhouse gas emissions and follows a transparent process culminating in a working plan that outlines the priorities for the development of implementing measures for the next years. Working plans are adopted by the Commission after consultation of the Ecodesign and Energy Labelling Consultation Forum[118]. The forum brings together representatives of Member States and relevant stakeholders.

[118] Register of Commission expert groups and other similar entities (europa.eu)

The Ecodesign Working Plan 2016-2019[119] identified tumble dryers as a priority. A new Ecodesign and Energy Labelling Working Plan 2022-24[120] has now been adopted and retains this product.

Once the product group has been selected, a preparatory study is undertaken by an independent consultant, also involving extensive technical discussions with interested stakeholders. The preparatory study follows the standard Methodology for the Ecodesign of Energy Related Products (MEErP)[121]. Subsequently, the Commission's first drafts of Ecodesign and energy labelling measures are submitted for discussion to the Consultation Forum.

After the Consultation Forum, the Commission drafts an impact assessment, which after RSB approval accompanies the draft implementing measures in the Commission's inter-service consultation. In this and subsequent steps, the Parliament's functional mailboxes for delegated/implementing acts are copied on each message from the Commission services. After the inter-service consultation, stakeholders are alerted when the draft measures are published in the WTO (World Trade Organisation) notification database.

After the WTO notification phase is completed, the two procedures follow different paths. The draft energy labelling delegated act is discussed in a Member States Expert Group where opinion(s) are expressed and consensus is sought but no vote is taken. The draft Ecodesign measure is submitted for vote to the Regulatory Committee.

The European Parliament and Council have the right of scrutiny for a period of up to four months. Within this time, the co-legislators can block the adoption process by the Commission. Parliament committees sometimes discuss draft objections to measures (light bulbs and fridges in 2009) or vote to reject a measure (vacuum cleaners in 2013[122]). On one occasion, an objection was adopted in plenary, blocking the measure for televisions in 2009[123].

Today, 29 Ecodesign Regulations, 14 Energy Labelling Regulations, 3 voluntary agreements and 2 tyre labelling regulations have been implemented. An overview of the measures in force in March 2022 is available on the Europa website[124].

The Market Surveillance Regulation[125] establishes the rules on Union market surveillance and control of products entering the Union market. Market Surveillance Authorities (MSAs), designated by Member States, will verify the conformity of the products with the requirements laid down in the implementing measures and delegated acts. Checks can be done on the product itself or by verifying the technical documentation.

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[119] Communication from the Commission Ecodesign Working Plan 2016-2019 (COM(2016) 773 final)
[120] Ecodesign and Energy Labelling Working Plan 2022-2024 (europa.eu)
[121] DocsRoom - European Commission (europa.eu)
[122] This objection was defeated in ENVI committee by 43 votes against and 4 in favour.
[123] The motivation of the objection was that the EP wanted to delay the discussion of the draft labelling measure so that it would have to become a delegated act under the recast post-Lisbon Energy Labelling Directive in 2010. The measure was subsequently adopted as a delegated act.
[124] summary_overview_of_ed-el_measures_v3.pdf (europa.eu)
[125] Regulation (EC) No 765/2008 of the European Parliament and of the Council of 9 July 2008 setting out the requirements for accreditation and market surveillance relating to the marketing of products and repealing Regulation (EEC) No 339/93 OJ L 218, 13.8.2008, p. 30